Regulating Discretion through Information*

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Abstract

An agent makes decisions for an organization and privately knows how different factors contribute to their shared objectives. The agent also pursues private objectives that the organization does not share. To limit the influence of private objectives while preserving the agent's discretion, the organization offers a menu of disclosure rules, each specifying what the agent learns about the factors relevant to the decision. Information useful for pursuing shared objectives can help serve private objectives. This creates incentives for the agent to misrepresent how different factors contribute to shared objectives when choosing a disclosure rule. I characterize the optimal menu. The optimal rules generally depart from the agent's judgment about the relative importance of factors contributing to shared objectives. The threshold for action can depend directly on a factor that enters only the agent's private objective; banning its direct use can distort the weights assigned to the other factors that contribute to the shared objective.

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1 Introduction

Organizations often entrust decisions to agents who are better informed about how different factors contribute to their shared objectives. A board of education, for example, wants each school to admit the applicants the school considers best suited to its programs. The school knows better than the board how different attributes of a student contribute to that fit. Similarly, a branch manager may know better than the firm's headquarters how different aspects of a worker's skill set contribute to the branch's performance. In these settings, the organization would like to let the agents choose how much weight to place on each factor in the decision making process. However, agents may also pursue private objectives that the organization does not share. A school may prefer applicants who need less financial aid. A branch manager may favor incumbents in the promotion process. In order to limit the influence of these private objectives while preserving discretion, the organization restricts what information agents can use in their decisions.

This paper studies how an organization should design information when an agent privately knows the relative importance of different factors for their shared objective. A screening problem arises: the organization wants decisions to reflect the agent's private knowledge but cannot directly verify the underlying weights of the relevant factors. To resolve this problem, the organization offers a menu of disclosure rules, each specifying what information the agent can use in making decisions. The incentive constraints capture a key trade-off. Information useful for pursuing the shared objective can also help the agent pursue private objectives. The agent can exploit correlation between factors relevant to the shared objective and his private benefit. By shifting weight toward factors that give him higher private benefits, he can serve his private objective while appearing to exercise his judgment about the shared objective. In a concrete example, suppose applicants with stronger extracurricular records tend to need less financial aid.[1] Even under need-blind admission, a school can reduce its aid costs by giving extracurriculars more weight in its admissions protocol. It can then justify this protocol as a reflection of its judgment about which applicants best fit its programs. I characterize the optimal menu of disclosure rules when the organization accounts for this incentive to misrepresent how different factors that contribute to the shared objective should be weighed.

Studies on ban-the-box policies demonstrate how a factor that a regulation seeks to exclude can continue to influence decisions through correlated variables. These policies prevent employers from asking about criminal records on job applications. Agan and Starr (2018) find that employers instead

[1] Chetty et al. (2026) find that applicants from high-income families receive higher nonacademic ratings, including assessments of extracurricular activities and leadership, conditional on test scores. The weight placed on these ratings contributes to their admissions advantage at highly selective private colleges.

used race as a proxy to infer the criminal records of job applicants. The callback advantage of white applicants over comparable black applicants rose from 7 to 43 percent after ban-the-box laws were implemented. In college admissions, Antonovics and Backes (2014) document that University of California campuses placed less weight on SAT scores and more weight on high school grades and family background following Proposition 209's ban on racial preferences. The organization may respond by further restricting the use of information about correlated attributes. Yet these attributes also contain information that both parties value for their shared objective. When the agent privately knows how much weight to place on each attribute, this stronger restriction will limit his ability to pursue the shared objective.

The model. An agent (he) makes a binary decision on behalf of a principal (she). His private information determines how the state contributes to their shared objective. Further, an additional component of the state enters only the agent's payoff as his private objective. For concreteness, the agent decides whether to accept a candidate. The candidate is described by (x_1, x_2, θ). The first two dimensions are the candidate's *mathc attributes*. The last dimension, θ, captures the agent's private benefit from accepting the candidate. In the financial aid example, x_1 and x_2 are the candidate's grades and extracurricular achievements, respectively. The dimension θ then captures the school's financial payoff determined by the candidate's aid status.

The agent has private information: his *type* τ captures the weight that both him and the principal placed on the first attribute. Formally, the candidate's *match quality* is determined by $x^{(\tau)} = \tau x_1 + (1-\tau)x_2$. The principal's payoff from accepting the candidate is given by $x^{(\tau)} - \kappa$, where κ captures the opportunity cost of accepting the candidate. The agent's payoff from accepting the candidate is given by $x^{(\tau)} - \kappa + \theta$, where θ represents the agent's private benefits from accepting the candidate. While both the principal and the agent evaluate a candidate by the same match quality, the agent is more willing to accept candidates that give him higher private benefits. Since the principal does not know the agent's type, she offers a menu of disclosure rules, providing a recommendation to the agent on whether to take the action. The agent observes his type and selects a rule, or opts out. In the baseline model, the agent commits to following the rule's recommendation.[2] In the binary-action setting, each rule is represented by an acceptance policy specifying the probability of recommending acceptance in each state.

[2]Section 5.1 gives a sufficient condition under which the optimal menu remains valid when the agent is free to disregard recommendations. This condition holds in the admissions example. Outside this case, an agent may benefit from selecting another type's rule and reversing its recommendations; the baseline analysis rules out this joint deviation.

Results. Consider first the principal's preferred disclosure rule for each type: accept when match quality exceeds the cost of acceptance, without regard to θ. I call this the first-best disclosure rule. Offering the menu of these disclosure rules is analogous to a blinding regulation: the agent can choose how the match attributes are weighed, but no policy conditions explicitly on θ. Even when every type is willing to participate under its first-best policy, this menu need not induce the agent to choose his own weights. The agent may misrepresent his type to obtain higher expected private benefits, at the expense of the shared objective. Proposition 2 formalizes this: every type finds it optimal to select its own first-best disclosure rule if and only if every rule in the menu delivers the same expected private benefit. This result captures the mechanism illustrated by the evidence discussed above: the characteristic the organization tries to exclude still enters the agent's decision making process through its correlation with attributes the agent is allowed to use.

The main result characterizes the optimal menu of disclosure rules. Every disclosure rule is a *linear scoring rule*: it scores each candidate by a weighted sum of the match attributes, $\sigma x_1 + (1-\sigma)x_2$, and accepts when the score exceeds a threshold that can depend on θ. Theorem 1 gives necessary and sufficient conditions that uniquely determine the scoring weights and threshold adjustments. Relative to the first-best policies, the optimal menu generally exhibits two distortions. First, the scoring weight σ can differ from the agent's own weight τ. Second, the acceptance threshold responds to the agent's private benefit. This adjustment can favor either high- or low-benefit candidates, even though the principal places no direct value on the private benefit itself.

The characterization in Theorem 1 and Corollary 1.1 identifies the economic constraint behind these distortions. When a rule's scoring weights differ from the agent's private type, the principal can improve her payoff by rotating the rule toward the agent's own scoring weights while adjusting the threshold's response to θ to hold the agent's payoff constant. However, such a rotation of the scoring rule also affects the payoff of the rule to other types. In turn, to maintain incentive compatibility, the principal must compensate other types by providing them with enough rent to keep them from switching to the improved rule. Unlike standard screening, the principal has no monetary transfers. Her only instrument to provide such rents is through the disclosure rules themselves. The optimal menu then balances the direct gain from improving a type's rule against the cost of the rent changes needed to preserve incentive compatibility. Theorem 1's necessary and sufficient conditions establish this trade-off as a complete characterization: together with participation and the boundary conditions, it fully determines the optimal menu and rules out other forms of gains from redesigning it.

I then apply the characterization to settings with ordered private benefits, where first-best rules that place more weight on one match attribute deliver higher expected private benefits. I call this the favored attribute. A regulation that bars conditioning on θ then fails in a predictable way. Each

agent places more weight on the favored attribute than his own preferences over match attributes warrant, as doing so raises his private benefit. Theorem 2 shows how the optimal menu responds. The optimal menu conditions directly on θ and compresses the range of expected private benefits across types. This compression preserves the ordering of private benefits across types but reduces the agent's incentive to overweight the favored attribute. The direction of the scoring weights' distortions remain one-sided: the assigned rule departs from the agent's own weights and places more weight on the favored attribute.

Finally, I ask how the principal's payoff changes when the match attributes reveal less about θ. I weaken this dependence while keeping the marginal distributions of match attributes and private benefits fixed. Theorem 3 shows that any incentive-compatible menu can then be redesigned to bring the principal closer to her first-best payoff at every type while preserving the agent's incentives to choose his assigned rule. Consequently, the principal's optimal payoff increases and the cost of screening falls. Less dependence makes it easier to provide information that serves the shared objective without also helping the agent pursue private benefits.

To solve the problem, I reformulate it as a monopoly screening problem in the spirit of Mussa and Rosen (1978). Identifying each disclosure rule with the surplus it generates on each match attribute and the private benefits it delivers, I characterize incentive compatibility by the standard monotonicity and envelope conditions. The reformulated problem departs from standard screening in two ways. First, a given allocation-transfer pair can generally be delivered by many disclosure rules. For any feasible allocation-transfer schedule, the principal can therefore select, for each type, the policy that gives her the highest payoff without affecting the agent's incentive constraints. This inner-layer problem already shows that every disclosure rule in the optimal menu is a linear scoring rule. Second, the reformulated problem features endogenous rent terms. More specifically, the principal's cost of conceding rent to the agent depends on how his threshold responds to θ, which is itself part of the solution. In Mussa and Rosen (1978), a unit of rent costs one unit of transfer, so the virtual type is known before the problem is solved; here the virtual criterion is determined along with the menu. The reformulation nonetheless makes it possible to treat the problem as a control problem over convex rent schedules and to characterize its solution by a first-order condition. Mapping that solution back to disclosure rules through the inner-layer problem delivers the main result: the first-order condition on the rent schedule becomes the first-order condition over weights and adjustments of Theorem 1.

The rest of this section discusses related literature (Section 1.1) and illustrates the mechanism in the admissions setting (Section 1.2). Section 2 sets up the general model. Section 3 characterizes the optimal menu and outlines the analysis behind the solution. Section 4 applies the characterization to show how the private benefits of the first-best policies shape the optimal menu, and derives

comparative statics in the dependence between the match attributes and θ. Section 5 discusses extensions, and Section 6 concludes. Proofs are deferred to a separate appendix.

1.1 Related Literature

This paper contributes to the literature on regulating decisions by restricting the information available to decision-makers. Chan and Eyster (2003) and Fryer et al. (2008) show how restrictions on the use of race can distort admissions decisions as institutions pursue diversity through other criteria. Ellison and Pathak (2021) quantify the efficiency costs of race-neutral alternatives in Chicago's exam schools. The evidence in Agan and Starr (2018) and Antonovics and Backes (2014), discussed above, likewise illustrates how decision-makers respond to restrictions by changing their use of permitted information. I study the design of the regulation itself when the principal does not know how the agent should weigh these permitted attributes. These weights determine an objective that both parties share, but the agent can misrepresent them to pursue private benefits. The optimal menu therefore balances preserving the agent's judgment about match quality against limiting his incentives to use correlated attributes as proxies.

The paper also relates to information design with fairness concerns. Liang, Lu, Mu, and Okumura (2026) characterize a fairness-accuracy frontier and study a designer who regulates the inputs available to an agent choosing an algorithm. They show that excluding group identity can be strictly suboptimal, and their input-design analysis takes the agent's objective as known. Here, the agent privately knows the weights defining the shared objective, so the principal must also account for his choice among disclosure rules. Conditioning on the characteristic that generates private benefits can improve the principal's payoff by reducing incentives to misrepresent these weights, even though that characteristic has no direct value to her. A related distinction concerns privacy. Strack and Yang (2024) characterize signals that leave beliefs about a protected characteristic unchanged. Such a restriction concerns what the agent can infer: a rule that does not explicitly condition on the characteristic need not preserve privacy when permitted attributes are correlated with it. My analysis asks how this dependence affects the optimal menu and the cost of screening.

The closest theoretical benchmarks are Kolotilin et al. (2017) and Candogan and Strack (2023). In the binary-action, scalar-state environment of Kolotilin et al. (2017), a single experiment can replicate the interim outcomes of any incentive-compatible menu. Candogan and Strack (2023) allow more general action spaces and multiple privately informed agents, with utilities affine in a common scalar state. They establish the existence of optimal laminar partitional mechanisms and derive bounds on their complexity. In my model, the agent's private information instead determines how several attributes contribute to an objective shared by both parties. Changing

the type changes the relative weights on these attributes, so payoff-relevant information cannot generally be summarized by the posterior mean of a common scalar state. The conflict comes from an additional state component that only the agent values. The screening problem is therefore to preserve the agent's judgment about match quality while limiting his incentive to select rules that deliver greater private benefits. I characterize the optimal menu through two jointly determined instruments: distortions to the scoring weights and adjustments to the acceptance threshold based on the private benefit.[3]

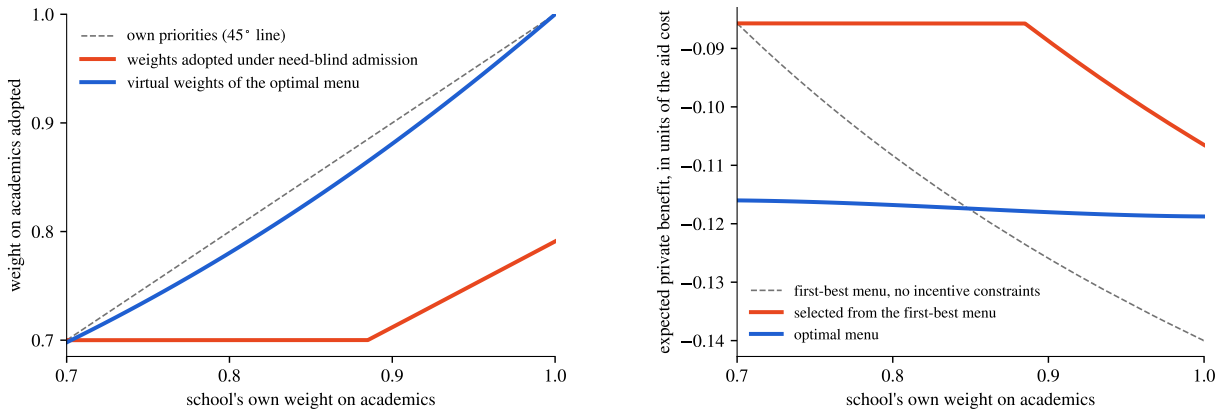
Other work uses menus of information to screen agents in different economic environments. Li and Xu (2026) study test design when a principal and an agent must unanimously agree to undertake a project and the principal has limited commitment in communicating test results. Hancart (2026) studies a decision-maker who learns about an agent from the agent's choice among feasible tests. Here, the private information concerns how to evaluate the state, rather than which state has occurred, and the principal designs the disclosure rules themselves. With monetary transfers, Bergemann et al. (2026) jointly design buyer information and a nonlinear pricing mechanism. In my setting, the principal must provide incentives entirely through the rules governing the decision.

Methodologically, I relate the information-design problem to monopoly screening in the tradition of Mussa and Rosen (1978). The difference between the surpluses generated on the two match attributes serves as the allocation, and the remaining component of the agent's payoff serves as a transfer. This representation yields the usual monotonicity and envelope conditions. The key difference is that the allocation and the transfer must be generated by the same disclosure rule. Providing additional rent changes which states lead to acceptance, so its marginal cost to the principal is determined jointly with the optimal menu. The possibility that participation binds at interior types also connects the analysis to countervailing incentives and type-dependent participation constraints (Lewis and Sappington, 1989; Maggi and Rodríguez-Clare, 1995; Jullien, 2000). I use this screening formulation to characterize both the weights on the shared attributes and the rule's response to private benefits.

1.2 An Illustration: Admissions with Financial Aid

To fix ideas, consider in more detail the admissions example introduced earlier. A board of education regulates the admission protocols of the schools under its authority. Applicants are described by grades, extracurriculars, and aid status. Each school evaluates an applicant's match quality as a weighted sum of grades and extracurriculars. One can interpret a school's weighting of

[3]Related analyses of optimal disclosure to privately informed receivers include the linear-programming approach of Kolotilin (2018) and the interval characterization of Guo and Shmaya (2019).



(a) Weights

(b) Expected private benefits

Figure 1: Panel (a): weight on academics: priorities vs adopted. Panel (b): expected private benefit

grades and extracurriculars as its admissions priorities. The board shares each school's priorities: it wants a school that prioritizes academics to admit applicants with strong academic achievements, and similarly for schools that prioritizes extracurriculars. The school, however, also takes into account its financial incentives when evaluating students who need financial aid. Suppose that applicants who need aid are, on average, as strong on grades as other applicants but weaker on extracurriculars.[4] A natural regulation is need-blind admission: each school may choose how to weigh grades and extracurriculars, but cannot condition their admissions decision on aid status.

Figure 1 shows the outcome of need-blind admission when schools choose freely among the admissions protocols it allows. In panel (a), the dashed line plots each school's own weight on grades, while the orange line plots the weight used by its chosen admissions protocol. Every school adopts weights that favor extracurriculars more than its own priorities do, and the majority of schools adopt the most extracurriculars-heavy protocol available. The distortion is intuitive: admissions protocols that weight extracurriculars more heavily admit fewer students who need financial aid. A school can then shift its weights toward extracurriculars to reduce its aid costs while sacrificing little match quality. Under need-blind admission, schools use extracurriculars as a proxy for aid status to serve their financial incentives. Panel (b) reinforces this intuition. The dashed line plots the expected private benefit a school receives from the admissions protocol matching its own priorities, and the orange line the private benefit it receives from the protocol it actually adopts. The

[4]Chetty et al. (2026) find that applicants from high-income families receive higher nonacademic ratings, including assessments of extracurricular activities and leadership, conditional on test scores. In the example below, 40 percent of applicants need aid; grades are uniformly distributed and unrelated to aid status; extracurriculars of aid applicants are concentrated at low values and those of other applicants at high values; and schools' weights on academics are spread evenly between 0.7 and 1.

gap between the two curves shows the expected aid savings a school obtains by shifting its weight toward extracurriculars in its admissions protocol.

The blue curves in Figure 1 preview the main characterization result by illustrating the optimal menu that solves the board's problem. In the optimal menu, admission protocols explicitly condition on aid status. The blue line in panel (b) shows the result: the expected private benefit is nearly the same across the whole menu. With private benefits nearly equal, schools now have reduced incentives to shift weight toward extracurriculars in their chosen admissions protocols. This is also reflected by the blue curve in panel (a): the school's weighting over academics and extracurriculars is now much more aligned with its priorities under the optimal menu than it was under the need-blind regulation. The rest of the paper studies the general model and makes these intuitions precise.

2 Framework

A principal (she) regulates the information an agent (he) can use in a binary decision problem. The agent has private information, his *type*, which determines the relative importance of two dimensions of the *state* for both parties' payoffs. Formally, the state space $\Omega = X_1 \times X_2 \times \Theta$ has three dimensions, $X_i = [0,1]$ and $\Theta \subseteq \mathbb{R}$. The first two dimensions, (x_1, x_2), are the *match attributes* of the state. The third dimension, θ, is the agent's *private benefit*: a component of the state that the agent values but the principal does not. In the leading examples, the state describes a candidate. The match attributes are the dimensions on which both parties evaluate the candidate. The private benefit captures the agent's additional payoff from accepting the candidate.

The agent's type $\tau \in \mathcal{T} = [0,1]$ determines the *match quality*

$$x^{(\tau)} \coloneqq \tau x_1 + (1-\tau)x_2,$$

a τ-weighted average of the two match attributes. The agent takes action $a = 0, 1$, with payoffs depending on the state ω, the type τ and the action a:

$$\begin{aligned} \text{Principal:} \qquad & v_\tau(\omega, a) = a\left(x^{(\tau)} - \kappa\right), \\ \text{Agent:} \qquad & u_\tau(\omega, a) = a\left(x^{(\tau)} - \kappa + \theta\right), \end{aligned}$$

where $\kappa \in (0,1)$ is a constant that captures the opportunity cost of taking action $a = 1$. Interpreting the payoffs, while both parties would like the agent to take the action when $x^{(\tau)}$ is high, the agent's decision under perfect information will be biased by θ.

The state is drawn from a common prior $\pi_0 \in \Delta(\Omega)$. The agent's type is drawn independently

of the state and uniformly over $\mathcal{T} = [0, 1]$.[5] I impose the following assumptions on the state space and the prior π_0:

Assumption 1.

(A1) $\Theta \subset \mathbb{R}$ is a compact interval with $0 \in \Theta$.

(A2) π_0 admits a density f on Ω that is bounded above, $f \leq f_{\max} < \infty$, and bounded away from zero near the pivotal state $(\kappa, \kappa, 0)$: $f \geq f_{\min} > 0$ on $B\big((\kappa, \kappa, 0), r_0\big) \cap \Omega$ for some $r_0 > 0$.

(A3) $\mathbb{E}[x_i + \theta] < \kappa$ for $i = 1, 2$.

(A1)–(A2) are regularity conditions. In particular, (A2) requires a bounded density that is positive near $(\kappa, \kappa, 0)$, where both parties are indifferent between the two actions for every type. (A3) ensures that an uninformed agent will choose $a = 0$, regardless of his type: his uninformed payoff from taking $a = 1$ is a convex combination of the two values in (A3).

Remark. The results extend to a model where both parties' payoffs depend linearly on θ:

$$\begin{aligned} \text{Principal:} \qquad & v_\tau(\omega, a) = a\left(x^{(\tau)} - \kappa + \beta_P \theta\right), \\ \text{Agent:} \qquad & u_\tau(\omega, a) = a\left(x^{(\tau)} - \kappa + \beta_A \theta\right). \end{aligned}$$

The key assumption for the model is that, for a given state and action, the payoff difference between the agent and the principal does not depend on the agent's type.[6]

Example 1 (Admissions with financial aid)**.** A board of education (the principal) regulates the admission protocols of the schools under its authority (the agents). Each applicant is characterized by their academic achievement x_1, extracurricular achievement x_2, and need for financial aid $\theta \leq 0$, with more negative θ's representing greater need. Schools weigh the two achievements differently, and the board shares each school's weighting: it wants a school that prioritizes academics to admit strong academic students, and a school that prioritizes extracurriculars to admit strong extracurricular students. The school, however, also bears the cost of aid its admitted students need, which the board does not.

Example 2 (Headquarters and branches)**.** A firm's headquarters (the principal) regulates the promotion decisions of a branch manager (the agent). Each worker has technical skill x_1 and client-facing

[5] The reduction to direct menus and the incentive analysis of Section 3 hold for any continuous, strictly positive type density. Uniformity is used in the characterization of the optimal menu in the main text, as it rules out bunching and simplifies the main result. Section 5 discusses general type densities.

[6] Put $\tilde{x}_i := x_i + \beta_P \theta$ and $\tilde{\theta} := (\beta_A - \beta_P)\theta$. Then $\tilde{x}^{(\tau)} = x^{(\tau)} + \beta_P \theta$, the principal's payoff is $a(\tilde{x}^{(\tau)} - \kappa)$ and the agent's is $a(\tilde{x}^{(\tau)} - \kappa + \tilde{\theta})$. The pivotal state is unchanged, a density is carried to a density by the linear change of variables, and (A1)–(A3) are imposed on $(\tilde{x}_1, \tilde{x}_2, \tilde{\theta})$, so that (A3) reads $\mathbb{E}[x_i + \beta_A \theta] < \kappa$.

skill x_2. The branch manager privately knows how each dimension of a worker's skill set contributes to the branch's performance. He also favors promoting workers who have worked in the branch for longer periods of time, with θ capturing a worker's incumbency status, a factor headquarters values less than the branch manager does. Suppose incumbents tend to have stronger client-facing skills because they have more experience with the branch's clients. Then, the branch manager can favor incumbents by placing greater weight on client-facing skills, even when promotion decisions cannot condition directly on incumbency.

The principal regulates the information the agent receives through a menu of binary recommendation rules. Each rule is represented by an acceptance policy, $q : \Omega \to [0, 1]$, where $q(\omega)$ is the probability that action $a = 1$ is recommended in state ω. "Acceptance" is a generic label for taking action $a = 1$. In the admissions example, an acceptance policy is an admissions protocol. The principal and agent's payoffs from implementing an acceptance policy are given by

$$V_\tau(q) = \mathbb{E}\big[(x^{(\tau)} - \kappa)\, q(\omega)\big], \qquad U_\tau(q) = \mathbb{E}\big[(x^{(\tau)} - \kappa + \theta)\, q(\omega)\big].$$

A *menu* is a set of acceptance policies. The principal commits to the menu. After observing his type and before the state is realized, the agent either selects a policy and commits to implementing its recommendation, or opts out and takes action $a = 0$. [7] A *direct mechanism* assigns an acceptance policy q_τ to each reported type τ. The mechanism satisfies incentive compatibility (IC) and individual rationality (IR) if truthful reporting and participation are optimal for every type:

$$U_\tau(q_\tau) \geq U_\tau(q_{\tau'}) \qquad \text{for all } \tau, \tau' \in [0, 1], \tag{IC}$$

$$U_\tau(q_\tau) \geq 0 \qquad \text{for all } \tau \in [0, 1]. \tag{IR}$$

The revelation principle allows the analysis to focus, without loss of generality, on direct mechanisms that satisfy incentive compatibility and individual rationality.

The timing is as follows.

1. The principal posts a menu of acceptance policies.

2. The agent observes his type and selects a policy, or opts out.

3. The state realizes, independently of the type, and the selected policy is applied to it.

[7] This assumption captures settings in which the agent has discretion over the choice of an evaluation protocol but must respect the decisions prescribed by the protocol he adopts. A school, for example, may choose another school's admissions protocol, but it may not then reject the applicants that protocol recommends admitting and admit those it recommends rejecting. Under (IR) and (A3), a type who selects his own policy follows its recommendations voluntarily. The additional restriction therefore rules out the joint deviation of selecting another type's policy and inverting its recommendations. Section 5.1 characterizes the additional incentive constraints when such deviations are allowed and gives a sufficient condition under which they hold for every incentive-compatible and individually rational menu. This condition holds in the admissions example.

4. The recommended action is implemented and payoffs realize.

The principal's problem is therefore

$$\max_{(q_\tau)_{\tau\in[0,1]}} \int_0^1 V_\tau(q_\tau)\, d\tau \quad \text{subject to (IC) and (IR).}$$

3 The Optimal Menu

This section characterizes the solution to the principal's problem. The analysis begins with the benchmark in which the principal knows the agent's type, then examines when a menu of first-best policies is incentive compatible under private information. I next establish that the optimal menu consists of linear scoring rules: each rule scores the state by a weighted sum of the match attributes and recommends acceptance exactly when that score exceeds a θ-dependent threshold. The optimal menu is then governed by two schedules: the scoring weight on the match attributes and the threshold adjustment assigned to each type. Theorem 1 characterizes these schedules as the unique solution to a system of necessary and sufficient optimality conditions. I then interpret the trade-offs underlying the principal's optimal choice of scoring weights and threshold adjustments. The section closes by illustrating the optimal menu for the admissions example discussed in the introduction. Proofs are deferred to the appendix.

Scoring rules and the private benefit of a policy. As a preliminary analysis, I rewrite the principal's and the agent's payoffs in terms of surpluses on match attributes and private benefits. For an acceptance policy q, let

$$S_i(q) \coloneqq \mathbb{E}\big[(x_i - \kappa)\, q(\omega)\big], \quad i = 1, 2, \qquad\qquad T(q) \coloneqq \mathbb{E}\big[\theta\, q(\omega)\big].$$

$S_i(q)$ is the surplus the policy generates on match attribute i, and $T(q)$ is the *expected private benefit* the policy delivers. Since $x^{(\tau)} - \kappa = \tau(x_1 - \kappa) + (1-\tau)(x_2 - \kappa)$, the payoffs of Section 2 can be rewritten as

$$V_\tau(q) = \tau S_1(q) + (1-\tau) S_2(q), \qquad U_\tau(q) = V_\tau(q) + T(q). \tag{1}$$

The agent's payoff from any policy is the principal's payoff plus the expected private benefit. The two parties therefore disagree about one thing only: how much private benefit a policy delivers.

A *linear scoring rule* with weight $\sigma \in \mathbb{R}$ and adjustment $\gamma \in \mathbb{R}$ is the policy

$$q^{\sigma,\gamma} \coloneqq \mathbb{1}\big\{\sigma x_1 + (1-\sigma) x_2 - \kappa + \gamma\theta > 0\big\}.$$

It scores the state by the weighted sum of the match attributes, with weight σ on the first attribute and $1-\sigma$ on the second, and takes the action when the score exceeds the threshold $\kappa - \gamma\theta$. The

adjustment determines how the threshold responds to the private benefit. A positive adjustment lowers the acceptance threshold as θ increases, while a negative adjustment raises it. In the admissions example, a negative adjustment makes the admissions protocol more lenient toward applicants who need financial aid.

Two instances of linear scoring rules are useful here. The principal's *first-best policy for type* τ is $q^{\tau,0}$: it scores the state by the type's own match quality and ignores the private benefit. It is the policy the principal would choose for the type if she faced neither incentive nor participation constraints, since $q^{\tau,0}$ maximizes V_τ over all policies. The agent's own rule, the policy he would follow under full information and no regulation, is $q^{\tau,1}$. More generally, for every γ, the rule $q^{\tau,\gamma}$ is the unique maximizer over all policies of

$$V_\tau(q) + \gamma\, T(q) = (1-\gamma)\, V_\tau(q) + \gamma\, U_\tau(q). \tag{2}$$

A rule with the agent's own weight, $\sigma = \tau$, and an adjustment in $[0,1]$ is thus the policy that maximizes a weighted sum of the two parties' payoffs, with Pareto weight γ on the agent.

Known type benchmark. Suppose first that the principal knows the agent's type τ. Her problem is then a persuasion problem with a single receiver: choose a policy to maximize $V_\tau(q)$ subject to the agent's willingness to participate, $U_\tau(q) \geq 0$.[8] Call the type *unconstrained* if its first-best policy is acceptable to the agent, $U_\tau(q^{\tau,0}) \geq 0$, and *constrained* otherwise.

Proposition 1 (Known type)**.** *Suppose the principal knows the agent's type τ. Her optimal policy is the linear scoring rule $q^{\tau,\gamma}$ with the agent's own scoring weight, $\sigma = \tau$, and adjustment $\gamma = \gamma^{\mathrm{kt}}(\tau)$.*

If τ is unconstrained, $\gamma^{\mathrm{kt}}(\tau) = 0$. If τ is constrained, $\gamma^{\mathrm{kt}}(\tau)$ is the unique adjustment in $(0,1)$ satisfying

$$U_\tau\big(q^{\tau,\gamma^{\mathrm{kt}}(\tau)}\big) = 0.$$

When the type is known, the principal retains the agent's scoring weight, $\sigma = \tau$, and adjusts the acceptance threshold only when participation requires it. An unconstrained type receives the principal's first-best policy. A constrained type receives a policy that accommodates the private benefit just enough to make the agent indifferent to opting out. By (2), this rule maximizes a weighted sum of the two payoffs, with the adjustment determining the weight on the agent's payoff needed to satisfy participation. In the admissions example at the parameters of Section 1.2, every school type is unconstrained: the board could impose its first-best protocol for each school if it knew the school's priorities.

[8] Following the recommendation is then voluntary, by the argument in Section 2.

Is first-best payoff achievable? Now suppose the agent's type is private. A natural candidate menu offers the first-best policy for each type. I call $\{q^{\tau,0}\}_{\tau\in[0,1]}$ the *first-best menu*. It ignores the private benefit entirely and is the model's analogue of the need-blind policy in the admissions example: each policy uses the corresponding type's own scoring weight, and no policy conditions on θ. If every type is unconstrained, the menu satisfies (IR), so the question is whether it satisfies (IC).

The answer depends on the private benefits delivered by the policies on this menu. Define the first-best benefits by $T^{\text{fb}}(\tau) \coloneqq T(q^{\tau,0})$, the expected private benefit under the first-best policy for type τ. Consider a type τ contemplating the first-best policy for a nearby type r. Selecting $q^{r,0}$ instead of $q^{\tau,0}$ changes his payoff by

$$U_\tau(q^{r,0}) - U_\tau(q^{\tau,0}) = \big[T^{\text{fb}}(r) - T^{\text{fb}}(\tau)\big] - \big[V_\tau(q^{\tau,0}) - V_\tau(q^{r,0})\big].$$

The second bracket is the loss in match quality, measured by his own weights, from scoring with a neighbor's weights. It is nonnegative, since $q^{\tau,0}$ maximizes V_τ, and it is of second order in $|r-\tau|$ by the envelope theorem. The first bracket is the change in private benefit, and it is of first order wherever T^{fb} has a nonzero slope. Therefore, whenever the first-best benefits vary, some type profits from selecting a neighbor's scoring rule, and the first-best menu is not incentive compatible. Conversely, when T^{fb} is constant, every policy on the menu delivers the same private benefit, say $\bar{T}$. Each type's payoff from any policy on it is $V_\tau(q) + \bar{T}$, and the agent ranks the policies exactly as the principal ranks them for his type: he selects $q^{\tau,0}$.

Proposition 2 (The first-best menu)**.** *Suppose every type is unconstrained. The first-best menu is incentive compatible if and only if T^{fb} is constant on $[0,1]$. In that case, it is the optimal menu.*

The second claim follows from the observation that the first-best menu allows the principal to achieve her payoff as if she was taking the action with full information. The first claim shows that, even when participation poses no obstacle, differences in the private benefits delivered by first-best policies prevent the principal from attaining her first-best payoff. In the admissions example, the first-best benefits are strictly decreasing in the school's weight on academics: academic-heavy cutoffs admit more applicants who need aid, because those applicants fall short on extracurriculars rather than on grades. Section 4 gives makes this intuition precise: every school then gains by moving toward the weight of a neighbor with a more extracurriculars-heavy priority. This distortion is also seen in Figure 1.

3.1 The optimal menu

Proposition 2 shows that variation in first-best benefits prevents the principal from attaining her first-best payoff, even when participation constraints are slack in the known-type benchmark. I now characterize the principal's optimal menu, subject to incentive compatibility and participation constraints. The first result simplifies the analysis by reducing the principal's problem to menus of linear scoring rules.

Proposition 3 (Linear scoring rules)**.** *Under Assumption 1, there exists an optimal direct mechanism satisfying (IC) and (IR) that assigns to each reported type $\tau \in [0,1]$ the acceptance policy*

$$q_\tau = \mathbb{1}\big\{\sigma(\tau)x_1 + [1-\sigma(\tau)]x_2 > \kappa - \gamma(\tau)\theta\big\},$$

where $\sigma : [0,1] \to \mathbb{R}$ and $\gamma : [0,1] \to (-1,1)$ are continuous.

Proposition 3 reduces the principal's problem to optimizing over two schedules $\sigma(\cdot)$ and $\gamma(\cdot)$. In the optimal direct mechanism, the policy assigned to type τ uses the scoring weights $(\sigma(\tau), 1-\sigma(\tau))$. The agent then takes the action if and only if the weighted score is greater than the θ-adjusted threshold $\kappa - \gamma(\tau)\theta$, where the adjustment rate $\gamma(\tau)$ depends on the agent's type as well. Relative to the first best, the principal distorts both the scoring weight and the acceptance threshold on the optimal menu. A threshold adjustment can already be required for participation when the type is known. With private types, the scoring weight may also depart from the agent's own weight (i.e. $\sigma(\tau) \neq \tau$). Further, the adjustment can take either sign. The main characterization determines how these two instruments are chosen jointly across types to maximize the principal's expected payoffs.

To characterize the optimal menu, I first describe the payoffs a candidate menu offers to the agent. For a pair of schedules $(\sigma, \gamma) \coloneqq (\sigma(\tau), \gamma(\tau))_{\tau\in[0,1]}$, let $q_\tau^{(\sigma,\gamma)} \coloneqq q^{(\sigma(\tau),\gamma(\tau))}$ denote the policy assigned to an agent of type τ. Define

$$U(\tau) \coloneqq U_\tau(q_\tau^{(\sigma,\gamma)})$$

to be type τ's payoffs under the policy assigned to him. Participation constraints imply that every IC and IR direct mechanism characterized by (σ, γ) must satisfy $U(\tau) \geq 0$

Incentive compatibility also depends on how each type values a different type's policy. First define the difference between the two match attribute surpluses, $S_1(q)$ and $S_2(q)$, by

$$\Delta(\tau) \coloneqq S_1(q_\tau^{(\sigma,\gamma)}) - S_2(q_\tau^{(\sigma,\gamma)}) = \mathbb{E}\big[(x_1 - x_2)q_\tau^{(\sigma,\gamma)}(\omega)\big].$$

Hence, $\Delta(\tau)$ captures how the policy $q_\tau^{(\sigma,\gamma)}$ differs in the surpluses it generates across the two match attributes. An agent of type τ who reports his type to be r then receives

$$U_\tau(q_r) = U(r) + (\tau - r)\Delta(r).$$

$\Delta(r)$ determines how the payoff from policy q_r varies with the agent's true type. Together, $U(\cdot)$ and $\Delta(\cdot)$ describe every type's payoff from every report, so both participation and incentive compatibility can be expressed in terms of these two schedules.

To understand the principal's optimal menu, consider a type τ whose assigned scoring weight $\sigma(\tau) \neq \tau$. The principal can rotate type τ's policy in the following manner: set $\hat{\sigma}(\tau) = \tau$ and choose $\hat{\gamma}(\tau)$ so that

$$U_\tau(q^{(\tau,\hat{\gamma}(\tau))}) = U(\tau).$$

The rotated test uses the agent's own scoring weight while holding his payoff constant. I show that this rotation strictly improves the principal's payoff for type τ. More specifically, define

$$H(\tau) \coloneqq \frac{\tau - \sigma(\tau)}{1 - \gamma(\tau)}.$$

Starting from the assigned policy $q^{(\sigma(\tau),\gamma(\tau))}$, the principal can move toward the rotated test $q^{(\tau,\hat{\gamma}(\tau))}$ while keeping $U(\tau)$ fixed. The magnitude $|H(\tau)|$ captures the principal's initial marginal gain from a rotation in this direction, normalized by the absolute change in the surplus difference $\Delta(\tau)$. Although the rotation preserves type τ's payoff, it changes $\Delta(\tau)$ and hence affects how other types value his policy. The principal must therefore account for its effects on reporting incentives across the menu. The main characterization balances these effects against the gain from correcting the scoring weight.

For any candidate pair of schedules (σ, γ), the induced policies and the prior fully determine U, Δ, and H. The main characterization uses these objects to give necessary and sufficient conditions for the scoring-weight and threshold-adjustment schedules to be optimal.

Theorem 1 (The optimal menu)**.** *Under Assumption 1, let $\sigma, \gamma : [0,1] \to \mathbb{R}$ be continuous schedules with $\gamma(\tau) \in (-1, 1)$ for every $\tau \in [0,1]$, and define U, Δ, and H as above.*

The direct mechanism assigning policy $q_\tau^{(\sigma,\gamma)}$ to each reported type $\tau \in [0,1]$ satisfies (IC) and (IR) and maximizes the principal's expected payoff among all direct mechanisms satisfying these constraints if and only if the following conditions hold:

(i) Incentive compatibility and participation. *Δ is nondecreasing, $U(\tau) \geq 0$ for every $\tau \in [0,1]$, and*

$$U(t) - U(s) = \int_s^t \Delta(z)\, dz \qquad \text{for all } 0 \leq s < t \leq 1.$$

(ii) First-order condition and complementary slackness. *H is absolutely continuous on $[0,1]$ and*

$$H'(\tau) \geq -\frac{\gamma(\tau)}{1 - \gamma(\tau)} \qquad \text{for almost every } \tau \in [0,1],$$

with equality for almost every τ such that $U(\tau) > 0$.

(iii) Boundary conditions.

$$H(0) \geq 0, \qquad H(1) \leq 0, \qquad U(0)H(0) = U(1)H(1) = 0.$$

There exists exactly one continuous pair (σ^*, γ^*) satisfying these conditions. Every optimal direct mechanism agrees with $q_\tau^{(\sigma^*,\gamma^*)}$ for almost every type τ, up to π_0-null sets of states.

The above discussion and Theorem 1 show why the principal distorts the weights on the match attributes along the optimal menu. Fixing a type τ, she improves her payoff when she rotates a policy with distorted weights $\sigma(\tau) \neq \tau$ in the direction of the true weight τ while adjusting $\gamma(\tau)$ to keep the agent's payoff constant. However, this rotation changes the value of type τ's policy to other types $\tau' \neq \tau$. The principal must then adjust other types' policies to preserve incentive compatibility. The costs of these adjustments are characterized by the term $\gamma(\tau)/[1-\gamma(\tau)]$. To see this, consider the principal's problem of increasing the rent of type τ' while holding surplus difference $\Delta(\tau')$ constant. For each promised rent, the principal chooses the policy that maximizes her payoff subject to this constraint. An additional marginal unit of expected private benefit delivered by the policy changes the principal's payoff by $-\gamma(\tau')$. The agent receives the additional private benefit but also shares in the change in the principal's payoff, resulting in a net marginal gain of $1-\gamma(\tau')$. Therefore, the principle's marginal cost of delivering an additional unit of rent to the agent is

$$\frac{\gamma(\tau')}{1-\gamma(\tau')}.$$

When $\gamma(\tau') > 0$, providing more rent costs the principal; when $\gamma(\tau') < 0$, it benefits her as well.

The first order condition relates the gain from rotating the scoring weight for a specific type with the cost of adjusting other type's rents. The following corollary describes the structure of the optimal menu and illustrates this trade-off more explicitly:

Corollary 1.1 (Pool and ramps)**.** *For the optimal schedules in Theorem 1, let*

$$P := \{\tau \in [0,1] : U(\tau) = 0\}.$$

Then:

(i) Pooling and separation. *P is a closed interval, possibly empty or a singleton. Two distinct types receive the same acceptance policy if and only if both belong to P. If P has positive length, its types receive a common policy $\bar{q}$ satisfying*

$$\bar{q} \in \arg\max_{q:\Omega\to[0,1]} \big\{S_1(q) : S_1(q) = S_2(q),\ S_2(q) + T(q) = 0\big\}.$$

(ii) Balance on the ramps. *If $P = [\tau_-, \tau_+] \neq \emptyset$, the positive-rent regions below and above P form*

the lower and upper ramps, respectively. When $\tau_- > 0$, the lower ramp satisfies

$$H(\tau) = -\int_0^\tau \frac{\gamma(s)}{1-\gamma(s)}\, ds, \qquad 0 \le \tau \le \tau_-. \tag{3}$$

When $\tau_+ < 1$, the upper ramp satisfies

$$H(\tau) = \int_\tau^1 \frac{\gamma(s)}{1-\gamma(s)}\, ds, \qquad \tau_+ \le \tau \le 1. \tag{4}$$

If $P = \emptyset$, both equations hold for every $\tau \in [0,1]$. In particular,

$$U(0) > 0 \implies \sigma(0) = 0, \qquad U(1) > 0 \implies \sigma(1) = 1.$$

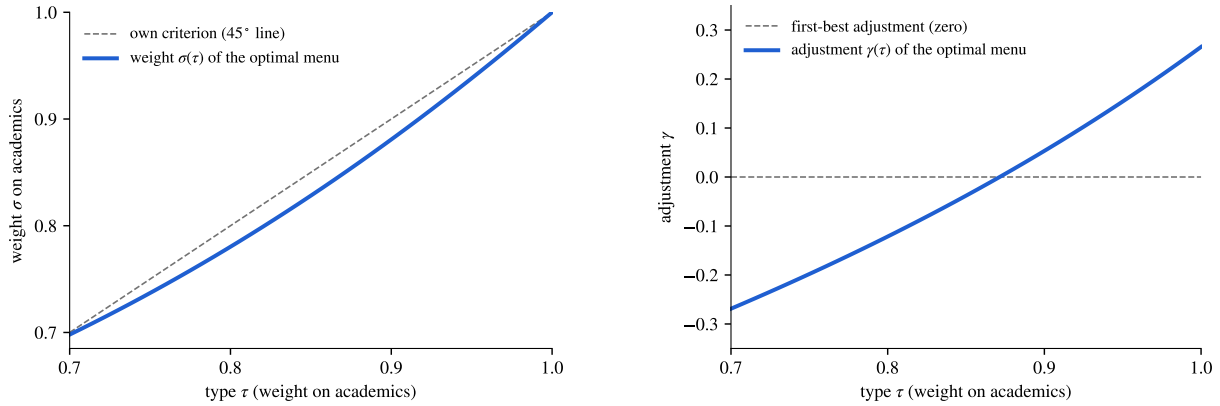
Corollary 1.1 shows that the optimal menu has the following structure: a (potentially empty) pool of types receiving zero rent, and a lower and upper ramp on each side of the pool satisfying their respective integral equations. On a positive-length pool of zero-rent types, IC implies $\Delta \equiv 0$: the shared policy equalizes the surplus of the two match attributes. Notice that the principal's payoffs across experiments that satisfies this condition is invariant over τ. Therefore, she simply chooses the policy that maximizes the common surplus across all policies that satisfies the pool's constraints.

The two integral on the ramps describes the key trade-off the principal faces. Focusing on the upper ramp, recall that $H(\tau)$ is the principal's marginal gain from rotating type τ's policy in the direction of his true scoring weights while holding his payoff constant. The right hand side of Equation 4 is the principal's cumulative marginal cost of rent for types above τ. Equation 4 expresses Theorem 1's first order conditions in integral form to demonstrate this trade-off. Further, Theorem 1's sufficiency statement establishes that balancing the gain from correcting the scoring weight against rent costs, subject to participation, fully determines the optimal menu. No other feasible change to the information policies can increase the principal's payoff.

The main characterization also addresses whether a menu of policies that do not condition directly on θ can be optimal.

Corollary 1.2 (Blind menus)**.** *If T^{fb} is not constant, then $\gamma(\tau) \neq 0$ for some τ: the optimal menu conditions on θ. Consequently, every menu of policies that do not condition on θ is strictly suboptimal, including the first-best menu offered as is, with agents choosing freely among its policies.*

The failure of the first-best menu leaves open whether the principal could attain her optimal payoff by offering a different menu that still excludes θ. The corollary rules this out whenever T^{fb} is not constant. Even a blind menu with weights optimized designed to account for the agent's incentives is strictly suboptimal: adjusting the scoring weights, or using more general policies based



(a) Weight

(b) Adjustment

Figure 2: The optimal admissions menu: (a) scoring weights $\sigma(\tau)$; (b) threshold adjustments $\gamma(\tau)$.

only on the match attributes, cannot attain the optimal payoff. Conditioning directly on the agent's private benefit is therefore essential to the optimal mechanism, even though the principal values only the match attributes.

3.2 The solved menu for the admissions example

I now apply Theorem 1 to the admissions example introduced in Section 1.2. Figure 2 displays the optimal scoring weights and threshold adjustments for the admissions example, using the parameters of Section 1.2.[9] Panel (a) plots the weight $\sigma(\tau)$ of each school's policy against the school's own weight τ on academics, and panel (b) plots the adjustment $\gamma(\tau)$. The menu has no pool at these parameters: every school earns a positive rent, both balance equations hold on the whole type space, and both ends are undistorted, $\sigma(0.7) = 0.7$ and $\sigma(1) = 1$.

Every policy on the menu is a scoring rule with an adjustment strictly between -1 and 1, as Theorem 1 requires. Notice that the optimal menu features negative adjustments. The main result implies that, even in settings where the board of education does not prioritize admitting low-income students, the optimal regulation it pursues may include admission protocols that are more lenient on those who need financial aid. Indeed, Panel (b) shows that the schools who are extracurriculars-heavy in their admissions protocol will accept students who need aid at a lower threshold than non-aid students. The opposite is true for academic-heavy schools: their assigned

[9]The schools' weights on academics lie in $[0.7, 1]$ rather than $[0, 1]$. A change of variables maps this case into the model: with $x_1' := x^{(1)}$ and $x_2' := x^{(0.7)}$ as the match attributes and $s := (\tau - 0.7)/0.3$ as the type, match quality is unchanged, $s\,x_1' + (1-s)\,x_2' = x^{(\tau)}$, and types are uniform on $[0, 1]$. The joint support of the new attributes is a parallelogram, which is why (A2) requires positivity only near the pivotal state. The figures keep the original scale.

admissions protocol raises the cutoff for aid applicants and increase their private benefits. The intuition is clear: the board uses both positive and negative adjustments to reduce differences in private benefits across schools. Panel (a) shows that every interior school is assigned a virtual scoring weight below its own type, $\sigma(\tau) < \tau$: the board scores the school's applicants with more weight on extracurriculars than the school itself would. This is the sign that the first-order condition (4) predicts. Section 4 makes these intuitions precise: it shows how the ordering of first-best private benefits determines the direction of the scoring-weight distortions and the signs of the threshold adjustments at the endpoints.

4 Private Benefits and the Shape of the Menu

Section 3 characterizes the optimal scoring weights and threshold adjustments. This section applies that characterization to explain how the prior shapes the optimal menu. To build intuition, I focus on settings in which first-best policies that place more weight on one match attribute deliver higher expected private benefits. The optimal menu responds by compressing these benefits across types, while retaining scoring weights that lean toward the same attribute. I interpret the two instruments through the incentives they create and return to the admissions example. The analysis then turns to how the principal's payoff changes when match attributes reveal less about the private benefit. The comparison holds the distribution of match quality fixed and shows how dependence raises the cost of screening. Proofs are deferred to the appendix.

4.1 Ordered private benefits

Proposition 2 shows that variation in the first-best benefits $T^{\text{fb}}(\tau) = T(q^{\tau,0})$ gives agents an incentive to select another type's policy. When policies that place more weight on one attribute deliver higher private benefits, these reporting incentives consistently point toward the same attribute. The following definition makes this idea formal:

Definition 1 (Ordered private benefits)**.** Private benefits are *ordered* if T^{fb} is monotone on $[0, 1]$. Label the match attributes so that T^{fb} is nonincreasing and call x_2 the *favored attribute*.

Under this labeling, a lower type's first-best policy places more weight on x_2 and delivers at least as much private benefit. Differences in these benefits create the incentive described in Section 3. Where T^{fb} has a negative slope, selecting a nearby lower type's policy brings a first-order gain in private benefit at a second-order loss in match quality. The agent therefore has an incentive to

overweight the favored attribute, regardless of his type. Notice that the ordering of T^{fb} is a property of the primitives: it simply considers how expected first-best benefits varies across different types.

The following theorem describes the shape of the optimal menu under ordered private benefits. In particular, I show that, in these settings, the distortion over scoring weights is one-sided and the optimal menu *compresses* private benefits across types. Write $q_\tau = q^{\sigma(\tau),\gamma(\tau)}$ for the policy assigned to type τ on the optimal menu.

Theorem 2 (Ordered private benefits)**.** *Suppose Assumption 1 holds, private benefits are ordered with x_2 the favored attribute, and every type earns positive rent on the optimal menu. Then:*

(a) One-sided scoring distortion. *Every type's scoring weight leans toward the favored attribute:*

$$\sigma(\tau) \leq \tau \qquad \textit{for every } \tau \in [0,1].$$

(b) Ordered and compressed private benefits. *The expected private benefit $T(q_\tau)$ is nonincreasing in τ, and its range is contained in the first-best range:*

$$T^{\text{fb}}(1) \leq T(q_1) \leq T(q_0) \leq T^{\text{fb}}(0).$$

Thus the optimal menu weakly lowers the highest private benefit and weakly raises the lowest.

If T^{fb} is strictly decreasing, then $T(q_\tau)$ is strictly decreasing. If T^{fb} is constant, the optimal menu is the first-best menu. The corresponding conclusions hold when x_1 is favored, after exchanging the labels of the two match attributes.

The threshold adjustments explain how the principal compresses the range of private benefits. Corollary 1.1 gives $\sigma(0) = 0$ and $\sigma(1) = 1$, so the endpoint types retain their own scoring weights. Their adjustments satisfy

$$\gamma(0) \leq 0 \leq \gamma(1). \tag{5}$$

At the bottom, where first-best benefits are highest, the threshold adjustment shifts acceptance toward states with lower private benefits. At the top, where first-best benefits are lowest, it shifts acceptance toward states with higher private benefits. In particular, the principal may adopt a scoring rule along the optimal menu that is biased *against* the agent's private benefits as a means to equalize expected private benefits across type. In the admissions example, this implies that a meritocratic principal may choose to include scoring rules that give students who need financial aid a more lenient threshold, even if all schools would rather accept non-aid students (all else equal). Since the scoring weights at these endpoints are unchanged, the adjustments lower the highest benefit and raise the lowest. Part (b) shows that the benefits delivered between the endpoints remain ordered, so the entire range is compressed.

The ordering of private benefits and the direction of the scoring distortion are linked by the surplus difference Δ introduced in Section 3. Along the optimal menu,

$$T(q_t) - T(q_s) = -\int_s^t H(r)\, d\Delta(r), \qquad 0 \le s < t \le 1, \tag{6}$$

where $H(r) = [r - \sigma(r)]/[1 - \gamma(r)]$. Incentive compatibility requires Δ to be nondecreasing, while part (a) gives $H \ge 0$. The identity therefore shows why expected private benefit cannot rise with type: increases in the surplus difference are accompanied by reductions in private benefit weighted by the scoring distortion.

To understand why the principal retains the scoring distortion, recall the trade-off in Corollary 1.1. Rotating a type's rule toward his own weight raises the principal's payoff from that type, holding his payoff constant. But it also makes the policy more attractive to the types above him, who must receive additional rents to preserve incentive compatibility. The principal stops correcting the scoring weight when the direct gain is offset by the cost of those rents.

Admissions with financial aid. The admissions example illustrates how the ordering follows from the prior. Let p denote the fraction of applicants who need aid, c the cost of aid, and $m \coloneqq \mathbb{E}[x_2 \mid \text{aid}]$ their mean extracurricular score. Using the schools' original weights on academics, $\tau \in [0.7, 1]$, the first-best benefits satisfy

$$T^{\text{fb}}(\tau) = -c\,p\left[(1-m) - \frac{\kappa - m}{\tau}\right], \qquad \frac{dT^{\text{fb}}(\tau)}{d\tau} = -\frac{c\,p\,(\kappa - m)}{\tau^2} < 0. \tag{7}$$

The sign follows from $m < \kappa$: applicants who need aid fall short on extracurriculars, so policies that place more weight on academics admit more of them. Policies that favor extracurriculars instead save on aid costs and deliver higher private benefits to the school. Thus extracurriculars are the favored attribute.[10]

Figure 2 shows how the board uses aid status to compress these financial incentives. It lowers aid applicants' acceptance thresholds at schools that place relatively more weight on extracurriculars and raises them at schools that place relatively more weight on academics. As Figure 1 shows, expected private benefits are nearly equal across the optimal menu. The assigned scoring weights still lean toward extracurriculars, but they are much closer to schools' own priorities than the weights schools choose under need-blind admission. Conditioning directly on aid status allows the board to address the financial incentive that otherwise leads schools to favor extracurriculars to avoid aid

[10] In the example, $\theta = -c$ for applicants who need aid and $\theta = 0$ otherwise. Grades are uniform and independent of aid status and extracurriculars. For the types in the example, $\tau \ge \kappa$ and $\tau \ge 1 - \kappa$, so the first-best cutoff on grades lies in $[0, 1]$ for every extracurricular score. Integrating the acceptance probability gives (7). The parameters of Section 1.2 give $p = 0.4$, $m = 1/3$, and $\kappa = 0.65$.

costs.

The direction of the scoring distortion and the ordering of private benefits also extend to menus in which only the lowest type earns zero rent. This is the regime of the admissions example at higher aid costs. The lowest school's participation constraint now binds, and its assigned scoring weight can depart from its own weight. The menu continues to lean toward extracurriculars and to deliver private benefits in the same order across schools.

4.2 Less informative attributes

This subsection explores how the principal's payoff changes when the dependence that generates these differences becomes weaker. The comparison keeps the distribution of match quality fixed, so any change in the principal's payoff comes from the incentives created by the private benefit.

Fix the marginal distributions of x_1, x_2, and θ, with x_1 independent of (x_2, θ) and $\mathbb{E}[\theta] = 0$. Let P_1 denote a prior with these properties, and let P_0 be the prior under which all three state variables are independent with the same marginals. For $\eta \in [0, 1]$, define

$$P_\eta \coloneqq (1 - \eta)P_0 + \eta P_1. \tag{8}$$

The parameter η measures the strength of dependence between x_2 and θ. At $\eta = 0$, the match attributes reveal nothing about the private benefit. As η rises, more of the dependence in P_1 is retained. The distribution of (x_1, x_2) stays fixed throughout the comparison, as does the distribution of θ.

This family also gives an information comparison. For $0 \leq \eta_1 < \eta_2$, retain the realization of x_2 under P_{η_2} with probability η_1/η_2 and otherwise replace it with an independent draw from its marginal distribution. The resulting joint law of x_2 and θ is the law under P_{η_1}. Thus reducing η makes x_2 less informative about θ while preserving the distribution of the match attributes.

To compare payoffs across priors, write

$$V_\tau^\eta(q) \coloneqq \mathbb{E}_\eta\big[(x^{(\tau)} - \kappa)q(\omega)\big], \qquad T_\eta(q) \coloneqq \mathbb{E}_\eta\big[\theta q(\omega)\big].$$

The first-best rules depend only on match quality, so their payoffs to the principal are unchanged:

$$V^{\mathrm{fb}}(\tau) \coloneqq V_\tau^\eta(q^{\tau,0}).$$

Their private benefits instead vary with dependence. Under P_0, the private benefit is independent of the rule's recommendation and has mean zero. Consequently,

$$T_\eta^{\mathrm{fb}}(\tau) \coloneqq T_\eta(q^{\tau,0}) = \eta T_1^{\mathrm{fb}}(\tau). \tag{9}$$

Increasing dependence scales up the differences in first-best benefits across types, while leaving

the value of match quality unchanged. If first-best benefits are ordered under P_1, that ordering is preserved throughout the family.

Let $V(\eta)$ be the principal's optimal expected payoff under P_η, and define

$$B \coloneqq \int_0^1 V^{\mathrm{fb}}(\tau)\, d\tau, \qquad C(\eta) \coloneqq B - V(\eta).$$

Here B is the principal's first-best payoff and $C(\eta)$ is her loss relative to that benchmark, termed the *cost of screening*.[11]

The independence benchmark. Under independence, every first-best rule delivers zero expected private benefit. Each type therefore ranks all the first-best rules by their value to the principal, and chooses his own rule. His payoff from doing so is $V^{\mathrm{fb}}(\tau) \geq 0$, so participation is satisfied as well. The principal can attain her first-best payoff:

Lemma 4 (Independence)**.** *Under P_0, the first-best menu satisfies (IC) and (IR), and*

$$V(0) = B, \qquad C(0) = 0.$$

Improving the menu as dependence falls. Fix $0 \leq \eta_1 < \eta_2 \leq 1$ and put $\lambda \coloneqq \eta_1/\eta_2$. The lower-dependence prior can be written as

$$P_{\eta_1} = (1-\lambda)P_0 + \lambda P_{\eta_2}. \tag{10}$$

There is a feasible menu under each component: the first-best menu under P_0, and any menu satisfying (IC) and (IR) under P_{η_2}. The principal can combine these menus to construct a feasible menu under P_{η_1}. The resulting payoff moves toward the first-best payoff for every type.

Theorem 3 (Less dependence, type by type)**.** *Let $0 \leq \eta_1 < \eta_2 \leq 1$ and $\lambda \coloneqq \eta_1/\eta_2$. For every menu $(q^2_\tau)_{\tau\in[0,1]}$ satisfying (IC) and (IR) under P_{η_2}, there is a menu $(\tilde{q}^1_\tau)_{\tau\in[0,1]}$ satisfying (IC) and (IR) under P_{η_1} such that*

$$V^{\mathrm{fb}}(\tau) - V^{\eta_1}_\tau(\tilde{q}^1_\tau) = \lambda\big[V^{\mathrm{fb}}(\tau) - V^{\eta_2}_\tau(q^2_\tau)\big] \qquad \textit{for every } \tau \in [0,1]. \tag{11}$$

The principal achieves this improvement by combining the two component menus state by state. Regard the prior decomposition in (10) as a draw of a component prior followed by a state. Conditional on the state, the principal applies each component's policy with the probability that the state came from that component. For every true type and every report, the resulting expected

[11] When $\mathbb{E}_1[\theta \mid x_2]$ is nondecreasing in x_2, every type is unconstrained in the known-type problem throughout this family. In that case $C(\eta)$ is exactly the cost of the agent's private information. A monotone likelihood ratio ordering of the conditional distributions of θ given x_2 implies this condition.

payoffs are the same convex combination of payoffs in the two environments. The resulting menu therefore satisfies incentive compatibility and participation. In particular, the principal's payoff from type τ is

$$V_\tau^{\eta_1}(\tilde{q}_\tau^1) = (1-\lambda)V^{\mathrm{fb}}(\tau) + \lambda V_\tau^{\eta_2}(q_\tau^2),$$

which gives the reduction in losses stated in the theorem.

Starting from an optimal menu at η_2, the constructed menu closes a fraction $1-\lambda$ of the principal's gap from the first best. Optimizing over menus at η_1 can only improve her expected payoff further. The same construction combines feasible menus under any two priors in the family. The principal can therefore attain the corresponding weighted average of their payoffs, which makes her optimal payoff concave in dependence. These comparisons yield the following result:

Corollary 3.1 (The cost of screening)**.** *The principal's optimal payoff V is nonincreasing and concave in η, and the cost of screening C is nondecreasing and convex. For every $0 \le \eta_1 < \eta_2 \le 1$,*

$$C(\eta_1) \le \frac{\eta_1}{\eta_2} C(\eta_2).$$

If $C(\eta_2) > 0$, then $V(\eta_1) > V(\eta_2)$.

Corollary 3.1 shows that dependence raises the cost of screening at an increasing rate. By (9), the same increase in dependence widens the differences in first-best benefits across types. The principal must accommodate the resulting reporting incentives through her acceptance policies, and the cost of doing so is convex in dependence. The theorem supplies the stronger constructive comparison behind this welfare result: a reduction in dependence allows the principal to reduce her loss from every type at once.

Correcting acceptance decisions. The payoff improvement can also be implemented by correcting acceptance decisions. Call a decision correct when it agrees with the first-best rule for that type. The following corollary shows that the principal can preserve every previously correct decision while reducing the probability of errors.

Corollary 3.2 (Correcting acceptance decisions)**.** *Under the hypotheses of Theorem 3, suppose x_1 has an atomless distribution and the original menu (q_τ^2) consists of linear scoring rules. There is a deterministic menu (q_τ^1) satisfying (IC) and (IR) under P_{η_1} that delivers the same payoffs for every true type and every report as the constructed menu $(\tilde{q}_\tau^1)$. For each type, define*

$$E_\tau^j := \{\omega : q_\tau^j(\omega) \neq q^{\tau,0}(\omega)\}, \qquad j = 1, 2.$$

The error sets satisfy

$$E_\tau^1 \subseteq E_\tau^2, \qquad P_{\eta_1}(E_\tau^1) = \lambda P_{\eta_2}(E_\tau^2) \qquad \text{for every } \tau \in [0,1]. \tag{12}$$

The deterministic implementation uses variation in x_1 to correct acceptance decisions while preserving the payoffs needed for incentive compatibility. To see how, fix x_2 and θ and consider the type's acceptance decision as x_1 varies. Compare it with the first-best cutoff on x_1. Within each interval of mistaken acceptances or mistaken rejections, retain a smaller interval of errors whose probability and integral of x_1 match those implied by the randomized construction. With x_2 and θ fixed, these two quantities determine every payoff component. Applying this adjustment across states and types produces the deterministic menu. Its acceptance boundaries can be nonlinear, allowing the principal to correct decisions while delivering the payoffs needed for incentive compatibility.

The welfare and decision comparisons isolate the same economic force. The principal and the agent agree on how match attributes contribute to match quality. Differences in the private benefits generated by first-best policies make other types' scoring weights attractive for reasons unrelated to the shared objective. Weakening this dependence lets the principal provide information that serves the shared objective at a lower incentive cost.

5 Extensions

For ease of exposition, the main analysis assumes that the agent commits to following his chosen policy's recommendations and that types are uniformly distributed. This section relaxes these assumptions in turn. I first allow the agent to choose how to respond to a policy's recommendations. The characterization in Section 3 solves a relaxation of this problem, and two inequalities determine whether its solution also solves the problem with unrestricted responses. I then consider a general distribution of types. The distribution changes the weight the principal places on the gains from correcting scoring weights and the rents required to preserve incentive compatibility. It also allows types who earn positive rents to share a policy. Proofs are deferred to the appendix.

5.1 Inverting recommendations

Suppose the agent can choose his action after observing a recommendation. In addition to following a policy, he can *invert* it: accept when it recommends rejection and reject when it recommends acceptance. The analysis in Section 3 relaxes the constraints against selecting another type's policy and inverting its recommendations. Solving this relaxed problem gives an upper bound on the principal's payoff. If the resulting menu satisfies the inversion constraints, it attains that bound and is optimal when the agent is free to choose his response.

To write these constraints, consider a type τ who selects a policy q and inverts it. His acceptance

policy is then $1 - q$, and his payoff is

$$U_\tau(1-q) = \mathbb{E}[x^{(\tau)} - \kappa + \theta] - U_\tau(q).$$

The first term is the payoff from accepting in every state; the second subtracts the payoff from the states in which q recommends acceptance. By (A3), the first term is negative. An agent who earns a nonnegative rent therefore prefers following his assigned policy to inverting it. Always rejecting yields zero, and always accepting yields a negative payoff, so (IR) also prevents either response from improving the agent's payoff. The remaining constraints concern selecting another type's policy and inverting it:

$$U(\tau) \geq \mathbb{E}[x^{(\tau)} - \kappa + \theta] - U_\tau(q_r) \qquad \text{for all } \tau, r \in [0,1]. \tag{13}$$

Recall from Section 3 that $U_\tau(q_r) = U(r) + (\tau - r)\Delta(r)$, where $\Delta(r) = S_1(q_r) - S_2(q_r)$. Thus the inversion constraints depend only on the rent and surplus-difference schedules already used to characterize the optimal menu. The following result reduces the entire family to two comparisons.

Proposition 5 (Verification of inversion constraints)**.** *Let the menu $(q_\tau)_{\tau\in[0,1]}$ satisfy (IC) and (IR). It satisfies* (13) *if and only if*

$$U(0) + U(1) - \Delta(1) \geq \mathbb{E}[x_2 + \theta] - \kappa, \tag{14}$$

$$U(0) + U(1) + \Delta(0) \geq \mathbb{E}[x_1 + \theta] - \kappa. \tag{15}$$

All inversion constraints hold with slack if both inequalities are strict.

Inequality (14) requires the lowest type to prefer his assigned policy to inverting the highest type's policy. Inequality (15) imposes the corresponding requirement on the highest type. These are the two most demanding comparisons. Under (IC), the slack for any other pair of types is at least the smaller of the two endpoint slacks. Verifying the additional constraints therefore requires only the rents of the two endpoint types and the surplus differences of their assigned policies.

Proposition 5 gives a direct way to apply the main characterization when responses are unrestricted. First solve the problem in Theorem 1, then evaluate (14) and (15) at its solution. When both hold, the same menu is feasible and optimal in the unrestricted problem. The relaxation then delivers the principal's attainable payoff, and the agent voluntarily follows the recommendations of his chosen policy.

A sufficient condition for the two inequalities to hold can also be stated directly in terms of the prior and the cutoff:

$$\max\left\{\mathbb{E}[(x_1 - x_2)^+], \mathbb{E}[(x_2 - x_1)^+]\right\} \leq \kappa - \mathbb{E}[\theta] - \max\{\mathbb{E}[x_1], \mathbb{E}[x_2]\}. \tag{16}$$

Here $z^+ \coloneqq \max\{z, 0\}$. The left side bounds the surplus difference a policy can generate, while the

right side is the smallest loss an uninformed type incurs from accepting. Together with nonnegative rents, this bound ensures that every menu satisfying (IC) and (IR) also satisfies the inversion constraints. A strict inequality guarantees slackness. It holds strictly in the admissions example, so the solved menu remains optimal when schools can reverse recommendations.[12]

Condition (16) applies to every menu satisfying (IC) and (IR). Verifying the solution instead requires only the two endpoint inequalities, evaluated at the rents and policies determined by Theorem 1. The next step is to use that characterization to obtain less restrictive primitive conditions guaranteeing that the inversion constraints are slack at the optimum. The dependence comparisons in Section 4.2 already preserve the inversion constraints when applied to menus satisfying them.

5.2 General type densities

Now let the type have a continuous density g, strictly positive on $[0, 1]$. The principal's objective becomes

$$\int_0^1 V_\tau(q_\tau) g(\tau)\, d\tau.$$

The agent's preferences over policies are unchanged. Incentive compatibility still requires a nondecreasing surplus difference Δ and the envelope condition

$$U(t) - U(s) = \int_s^t \Delta(z)\, dz, \qquad 0 \le s < t \le 1,$$

while participation requires $U \ge 0$. The density changes how the principal values the policy and rent adjustments needed to satisfy these constraints.

The weight on gains and rent costs. Recall the trade-off in Corollary 1.1. Correcting a type's scoring weight raises the principal's payoff from that type, but changes the rents required by other types. With a general density, each effect is weighted by the frequency of the types it affects. For an optimal menu with the scoring representation $q_\tau = q^{\sigma(\tau),\gamma(\tau)}$, with $\gamma(\tau) < 1$, the balance on a separating part of the upper ramp is

$$\frac{\tau - \sigma(\tau)}{1 - \gamma(\tau)}\, g(\tau) = \int_\tau^1 \frac{\gamma(s)}{1 - \gamma(s)}\, g(s)\, ds. \tag{17}$$

The left side is the marginal gain from changing the surplus difference at type τ, holding his rent fixed, multiplied by his density. The right side is the cumulative marginal cost of the rents passed to types above him, weighted by their densities. On a separating part of the lower ramp, the

[12] After the change of variables in the footnote to Figure 2, the attributes are $x'_1 = x_1$ and $x'_2 = 0.7x_1 + 0.3x_2$. Both $\mathbb{E}[(x'_1 - x'_2)^+]$ and $\mathbb{E}[(x'_2 - x'_1)^+]$ are at most 0.055, while $\mathbb{E}[x'_1] = 0.5$, $\mathbb{E}[x'_2] = 0.51$, $\kappa = 0.65$, and $\mathbb{E}[\theta] = -0.4c$. The right side of (16) is therefore $0.14 + 0.4c$, so the inequality is strict for every aid cost $c \ge 0$.

corresponding balance is

$$\frac{\tau - \sigma(\tau)}{1 - \gamma(\tau)}\, g(\tau) = -\int_0^\tau \frac{\gamma(s)}{1 - \gamma(s)}\, g(s)\, ds. \tag{18}$$

Setting $g \equiv 1$ recovers the equations of Section 3.

The density determines the relative importance of the direct gain and the rent effects. Correcting the policies of a large group of types has a large direct payoff, while the rents needed to sustain that correction are costly when they accrue to a large group of other types. The optimal scoring weights reflect this comparison. The same economic trade-off governs the choice of policies; the type distribution determines how the principal aggregates its two sides.

Bunching at positive rents. A general density also permits intervals of types to share a policy while earning positive rents. Call such an interval a *bunch*. If the types in $[a, b]$ receive a common policy $\bar{q}$, then their surplus difference is constant, $\Delta(\tau) = \bar{\Delta} \coloneqq S_1(\bar{q}) - S_2(\bar{q})$, and their rents satisfy

$$U(\tau) = U(a) + (\tau - a)\bar{\Delta}, \qquad \tau \in [a, b].$$

The rent schedule is affine on the bunch. These types share a policy even though their participation constraints are slack. In the zero-rent pool of Section 3, by contrast, the common policy must deliver $\bar{\Delta} = 0$ and zero rent.

Bunching arises when the policies selected by the pointwise balance would violate the monotonicity of Δ required by incentive compatibility. The principal then chooses a common policy for an interval of types. The relevant balance compares the gains and rent costs over that interval, rather than at each type separately. For the upper-ramp balance, define the difference between the two sides of (17) by

$$R(\tau) \coloneqq \frac{\tau - \sigma(\tau)}{1 - \gamma(\tau)}\, g(\tau) - \int_\tau^1 \frac{\gamma(s)}{1 - \gamma(s)}\, g(s)\, ds.$$

On an upper bunch $[a, b]$, the integrated conditions are

$$\int_a^b R(\tau)\, d\tau = 0, \qquad \int_a^c R(\tau)\, d\tau \geq 0 \quad \text{for every } c \in (a, b). \tag{19}$$

The first condition balances the gains and rent costs of adjusting the common policy. The second rules out a profitable separation within the bunch. This is the ironing logic of Mussa and Rosen (1978): monotonicity replaces the pointwise balance with a balance across the types who share a policy.

General type densities thus extend the pool-and-ramps structure by allowing positive-rent bunches. The uniform density used in Section 3 rules out these bunches and gives the simpler

structure of Corollary 1.1. The welfare comparisons in Section 4.2 hold for a general type density: the constructed menu improves the principal's payoff type by type, and integrating those gains against g preserves the comparison.

6 Conclusion

This paper studies how an organization should design information when an agent privately knows how to weigh the attributes that contribute to their shared objective. I characterize the optimal menu of linear scoring rules, in which both the scoring weights and the acceptance thresholds respond to the agent's incentives. The principal would like to align each rule with the agent's own weights, but must compensate other types to keep them from selecting the improved rule. The optimal menu balances this gain against the cost of providing those rents through information.

The results highlight the role of information about private benefits in regulating decisions. Conditioning directly on that information allows the principal to address the incentives that lead agents to use match attributes as proxies. When first-best benefits are ordered, the optimal menu compresses their range across types. Less dependence between match attributes and private benefits reduces the cost of screening.

One direction for future research is to study simpler menus that equalize expected private benefits across policies. The optimal admissions menu nearly does so, raising the question of when this restriction captures most of the gain over need-blind admission.

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Appendix A Recasting the Problem

I first reduce the principal's problem to a screening problem in the agent's surplus difference and utility. The reduction separates the incentive constraints across types from the choice of a policy delivering a given surplus difference and utility to a particular type.

A.1 Policy statistics and the feasible set

For any measurable acceptance policy $q : \Omega \to [0, 1]$, define

$$S_i(q) \coloneqq \mathbb{E}[(x_i - \kappa)q(\omega)], \quad i = 1, 2, \qquad T(q) \coloneqq \mathbb{E}[\theta q(\omega)]. \tag{20}$$

Write $s(q) = (S_1(q), S_2(q), T(q))$ and $z(\omega) = (x_1 - \kappa, x_2 - \kappa, \theta)$, so that $s(q) = \mathbb{E}[zq]$. The set of feasible policy statistics is

$$\mathcal{S} \coloneqq \big\{\mathbb{E}[zq] : q : \Omega \to [0, 1] \text{ measurable}\big\} \subset \mathbb{R}^3. \tag{21}$$

These three statistics determine both parties' payoffs for every type:

$$V_\tau(q) = \tau S_1(q) + (1 - \tau)S_2(q), \qquad U_\tau(q) = V_\tau(q) + T(q). \tag{22}$$

Lemma 6 (Feasible statistics and linear thresholds)**.** *Under Assumption 1, $\mathcal{S}$ is a compact, full-dimensional, strictly convex set. For every $w \in \mathbb{R}^3 \setminus \{0\}$, its support function is*

$$h_{\mathcal{S}}(w) \coloneqq \max_{s \in \mathcal{S}} w \cdot s = \mathbb{E}[(w \cdot z)^+], \tag{23}$$

and the maximizing policy is, up to π_0-null sets, uniquely

$$q^w(\omega) = \mathbb{1}\{w_1(x_1 - \kappa) + w_2(x_2 - \kappa) + w_\theta\theta > 0\}. \tag{24}$$

Every boundary point of $\mathcal{S}$ is implemented by a unique policy of this form, up to π_0-null sets.

Proof. The set $\{q \in L^\infty(\pi_0) : 0 \le q \le 1\}$ is convex and weak-$*$ compact. Since the coordinates of z are bounded, the map $q \mapsto \mathbb{E}[zq]$ is weak-$*$ continuous. Its image $\mathcal{S}$ is therefore convex and compact.

For any $w \neq 0$, maximizing $\mathbb{E}[(w \cdot z)q]$ pointwise gives (23) and (24). The prior has a Lebesgue density, so $\pi_0\{w \cdot z = 0\} = 0$. Hence the maximizing policy, and thus its statistic, is unique. Moreover,

$$h_{\mathcal{S}}(w) + h_{\mathcal{S}}(-w) = \mathbb{E}[|w \cdot z|] > 0 \qquad (w \neq 0),$$

so $\mathcal{S}$ is not contained in any affine hyperplane and is full-dimensional. Every boundary point has a nonzero supporting normal w. Any policy implementing that point maximizes $w \cdot s(q)$ and must coincide almost surely with (24).

To see strict convexity directly, take two distinct boundary statistics s_0, s_1 and their threshold policies q_0, q_1. The policies disagree on a set of positive probability. Choose a state in this set that is interior to Ω and lies on neither threshold plane. Moving it toward the pivotal state preserves the strict disagreement, because both threshold planes pass through the pivot. Hence there is an open region $E \subset \Omega$ near the pivot on which the policies disagree. Assumption (A2) gives $\pi_0(E) > 0$.

For $0 < \lambda < 1$, the mixed policy $q_\lambda = (1-\lambda)q_0 + \lambda q_1$ implements $s_\lambda = (1-\lambda)s_0 + \lambda s_1$. On E, its acceptance probability is strictly between zero and one, leaving room to adjust the policy in either direction. If s_λ were a boundary point, let $w \neq 0$ be a supporting normal there. For $0 < \varepsilon < \min\{\lambda, 1-\lambda\}$, the perturbed policy

$$\widetilde{q}(\omega) = q_\lambda(\omega) + \varepsilon \, \operatorname{sgn}(w \cdot z(\omega)) \mathbb{1}_E(\omega)$$

still takes values in $[0,1]$, but

$$w \cdot \big(s(\widetilde{q}) - s_\lambda\big) = \varepsilon \mathbb{E}\big[|w \cdot z| \mathbb{1}_E\big] > 0.$$

The inequality is strict because E has positive probability and $\{w \cdot z = 0\}$ is prior-null. Thus the perturbation pushes the mixed statistic beyond its proposed supporting plane, a contradiction. Every nontrivial mixture of distinct boundary points is therefore interior. If either original point is interior, the same conclusion follows immediately from convexity. This proves strict convexity of $\mathcal{S}$. ■

A.2 Incentive compatibility in (Δ, U)

Given a direct menu $(q_\tau)_{\tau \in [0,1]}$, let

$$\Delta(\tau) \coloneqq S_1(q_\tau) - S_2(q_\tau), \qquad U(\tau) \coloneqq U_\tau(q_\tau). \tag{25}$$

For a fixed policy, write $\delta = S_1 - S_2$ and $y = S_2 + T$. Its payoff to type τ is $\tau\delta + y$. This is the allocation–transfer form of Mussa and Rosen (1978), with δ playing the role of allocation and y the role of a transfer to the agent. Here both are generated by the acceptance policy and constrained by $\mathcal{S}$. In particular, the payoff to a type τ who reports r is

$$U_\tau(q_r) = U(r) + (\tau - r)\Delta(r). \tag{26}$$

Lemma 7 (Incentive compatibility and participation)**.** *A direct menu satisfies (IC) and (IR) if and only if its induced schedules satisfy*

(i) Δ *is nondecreasing on* $[0,1]$*;*

(ii) $U(t) - U(s) = \int_s^t \Delta(r)\, dr$ *for every* $0 \le s < t \le 1$*;*

(iii) $U(\tau) \ge 0$ *for every* $\tau \in [0,1]$.

In this case U is convex and Lipschitz, and $U' = \Delta$ almost everywhere.

Proof. By (26), (IC) is equivalent to

$$U(\tau) \geq U(r) + (\tau - r)\Delta(r) \qquad \text{for all } \tau, r \in [0,1]. \tag{27}$$

Applying this inequality in both directions gives, for $s < t$,

$$(t-s)\Delta(s) \leq U(t) - U(s) \leq (t-s)\Delta(t). \tag{28}$$

It follows that Δ is nondecreasing. The gaps are uniformly bounded because $|\Delta(\tau)| \leq \mathbb{E}[|x_1 - x_2|]$. Thus (28) makes U Lipschitz. Also, (27) expresses U as the supremum of the affine functions $U(r) + (\tau - r)\Delta(r)$, so U is convex. At every interior differentiability point, (27) gives $U' = \Delta$. Absolute continuity of U then gives (ii), and (IR) is exactly (iii).

Conversely, monotonicity and (ii) imply, for $s < t$,

$$(t-s)\Delta(s) \leq \int_s^t \Delta(r)\, dr \leq (t-s)\Delta(t).$$

These inequalities give (27) for both orders of the two types, hence (IC). Condition (iii) gives (IR). ■

This characterization does not require differentiability. The assigned gaps satisfy

$$\Delta(\tau) \in [U'_-(\tau), U'_+(\tau)] \quad (0 < \tau < 1), \qquad \Delta(0) \leq U'_+(0), \quad \Delta(1) \geq U'_-(1).$$

To express the remaining feasibility restriction, define

$$L_\tau(S_1, S_2, T) \coloneqq \big(S_1 - S_2,\ \tau S_1 + (1-\tau)S_2 + T\big), \qquad \mathcal{D}_\tau \coloneqq L_\tau(\mathcal{S}). \tag{29}$$

The set $\mathcal{D}_\tau$ is compact and convex. A pair (δ, u) is deliverable to type τ exactly when $(\delta, u) \in \mathcal{D}_\tau$. Consequently, a pair of schedules (Δ, U) is induced by an (IC), (IR) menu if and only if it satisfies the three conditions in Lemma 7 and

$$(\Delta(\tau), U(\tau)) \in \mathcal{D}_\tau \qquad \text{for every } \tau \in [0,1]. \tag{30}$$

The construction below supplies an implementing menu and chooses, among policies with the same delivered pair, the one the principal prefers.

A.3 The principal's inner problem

Fix τ and a deliverable pair $(\delta, u) \in \mathcal{D}_\tau$. The principal's inner problem is

$$\begin{aligned} \Psi(\tau, \delta, u) \coloneqq \max_{0 \le q \le 1} \quad & \tau S_1(q) + (1-\tau) S_2(q) \\ \text{subject to} \quad & S_1(q) - S_2(q) = \delta, \\ & \tau S_1(q) + (1-\tau) S_2(q) + T(q) = u. \end{aligned} \tag{31}$$

The maximization is over measurable policies. Solving the two equalities for the match surpluses gives

$$S_1 = u + (1-\tau)\delta - T, \qquad S_2 = u - \tau\delta - T.$$

Thus the feasible statistics form the fiber

$$\Phi_\tau(\delta, u) \coloneqq \mathcal{S} \cap L_\tau^{-1}(\delta, u) = \big\{(u + (1-\tau)\delta - t,\ u - \tau\delta - t,\ t) \in \mathcal{S}\big\}. \tag{32}$$

It is a nonempty compact segment, possibly a singleton, parallel to $v = (1, 1, -1)$, the common kernel direction of the maps L_τ. Define

$$T_{\min}(\tau, \delta, u) \coloneqq \min\{t : (u + (1-\tau)\delta - t,\ u - \tau\delta - t,\ t) \in \mathcal{S}\}. \tag{33}$$

Since the principal's payoff on this fiber is $u - T$, the solution value is

$$\Psi(\tau, \delta, u) = u - T_{\min}(\tau, \delta, u). \tag{34}$$

Lemma 8 (Solution of the inner problem)**.** *For every deliverable (δ, u), the unique optimal statistic is*

$$s^*(\tau, \delta, u) = \big(u + (1-\tau)\delta - T_{\min},\ u - \tau\delta - T_{\min},\ T_{\min}\big), \tag{35}$$

where $T_{\min} = T_{\min}(\tau, \delta, u)$. It is the endpoint of $\Phi_\tau(\delta, u)$ in direction v and lies on $\partial\mathcal{S}$. The policy implementing it is unique up to π_0-null sets and is a linear threshold of the form (24)*. For each τ, $T_{\min}(\tau, \cdot, \cdot)$ is strictly convex and $\Psi(\tau, \cdot, \cdot)$ is strictly concave on $\mathcal{D}_\tau$.*

Proof. Compactness gives a minimum in (33); its value fixes all three coordinates in (35). Moving in direction v keeps δ and u unchanged, decreases T at unit rate, and increases the principal's payoff at unit rate. Hence the minimizer is the endpoint in this direction. It cannot be interior to $\mathcal{S}$, since then a further small movement in direction v would be feasible and profitable. Lemma 6 gives its unique threshold implementation.

For strict convexity, take distinct delivered pairs $p_0, p_1 \in \mathcal{D}_\tau$ and their minimizing statistics s_0, s_1. These statistics are distinct because their images under L_τ differ. For $\lambda \in (0, 1)$, strict convexity of $\mathcal{S}$ places $s_\lambda = (1-\lambda)s_0 + \lambda s_1$ in its interior. It delivers $p_\lambda = (1-\lambda)p_0 + \lambda p_1$. A small

movement from s_λ in direction v therefore strictly reduces its private benefit while preserving p_λ, so

$$T_{\min}(\tau, p_\lambda) < (1-\lambda)T_{\min}(\tau, p_0) + \lambda T_{\min}(\tau, p_1).$$

Strict concavity of Ψ follows from (34). ■

A.4 The reduced screening problem

The principal's problem is equivalent to

$$\begin{aligned} \sup_{\Delta, U} \quad & J(U) \coloneqq \int_0^1 \Psi\big(\tau, U'(\tau), U(\tau)\big)\, d\tau \\ \text{subject to} \quad & \Delta \text{ is nondecreasing on } [0,1], \\ & U(t) - U(s) = \int_s^t \Delta(r)\, dr \quad \text{for all } 0 \le s < t \le 1, \\ & U(\tau) \ge 0 \quad \text{for all } \tau \in [0,1], \\ & (\Delta(\tau), U(\tau)) \in \mathcal{D}_\tau \quad \text{for all } \tau \in [0,1]. \end{aligned} \tag{36}$$

The envelope constraint gives $U' = \Delta$ almost everywhere. Every (IC), (IR) menu induces a pair satisfying these constraints, and its payoff is at most

$$\int_0^1 \Psi\big(\tau, \Delta(\tau), U(\tau)\big)\, d\tau = J(U).$$

Conversely, take any feasible pair (Δ, U) in (36) and solve the inner problem at each type's delivered pair. The resulting assignment preserves the two schedules and therefore every payoff from every report in (26). It satisfies (IC) and (IR) and attains $J(U)$.

The assignment is measurable by standard selection results. The compact-valued, closed-graph fiber correspondence has a Borel minimizing statistic by the measurable maximum theorem. Selecting a unit supporting normal measurably then yields a jointly measurable threshold policy through (24).

Appendix B Solving the Principal's Schedule Problem

B.1 Uniform interiority

I first show that an optimal schedule remains uniformly inside the deliverable sets. This permits differentiating the objective along feasible schedule deformations. The argument allows kinks in utility and does not impose a sign restriction on the private benefit.

For a feasible utility schedule U, define its completed slope correspondence by

$$\Gamma_U(\tau) = \begin{cases} [U'_-(\tau), U'_+(\tau)], & 0 < \tau < 1, \\ \{U'_+(0)\}, & \tau = 0, \\ \{U'_-(1)\}, & \tau = 1, \end{cases} \qquad P_U(\tau) = \{(\delta, U(\tau)) : \delta \in \Gamma_U(\tau)\}.$$

Thus at a kink the correspondence includes every slope between the two one-sided derivatives, and at an endpoint it contains the limiting slope from the interior. For a deliverable pair $p \in \mathcal{D}_\tau$, write

$$\mu_\tau(p) = \operatorname{dist}(p, \partial\mathcal{D}_\tau), \qquad \mu_U(\tau) = \min_{p \in P_U(\tau)} \mu_\tau(p).$$

Lemma 9 (Uniform interiority)**.** *Under Assumption 1, the reduced problem* (36) *has an optimal utility schedule. Every optimal utility schedule U^* satisfies*

$$\inf_{\tau \in [0,1]} \mu_{U^*}(\tau) > 0. \tag{37}$$

In particular, the conclusion includes both limiting slopes at every interior kink and the one-sided limiting slopes at the endpoints.

The proof proceeds in steps. The geometry of feasible statistics yields an unbounded marginal payoff gain from moving inward near the deliverable frontier. Scaling an optimal schedule toward zero then implies integrability of the inverse square-root distance to that frontier. Incentive compatibility rules out interior contacts, and a slope truncation rules out the remaining endpoint contacts. Compactness gives the uniform bound.

Lemma 10 (Existence and completed-graph feasibility)**.** *The reduced problem has a maximizer. For every feasible utility schedule U, every point of $P_U(\tau)$ belongs to $\mathcal{D}_\tau$.*

Proof. Bounded state payoffs give a common bound on the utility and surplus difference delivered by any policy. The envelope relation therefore makes all feasible utility schedules uniformly bounded and Lipschitz. They are also convex and nonnegative by Lemma 7, and the feasible class is nonempty because the reject-all policy delivers $U \equiv 0$ and $\Delta \equiv 0$. These bounds ensure that every maximizing sequence has a uniformly convergent subsequence.

This class is closed for uniform convergence. Indeed, a uniform limit U of feasible schedules U_n is convex, nonnegative, and Lipschitz. Fix a type τ and take a convergent subsequence of its bounded witnessing gaps, $\Delta_n(\tau) \to \delta$. The inequalities

$$U_n(t) \geq U_n(\tau) + (t - \tau)\Delta_n(\tau) \qquad (t \in [0, 1])$$

pass to the limit. Closedness of $\mathcal{D}_\tau$ also gives $(\delta, U(\tau)) \in \mathcal{D}_\tau$. Choosing such a supporting slope at each type yields a bounded nondecreasing, hence measurable, gap schedule with the envelope property. Lemma 7 and (30) therefore make U feasible. The feasible utility schedules are consequently compact in the uniform norm.

The feasible-pair graph

$$\mathcal{G} = \{(\tau, p) : \tau \in [0, 1],\ p \in \mathcal{D}_\tau\}$$

is compact: it is the image of $[0, 1] \times \mathcal{S}$ under $(\tau, s) \mapsto (\tau, L_\tau s)$. The function $T_{\min}$ is lower semicontinuous on this graph. To see this, for a convergent sequence of feasible pairs choose minimizing statistics, pass to a subsequence attaining the lower limit of their loadings, and then to a convergent subsequence in $\mathcal{S}$. Its limit delivers the limiting pair, whose minimum loading can only be lower. Hence Ψ is upper semicontinuous on $\mathcal{G}$. As the payoff of an acceptance policy, it also satisfies a uniform bound

$$|\Psi(\tau, p)| \leq C \qquad ((\tau, p) \in \mathcal{G}) \tag{38}$$

for some finite constant C.

If $U_n \to U$ uniformly, convexity implies $U_n'(\tau) \to U'(\tau)$ at every point where U and all the U_n are differentiable: bound each derivative between the backward and forward secant slopes, first pass to the limit in n, and then shrink the secant interval. These points have full measure. Upper semicontinuity of Ψ and the reverse Fatou inequality, using (38), give $\limsup_n J(U_n) \leq J(U)$. Thus J attains its supremum on the compact feasible class.

Finally, approach any type from either available side through differentiability points of a feasible U. The corresponding derivatives converge to the one-sided slope, while utilities converge to $U(\tau)$. Closedness of $\mathcal{G}$ makes these limiting pairs deliverable. Convexity of $\mathcal{D}_\tau$ fills the interval between them at a kink. This proves the completed-graph assertion, including the endpoint convention. ■

Geometry of feasible statistics. Work in centered coordinates $z = (x_1 - \kappa, x_2 - \kappa, \theta)$, and extend their density f by zero outside $K = [-\kappa, 1 - \kappa]^2 \times \Theta$. Write $e_\theta = (0, 0, 1)$, $v = (1, 1, -1)$, $V = |v|$, and $e_v = v/V$. All distances and norms are Euclidean. The bounds below depend on the fixed primitives, but are uniform over types and feasible statistics.

Lemma 11 (Uniform geometric bounds)**.** *There are constants $0 < \rho \le R < \infty$ such that, at every $s \in \partial\mathcal{S}$ and every outward unit normal n,*

$$\overline{B}(s - \rho n, \rho) \subseteq \mathcal{S} \subseteq \overline{B}(s - Rn, R). \tag{39}$$

Every boundary statistic has a unique outward unit normal. If $s(n)$ is the statistic exposed by a unit vector n, then

$$|s(n') - s(n)| \le R|n' - n|. \tag{40}$$

There is also $\rho_D > 0$ such that $\overline{B}(0, \rho_D) \subseteq \mathcal{D}_\tau$ for every type. The sets $\mathcal{D}_\tau$ vary Lipschitz continuously in Hausdorff distance. A pair belongs to $\partial\mathcal{D}_\tau$ exactly when its fiber is a singleton. At a frontier pair with outward unit normal λ, that singleton has normal

$$n_0 = \frac{L_\tau^\top \lambda}{|L_\tau^\top \lambda|}, \qquad n_0 \cdot v = 0. \tag{41}$$

The normal λ is unique.

The two tangent balls control how sharply the boundary can bend. The inner ball provides feasible movements from a boundary statistic; the outer ball bounds their size. The density bounds in Assumption 1 supply these balls directly.

Proof. For a nonnegative, integrable function g of compact support, let

$$\mathcal{S}(g) = \left\{ \int zq(z)g(z)\,dz : 0 \le q \le 1 \right\}.$$

The compactness argument of Lemma 6 does not require $\int g = 1$. Pointwise maximization gives its support function, and support functions add when densities add:

$$h_{\mathcal{S}(g)}(w) = \int (w \cdot z)^+ g(z)\,dz, \qquad \mathcal{S}(g_1 + g_2) = \mathcal{S}(g_1) + \mathcal{S}(g_2).$$

Choose a small half-ball $H_\varepsilon = \overline{B}(0, r) \cap \{\varepsilon z_\theta \ge 0\}$, $\varepsilon \in \{-1, 1\}$, contained in K, on which $f \ge f_{\min}$ almost everywhere. It exists whether zero is interior to Θ or either endpoint: choose a side of Θ of positive length and take r sufficiently small. For a constant density $a > 0$ on this half-ball, symmetry and $b^+ = (|b| + b)/2$ give

$$h_{\mathcal{S}(a\mathbb{1}_{H_\varepsilon})}(w) = \rho_a(|w| + \varepsilon w_\theta), \qquad \rho_a = \frac{a}{2}\int_{H_\varepsilon} |z_\theta|\,dz > 0.$$

To see the symmetry calculation, the integral of $|w \cdot z|$ over a half-ball is half its integral over the full ball and is proportional to $|w|$; the vector $\int_{H_\varepsilon} z\,dz$ equals $\varepsilon e_\theta \int_{H_\varepsilon} |z_\theta|\,dz$. Thus the half-ball generates the translated ball $\overline{B}(\varepsilon\rho_a e_\theta, \rho_a)$. Taking $a = f_{\min}$ and writing $\rho = \rho_a$, additivity yields $\mathcal{S} = K_0 + \overline{B}(0, \rho)$ for a compact convex K_0. At a support point s with normal n, the ball contributes ρn; therefore $s - \rho n \in K_0$, proving the inner inclusion.

For the outer inclusion, choose a centered ball $\mathcal{B}$ containing K. The complementary density $g = f_{\max}\mathbb{1}_{\mathcal{B}} - f$ is nonnegative almost everywhere. The support function generated by a constant density on the full ball is $R|w|$ for a finite $R > 0$, so $\mathcal{S} + \mathcal{S}(g) = \overline{B}(0, R)$. If y supports $\mathcal{S}(g)$ in direction n, then $s + y = Rn$. Consequently $\mathcal{S} \subseteq \overline{B}(s - Rn, R)$. Both balls touch at s with normal n. Any other normal of $\mathcal{S}$ at s would support the inner ball there, and hence must equal n. The outer ball meets that supporting plane only at s, also giving uniqueness of the exposed point. For $h = s(n') - s(n)$, the two outer-ball inequalities imply $|h|^2 + 2Rn \cdot h \leq 0$ and $|h|^2 - 2Rn' \cdot h \leq 0$. Adding and using Cauchy–Schwarz proves (40).

For the projection claims, put

$$B_\tau = L_\tau^\top (L_\tau L_\tau^\top)^{-1}, \qquad \sup_{\tau \in [0,1]} (\|L_\tau\| + \|B_\tau\|) < \infty. \tag{42}$$

These bounds follow because L_τ has two independent rows and varies continuously on the compact type interval. Also $L_\tau B_\tau = I$ and $\operatorname{range} B_\tau = v^\perp$. For unit λ, the identity $\lambda = B_\tau^\top L_\tau^\top \lambda$ shows that $|L_\tau^\top \lambda|$ is uniformly bounded away from zero. The translated half-ball statistic set $\overline{B}(\varepsilon\rho e_\theta, \rho)$ lies in $\mathcal{S}$. The line $\mathbb{R}v$ enters its interior: for small $t > 0$,

$$|-\varepsilon t v - \varepsilon\rho e_\theta|^2 = \rho^2 - 2\rho t + |v|^2 t^2 < \rho^2.$$

Choose a ball $\overline{B}(s_*, r_*) \subseteq \mathcal{S}$ centered at such an $s_* \in \mathbb{R}v$. The uniform bound on B_τ then gives a common $\rho_D > 0$ such that every $|p| \leq \rho_D$ is delivered by $s_* + B_\tau p$ in that ball. Finally, $\mathcal{S}$ is bounded and L_τ is affine in τ. Matching $L_\tau s$ with $L_t s$ therefore proves Hausdorff Lipschitz continuity of the deliverable sets.

At a frontier pair, a supporting normal λ lifts to the nonzero supporting vector $L_\tau^\top \lambda$ of $\mathcal{S}$. Every statistic in its fiber maximizes this functional, so the fiber is a singleton and (41) follows. Conversely, an interior pair p is the midpoint of two distinct feasible pairs. Choose lifts of those pairs; their midpoint is interior to $\mathcal{S}$ by strict convexity and delivers p. The fiber therefore has positive length. Two normals at a frontier pair lift to normals at its singleton fiber. Uniqueness there and injectivity of $L_\tau^\top$ imply uniqueness in $\mathcal{D}_\tau$. ■

The gradient and the gain from moving inward. A movement in direction v substitutes shared surplus for private benefits while holding both delivery coordinates fixed. At a frontier pair its fiber has collapsed to a point. Moving inward opens room for this substitution; the tangent balls will show that this room grows at a square-root rate. For an interior pair $p = (\delta, u)$, let $s_+(\tau, p)$ denote its minimum-loading statistic.

Lemma 12 (Interior gradient)**.** *At every $p \in \operatorname{int} \mathcal{D}_\tau$, $T_{\min}(\tau, \cdot)$ is differentiable. If n is the unit normal at $s_+(\tau, p)$ and $a = n \cdot e_v$, then $a > 0$ and*

$$L_\tau^\top \nabla_p T_{\min} - e_\theta = \frac{n}{Va}, \qquad D_p T_{\min}[e] = e_\theta \cdot B_\tau e + \frac{n \cdot B_\tau e}{Va}. \tag{43}$$

For every $\eta > 0$, the gradients are uniformly bounded away from the frontier:

$$\sup_{\tau, p:\, \mu_\tau(p) \geq \eta} \big(|\nabla_p T_{\min}(\tau, p)| + |\nabla_p \Psi(\tau, p)| \big) < \infty. \tag{44}$$

Proof. The fiber has positive length. The support inequality between its two endpoints gives $n \cdot e_v \geq 0$; equality would expose both endpoints, so $a > 0$. The vector $w = n/(Va)$ satisfies $w \cdot v = 1$, whence $w + e_\theta \in v^\perp = \operatorname{range} L_\tau^\top$. Let g be the unique vector satisfying $L_\tau^\top g = w + e_\theta$. The support inequality at s_+, minimized over the fiber of p', gives

$$T_{\min}(\tau, p') \geq T_{\min}(\tau, p) + g \cdot (p' - p).$$

Conversely, any subgradient g' gives a supporting vector $L_\tau^\top g' - e_\theta$ at s_+ with dot product one with v. Normal uniqueness forces it to equal w, and hence $g' = g$. A finite convex function with a unique subgradient at an interior point is differentiable there. This proves the first identity in (43); taking its inner product with $B_\tau e$ proves the second. Finally, $T_{\min}$ is uniformly bounded because Θ is compact. For $\mu_\tau(p) \geq \eta$, apply its subgradient inequality at $p + \eta g/|g| \in \mathcal{D}_\tau$. The resulting bound on $\eta|g|$ is uniform in τ, p; the case $g = 0$ is immediate. Since $\Psi = u - T_{\min}$, this proves (44). ■

Lemma 13 (Uniform inward gain)**.** *There are constants $\bar{\mu}, c, C > 0$ such that, if $0 < \mu = \mu_\tau(p) \leq \bar{\mu}$, p_0 is any nearest frontier point with outward unit normal λ, and $|e| = 1$, $\lambda \cdot e \leq -c_0 < 0$, then*

$$D_p \Psi(\tau, p)[e] \geq c c_0 \mu^{-1/2} - C. \tag{45}$$

The constants are uniform over τ, p, p_0, e and c_0. At any frontier pair p_0, every direction with $\lambda \cdot e < 0$ enters the interior for sufficiently small positive steps and satisfies $D_e^+ \Psi(\tau, p_0) = +\infty$.

Proof. In this proof c, C denote positive constants depending only on the fixed primitives; their values may change between inequalities. The contained ball $\overline{B}(p, \mu)$ touches a supporting line at p_0, so $p - p_0 = -\mu\lambda$. Let s_0 be the singleton fiber at p_0, and set

$$b = |L_\tau^\top \lambda|, \quad n_0 = L_\tau^\top \lambda / b, \quad z_0 = B_\tau(p - p_0), \quad h = -n_0 \cdot z_0 = \mu / b.$$

The linear-algebra bounds make b bounded above and away from zero, uniformly in the type and normal. Thus h and μ are comparable, and $|z_0| \leq Ch$. Both z_0 and n_0 are perpendicular to e_v; the fiber line is $s_0 + z_0 + \mathbb{R} e_v$.

The tangent balls give elementary circle comparisons on this line. For a ball of radius $r' = \rho, R$

tangent at s_0, its intersection with the line has coordinates $|t| \le \sqrt{2r'h - |z_0|^2}$ along e_v, since

$$|z_0 + te_v + r'n_0|^2 = (r')^2 - 2r'h + |z_0|^2 + t^2.$$

Because $|z_0|^2 \le Ch^2$, the inner radicand is positive for uniformly small h. Writing $s_+ = s_0 + z_0 + te_v$, containment between the two ball intersections gives

$$c\sqrt{h} \le t \le C\sqrt{h}. \tag{46}$$

Thus an inward delivery change of order h permits a favorable movement of order $\sqrt{h}$ along the fiber. The next step is to bound the actual normal and turn this comparison of values into a marginal-gain bound.

Let n be the normal at s_+ and $a = n \cdot e_v > 0$. The inner tangent ball at s_+ lies below the supporting plane at s_0, which gives $\rho(1 - n_0 \cdot n) \le h$ and hence $|n - n_0| \le C\sqrt{h}$. Since $n_0 \cdot e_v = 0$, this also gives $a \le C\sqrt{h}$. For the reverse bound, the supporting plane at s_+ evaluated at s_0 gives

$$ta \ge -n \cdot z_0 \ge h - |n - n_0||z_0| \ge h - Ch^{3/2} \ge h/2$$

for uniformly small h. The upper bound on t then gives

$$|n - n_0| \le C\sqrt{h}, \qquad c\sqrt{h} \le a \le C\sqrt{h}. \tag{47}$$

All the smallness conditions can therefore be met by a single $\bar{\mu} > 0$ independent of τ, p, p_0, e, c_0.

The exact gradient identity now separates the inward gain from bounded terms:

$$D_p T_{\min}[e] = e_\theta \cdot B_\tau e + \frac{\lambda \cdot e}{Vba} + \frac{(n - n_0) \cdot B_\tau e}{Va} \le -cc_0\mu^{-1/2} + C. \tag{48}$$

The first term is bounded by the uniform norm of B_τ. The last is bounded because both $|n - n_0|$ and a have square-root order. The middle term has the displayed negative bound because $\lambda \cdot e \le -c_0$ and $a \le C\sqrt{\mu}$. Since $\Psi = u - T_{\min}$ and $|e_u| \le 1$, this proves (45) with uniform constants, including as c_0 varies.

At exact contact, fix e with $\lambda \cdot e < 0$, and put $k = B_\tau e$ and $\alpha = -\lambda \cdot e/b > 0$. For sufficiently small $s > 0$, the inner tangent ball contains

$$s_0 + sk + \sqrt{2\rho\alpha s - s^2|k|^2}\, e_v,$$

which delivers $p_0 + se$. Its chord midpoint $s_0 + sk$ is strictly inside the ball, so the delivered pair is interior. The original fiber is a singleton. Comparing its loading with this feasible statistic gives

$$\frac{T_{\min}(\tau, p_0 + se) - T_{\min}(\tau, p_0)}{s} \le e_\theta \cdot k - \frac{1}{V}\sqrt{\frac{2\rho\alpha}{s} - |k|^2} \longrightarrow -\infty.$$

The utility coordinate changes at the finite rate e_u, so the inward derivative of Ψ is $+\infty$. ■

Scaling the utility schedule. Let U^* be an optimal schedule in (36), whose existence is established in Lemma 10. Write $\mu(\tau) = \mu_{U^*}(\tau)$ for the margin of its completed slope graph.

Lemma 14 (Integrability of the boundary singularity)**.** *The completed margin of an optimal schedule satisfies*

$$\int_0^1 \mu(\tau)^{-1/2}\, d\tau < \infty, \tag{49}$$

with $0^{-1/2} = +\infty$. Consequently, on no nondegenerate one-sided interval adjacent to a type τ_0 can $\mu(\tau) \le K|\tau - \tau_0|^2$ hold almost everywhere for a finite $K > 0$.

Proof. By Lemma 10, Ψ is measurable and uniformly bounded on the compact feasible-pair graph $\mathcal{G}$. The pointwise margin is continuous on the graph, by the formula

$$\mu_\tau(p) = \min_{|\lambda|=1} \{h_{\mathcal{S}}(L_\tau^\top \lambda) - \lambda \cdot p\}.$$

Indeed, $\overline{B}(p, r) \subseteq \mathcal{D}_\tau$ exactly when every unit-direction support slack is at least r; the largest such radius is the distance to the frontier. At differentiability points of U^* put $p(\tau) = (U^{*\prime}(\tau), U^*(\tau))$. The path is measurable, and its margin equals $\mu(\tau)$ except at the at most countably many kinks and the endpoints. All subsequent integrals can therefore be evaluated on this path.

Scale every promised utility and surplus difference by $1 - t$, where $t \in [0, 1]$. Mixing each assigned policy with rejection delivers these new promises; monotonicity, the envelope relation, and participation are preserved. Thus $U_t = (1 - t)U^*$ is feasible. The principal then chooses the best policy within each new fiber, obtaining $J(U_t)$. For $t > 0$ define

$$Q_t(\tau) = \frac{\Psi(\tau, (1-t)p(\tau)) - \Psi(\tau, p(\tau))}{t}.$$

Concavity makes Q_t nondecreasing as $t \downarrow 0$, while bounded payoffs give $Q_t \ge Q_1 \ge -C$ for a uniform constant C. Optimality gives $\int_0^1 Q_t\, d\tau \le 0$. Monotone convergence applied to $Q_t + C$ shows that the limiting marginal gain $G = \lim_{t \downarrow 0} Q_t$ is integrable and satisfies

$$G \ge -C, \qquad \int_0^1 G(\tau)\, d\tau \le 0. \tag{50}$$

The common ball around zero makes this scaling direction uniformly inward near the frontier. Indeed, at an interior pair p with nearest frontier point p_0 and outward unit normal λ,

$$\lambda \cdot p_0 \ge \rho_D, \qquad \lambda \cdot p = \lambda \cdot p_0 - \mu_\tau(p).$$

For sufficiently small margin, $\lambda \cdot p$ is therefore bounded below by a positive constant. Since feasible pairs are uniformly bounded, apply Lemma 13 in direction $-p/|p|$ and multiply by $|p|$. For uniform constants $c, C > 0$, this gives

$$G(\tau) \ge c\, \mu(\tau)^{-1/2} - C \tag{51}$$

whenever the margin is positive and sufficiently small.

At an exact frontier pair, the direction $-p$ is strictly inward, so Lemma 13 gives $G = +\infty$. Such contacts at differentiability points form a null set because G is integrable. The completed margin is consequently positive almost everywhere. Away from the frontier, $\mu^{-1/2}$ is bounded and $G \geq -C$, so enlarging the constant extends (51) to almost every type. Integrating and using (50) proves (49). If $\mu(\tau) \leq K|\tau - \tau_0|^2$ almost everywhere on a nondegenerate one-sided interval, its inverse square root dominates $K^{-1/2}|\tau - \tau_0|^{-1}$ there, whose integral diverges. ■

Lemma 15 (Interior contacts are impossible)**.** *Let U^* be optimal. Then $\mu_{U^*}(\tau) > 0$ for every $0 < \tau < 1$. A contact at $\tau = 1$ can only have an outward normal with positive gap coordinate; a contact at $\tau = 0$ can only have an outward normal with negative gap coordinate.*

Proof. Fix a unit vector $\lambda = (\lambda_\delta, \lambda_u)$ and define the moving support value

$$H_\lambda(\tau) = h_{\mathcal{S}}(L_\tau^\top \lambda).$$

Let $s(w)$ denote the unique statistic exposed by a nonzero vector w. The support-point bound in Lemma 11, together with the uniform lower bound on $|L_\tau^\top \lambda|$, implies that H_λ has a Lipschitz derivative. In fact,

$$H'_\lambda(\tau) = \lambda_u\big(s_1(L_\tau^\top \lambda) - s_2(L_\tau^\top \lambda)\big).$$

This follows by differentiating the support function; its derivative at a nonzero vector is its unique exposed statistic. The displayed derivative is Lipschitz because $L_\tau^\top \lambda$ is affine in τ, is bounded away from zero, and the exposed statistic is Lipschitz as a function of the unit normal.

Suppose $p_0 = (\delta_0, U^*(\tau_0)) \in P_{U^*}(\tau_0)$ is a frontier point, with outward unit normal λ. Its unique fiber statistic is exposed by $L_{\tau_0}^\top \lambda$, so

$$H_\lambda(\tau_0) = \lambda \cdot p_0, \qquad H'_\lambda(\tau_0) = \lambda_u \delta_0.$$

The derivative identity cancels the first-order utility change in the comparison between the schedule and the moving support line. The remaining Taylor error is quadratic. Writing $h = \tau - \tau_0$, the Lipschitz derivative gives a finite constant C such that

$$H_\lambda(\tau) \leq \lambda \cdot p_0 + \lambda_u \delta_0 h + Ch^2. \tag{52}$$

For any $\delta \in \Gamma_{U^*}(\tau)$, set

$$E = U^*(\tau) - U^*(\tau_0) - \delta_0 h.$$

The supporting-slope inequalities at τ_0 and τ give

$$0 \leq E \leq (\delta - \delta_0)h. \tag{53}$$

Consider the possible signs of the normal's gap coordinate.

If $\lambda_\delta > 0$, the supporting inequality makes δ_0 the largest slope in $\Gamma_{U^*}(\tau_0)$. On the available right side, $\delta - \delta_0 \geq 0$. For sufficiently small $h \geq 0$, (53) implies

$$\lambda_\delta(\delta - \delta_0) + \lambda_u E \geq 0.$$

For $\lambda_u \geq 0$ both terms are nonnegative; for $\lambda_u < 0$, their sum is at least $(\lambda_\delta + \lambda_u h)(\delta - \delta_0)$, which is nonnegative for small h. If $\lambda_\delta < 0$, the same argument on the available left side uses $\delta - \delta_0 \leq 0$ and $h \leq 0$. If $\lambda_\delta = 0$ and $\lambda_u > 0$, the inequality holds on both available sides directly from $E \geq 0$.

In each of these cases, combining the inequality with (52) yields

$$0 \leq H_\lambda(\tau) - \lambda \cdot (\delta, U^*(\tau)) \leq Ch^2.$$

The pointwise margin is no larger than this support slack: moving from $(\delta, U^*(\tau))$ in direction λ must leave $\mathcal{D}_\tau$ by the time this slack is exceeded. Thus, on the indicated side,

$$\mu_{U^*}(\tau) \leq C|\tau - \tau_0|^2. \tag{54}$$

The remaining case, $\lambda = (0, -1)$, is impossible. The common ball $\overline{B}(0, \rho_D) \subseteq \mathcal{D}_{\tau_0}$ implies $\lambda \cdot p_0 \geq \rho_D$, whereas $\lambda \cdot p_0 = -U^*(\tau_0) \leq 0$.

Every interior contact therefore yields (54) on a nondegenerate one-sided interval. This contradicts Lemma 14, because the integral of $|\tau - \tau_0|^{-1}$ on that interval diverges. At $\tau_0 = 1$, the same contradiction excludes $\lambda_\delta \leq 0$; at $\tau_0 = 0$, it excludes $\lambda_\delta \geq 0$. These are exactly the claimed endpoint restrictions. ■

Lemma 16 (Endpoint contacts are impossible)**.** *For an optimal schedule U^*, both completed endpoint pairs are interior to their deliverable sets.*

Proof. Consider the upper endpoint. Suppose

$$p_* = (\bar{\Delta}, U^*(1)) \in \partial\mathcal{D}_1, \qquad \bar{\Delta} = U_-^{*\prime}(1).$$

By Lemma 15, its outward unit normal λ has $\lambda_\delta > 0$. If $U^*(1) = 0$, nonnegativity gives $\bar{\Delta} \leq 0$. Evaluating the supporting inequality at $+\rho_D\lambda \in \mathcal{D}_1$ gives

$$\lambda_\delta\bar{\Delta} = \lambda \cdot p_* \geq \rho_D > 0,$$

a contradiction. Hence $U^*(1) > 0$, and $U^* \geq 2\eta > 0$ on a terminal interval.

Choose $\Delta(\tau) = U_+^{*\prime}(\tau)$ for $\tau < 1$ and $\Delta(1) = \bar{\Delta}$. These slopes are feasible by Lemma 10. They satisfy $\Delta(\tau) < \bar{\Delta}$ for every $\tau < 1$. Otherwise monotonicity would make the slope constant on a terminal interval. If s_* is the unique statistic delivering p_*, the pairs on that interval would be

$$(\bar{\Delta}, U^*(1) + (\tau - 1)\bar{\Delta}) = L_\tau s_*.$$

Their fiber is the same singleton for every τ, because all the maps L_τ have the same kernel. They would all be frontier points, contrary to Lemma 14.

For small $t > 0$, cap the slope at $\bar{\Delta} - t$:

$$\Delta_t(\tau) = \min\{\Delta(\tau), \bar{\Delta} - t\}, \quad a_t(\tau) = \Delta(\tau) - \Delta_t(\tau),$$
$$b_t(\tau) = \int_0^\tau a_t(r)\, dr, \qquad\qquad U_t = U^* - b_t.$$

Let $\tau_t = \inf\{\tau : \Delta(\tau) > \bar{\Delta} - t\}$ and $\ell_t = 1 - \tau_t$. Then $\tau_t < 1$ and $\tau_t \uparrow 1$ as $t \downarrow 0$. Monotonicity of a_t gives

$$0 \le a_t \le t, \qquad 0 \le b_t(\tau) \le \ell_t a_t(\tau). \tag{55}$$

The cap preserves monotonicity, and U_t has slope Δ_t almost everywhere. Since $b_t \le t\ell_t$ and only a terminal interval is affected, participation holds for small t. The utility reduction is the integral of the slope reduction over a shrinking interval, so it is small relative to that slope reduction. The change in the delivered pair, $(-a_t, -b_t)$, is therefore nearly horizontal, pointing into the deliverable set at the endpoint.

The following convex combination makes this feasibility argument precise. The direction $(-1, 0)$ is strictly inward at p_*. Lemma 13 therefore permits a choice of $s > 0$ with $z = p_* - s(1, 0) \in \operatorname{int} \mathcal{D}_1$. Choose $r > 0$ so that $\overline{B}(z, 2r) \subseteq \mathcal{D}_1$. By continuity of support functions, $\overline{B}(z, r) \subseteq \mathcal{D}_\tau$ for all τ sufficiently close to 1.

Write $p(\tau) = (\Delta(\tau), U^*(\tau))$ and $p_t(\tau) = (\Delta_t(\tau), U_t(\tau))$. At a changed type, $a = a_t(\tau) > 0$, and for $t < s$ define

$$\alpha = \frac{a}{s} \in (0, 1), \qquad z_\tau = p(\tau) - s\left(1, \frac{b_t(\tau)}{a}\right).$$

Then

$$p_t(\tau) = (1 - \alpha)p(\tau) + \alpha z_\tau, \qquad |z_\tau - z| \le |p(\tau) - p_*| + s\ell_t.$$

The last bound tends uniformly to zero over changed types: $\tau_t \to 1$, the monotone slope converges to $\bar{\Delta}$, and U^* is continuous. Consequently $z_\tau \in B(z, r)$ and $p_t(\tau) \in \operatorname{int} \mathcal{D}_\tau$ for small t. If $a_t(\tau) = 0$, (55) gives $b_t(\tau) = 0$, so that pair is unchanged. All constraints of (36) are satisfied.

Finally, consider the payoff effect. The changed pairs $p_t(\tau)$ converge uniformly to the endpoint contact, so their margins tend uniformly to zero. Every nearest frontier point also converges to p_*, and its outward unit normal converges to λ. To see the latter claim, pass to the limit in the support equality along any sequence of approaching types and frontier points: every limiting normal supports $\mathcal{D}_1$ at p_*, where the normal is unique.

By (55), the unit displacement

$$e_t(\tau) = \frac{p_t(\tau) - p(\tau)}{|p_t(\tau) - p(\tau)|}$$

converges uniformly to $(-1, 0)$ on changed types. These directions therefore have a common positive inward component relative to the nearby frontier normals. Lemma 13 implies that the directional derivatives at the changed pairs tend uniformly to $+\infty$ as $t \downarrow 0$. In particular, they are all positive for sufficiently small t. Concavity then gives

$$\begin{aligned}\Psi(\tau, p_t(\tau)) - \Psi(\tau, p(\tau)) &\geq \nabla_p \Psi(\tau, p_t(\tau)) \cdot (p_t(\tau) - p(\tau)) \\ &= |p_t(\tau) - p(\tau)|\, D_p \Psi(\tau, p_t(\tau))[e_t(\tau)] > 0.\end{aligned}$$

The changed types contain a nondegenerate terminal interval, and all other payoffs are unchanged. Integrating yields $J(U_t) > J(U^*)$, contradicting optimality.

Interchanging x_1 and x_2 and replacing τ by $1 - \tau$ gives the lower-endpoint argument. This transformation preserves the assumptions, so it does not require a symmetric prior. ■

Proof of Lemma 9. Existence is established in Lemma 10. Lemmas 15 and 16 exclude every frontier contact of the completed slope graph. This graph is compact. It is bounded by feasibility; at an interior type, closedness follows by passing to the limit in the supporting-slope inequalities. At an endpoint, both one-sided derivatives at approaching interior types converge to the prescribed limiting endpoint slope, so every intervening slope has that same limit.

The function $(\tau, p) \mapsto \mu_\tau(p)$ is continuous on $\mathcal{G}$, as shown by the support representation in Lemma 14. It is strictly positive at every point of the compact completed slope graph, and therefore has a strictly positive minimum there. This is (37). ■

B.2 Schedule deformations and necessary conditions

Fix an optimal utility schedule U^*. Throughout this subsection, use the gap representative

$$\Delta(\tau) = U_+^{*\prime}(\tau) \quad (0 \le \tau < 1), \qquad \Delta(1) = U_-^{*\prime}(1), \qquad p(\tau) = (\Delta(\tau), U^*(\tau)).$$

Lemma 10 makes these pairs feasible, and Lemma 9 places their completed graph uniformly inside the deliverable sets. Define the principal's marginal values of utility and surplus difference by

$$m(\tau) \coloneqq \Psi_u(\tau, p(\tau)), \qquad \pi(\tau) \coloneqq \Psi_\delta(\tau, p(\tau)). \tag{56}$$

The type density is uniform, so these are also the marginal values entering the principal's integrated objective. I first derive the restrictions obtained by shifting utility and adding or reversing kinks. These restrictions also establish continuity of the optimal gap.

Marginal values and feasible deformations.

Lemma 17 (Continuity of the marginal values)**.** *The map $(\tau, p) \mapsto \nabla_p \Psi(\tau, p)$ is continuous whenever $p \in \operatorname{int} \mathcal{D}_\tau$. The functions m and π are bounded and have one-sided limits at every interior type a, given by*

$$m(a^\pm) = \Psi_u\big(a, U_\pm^{*\prime}(a), U^*(a)\big),$$
$$\pi(a^\pm) = \Psi_\delta\big(a, U_\pm^{*\prime}(a), U^*(a)\big).$$

Their one-sided limits at 0 and 1 equal their defined endpoint values. In particular, both marginal values are continuous wherever Δ is continuous.

Proof. I first show continuity of the minimum-loading statistic. Suppose $(\tau_n, p_n) \to (\tau, p)$ through interior pairs, with $p \in \operatorname{int} \mathcal{D}_\tau$. The fiber lines are parallel to v and converge to the fiber line at (τ, p): each contains $B_{\tau_n} p_n$, where B_τ is the continuous right inverse in (42). The limiting fiber has positive length by Lemma 11, and strict convexity of $\mathcal{S}$ puts the relative interior of this segment in $\operatorname{int} \mathcal{S}$.

The minimizing statistics s_n lie in the compact set $\mathcal{S}$. Any subsequential limit $\bar{s}$ belongs to the limiting fiber. If its loading exceeded the minimum on that fiber, choose a point s' in the relative interior of the fiber with strictly smaller loading than $\bar{s}$. Since s' is interior to $\mathcal{S}$, the nearby fiber lines contain feasible points converging to s'. For large n their loadings would be strictly below that of s_n, contrary to its minimality. Thus every subsequential limit is the unique minimum-loading statistic $s^*(\tau, p)$, proving continuity.

The outward unit normal at this statistic is continuous as well. Indeed, any convergent subsequence of normals to s_n supports $\mathcal{S}$ at $s^*(\tau, p)$ in the limit, and Lemma 11 gives uniqueness of that

normal. Its dot product with v is strictly positive by Lemma 12. The gradient identity (43), together with continuity of B_τ, therefore gives joint continuity of $\nabla_p T_{\min}$ and of $\nabla_p \Psi$.

Boundedness along the optimal path follows from the uniform margin and (44). Monotonicity of Δ gives its one-sided limits, while U^* is continuous. The limiting pairs belong to the completed graph and hence are interior. Composing the jointly continuous gradient with these limits proves the remaining assertions. ■

Call a Lipschitz function $h : [0,1] \to \mathbb{R}$ an *admissible deformation* if $U^* + th$ is a feasible utility schedule for every sufficiently small $t > 0$. Let $\nu = U^{*\prime\prime}$ denote the distributional second derivative on $(0,1)$, a finite nonnegative measure.

Lemma 18 (First variation and admissibility)**.** *For every admissible deformation h,*

$$DJ(U^*)[h] := \lim_{t \downarrow 0} \frac{J(U^* + th) - J(U^*)}{t} = \int_0^1 \left[m(\tau)h(\tau) + \pi(\tau)h'(\tau) \right] d\tau \leq 0. \tag{57}$$

If h' has bounded variation, then h is admissible if and only if there is a finite constant $C \geq 0$ such that

$$h'' \geq -C\nu \quad \text{as measures on } (0,1), \qquad h \geq -CU^* \quad \text{on } [0,1]. \tag{58}$$

Proof. At almost every type the delivered pair changes by $t(h', h)$. For small t, the entire segment of such pairs remains at a common positive distance from the frontier, because h and h' are bounded and the optimal completed graph has a uniform margin. The gradient bound in Lemma 12 therefore dominates the difference quotient by a constant times $|h| + |h'|$. Differentiability and dominated convergence give the integral formula in (57); optimality gives its sign.

For the second assertion, convexity of $U^* + th$ is equivalent to $\nu + th'' \geq 0$, and participation is equivalent to $U^* + th \geq 0$. If the deformation is feasible for some $t > 0$, both inequalities in (58) follow with $C = 1/t$.

Conversely, (58) makes $U^* + th$ convex and nonnegative for small t. Its right-hand slopes, with the limiting slope used at 1, differ from the corresponding slopes of U^* by at most $t\|h'\|_\infty$. Its utility differs by at most $t\|h\|_\infty$. Uniform interiority therefore makes all these pairs deliverable for small t. The envelope relation and Lemma 7 give the remaining feasibility conditions. ■

The two inequalities in (58) have separate roles. The first permits removing curvature only where the schedule already has curvature. The second permits lowering utility only where the participation constraint leaves enough room. Uniform interiority ensures that sufficiently small movements satisfying these two requirements remain deliverable.

Shifts and kink deformations. For $c \in [0,1]$, define the upward and downward kink functions

$$h_c(\tau) = (\tau - c)^+, \qquad \widetilde{h}_c(\tau) = (c - \tau)^+.$$

Both are convex and nonnegative, hence admissible. An upward kink increases the gap of every type above c by one unit and increases type τ's utility by $\tau - c$. Its first-order payoff effect, and that of the mirrored kink, are

$$\begin{aligned} \Lambda(c) &\coloneqq \int_c^1 \big[m(\tau)(\tau - c) + \pi(\tau)\big]\, d\tau, \\ \widetilde{\Lambda}(c) &\coloneqq \int_0^c \big[m(\tau)(c - \tau) - \pi(\tau)\big]\, d\tau. \end{aligned} \tag{59}$$

Write the cumulative marginal values of utility as

$$\lambda_\uparrow(c) \coloneqq \int_c^1 m(\tau)\, d\tau, \qquad \lambda_\downarrow(c) \coloneqq -\int_0^c m(\tau)\, d\tau.$$

The first integral measures the value of raising utility equally for every type above c; the sign in the second is chosen so that both functions have derivative $-m$ almost everywhere.

Lemma 19 (Shift, kink, and endpoint inequalities)**.** *The optimal schedule satisfies*

$$\int_0^1 m\, d\tau \le 0, \qquad \Lambda(c) \le 0, \qquad \widetilde{\Lambda}(c) \le 0 \quad (c \in [0,1]). \tag{60}$$

The first inequality is an equality if $U^* > 0$ everywhere. The kink functionals are Lipschitz and satisfy

$$\begin{aligned} \Lambda(c) &= \int_c^1 (\pi + \lambda_\uparrow)(s)\, ds, \\ \widetilde{\Lambda}(c) &= -\int_0^c (\pi + \lambda_\downarrow)(s)\, ds, \\ \Lambda(c) - \widetilde{\Lambda}(c) &= \int_0^1 \big[\tau m(\tau) + \pi(\tau)\big]\, d\tau - c\int_0^1 m(\tau)\, d\tau. \end{aligned} \tag{61}$$

In particular, $\Lambda(1) = \widetilde{\Lambda}(0) = 0$. Their one-sided derivatives at every interior type a are

$$\Lambda'(a^\pm) = -\pi(a^\pm) - \lambda_\uparrow(a), \qquad \widetilde{\Lambda}'(a^\pm) = -\pi(a^\pm) - \lambda_\downarrow(a), \tag{62}$$

with the corresponding one-sided formulas at the endpoints. Consequently,

$$\pi(0) \ge 0, \qquad \pi(1) \le 0. \tag{63}$$

Proof. The constant deformation $h \equiv 1$ is admissible, so (57) gives $\int_0^1 m \le 0$. If $U^* > 0$ everywhere, continuity gives a strictly positive minimum and $h \equiv -1$ is also admissible, yielding equality. Applying (57) to h_c and $\widetilde{h}_c$ gives the two kink inequalities.

Changing the order of integration gives

$$\int_c^1 m(\tau)(\tau - c)\, d\tau = \int_c^1 \int_s^1 m(\tau)\, d\tau\, ds = \int_c^1 \lambda_\uparrow(s)\, ds.$$

The same calculation for the mirrored kink proves the first two identities in (61). Subtracting the defining integrals gives the third. Boundedness of m, π makes both kink functionals Lipschitz. Lemma 17 gives one-sided limits of their integrands, proving (62).

Since $\Lambda \leq 0$ and $\Lambda(1) = 0$, its left derivative at 1 is nonnegative. That derivative is $-\pi(1)$ because $\lambda_\uparrow(1) = 0$. Similarly, $\widetilde{\Lambda} \leq 0$ and $\widetilde{\Lambda}(0) = 0$ imply $\widetilde{\Lambda}'(0^+) = -\pi(0) \leq 0$. These are (63). ■

The optimal gap is continuous. A jump in the gap creates a kink in utility that can be partially removed. This makes the corresponding kink functional attain its maximum of zero. Strict concavity of the inner value then gives the opposite pattern of one-sided derivatives at that maximum, ruling out the jump.

Lemma 20 (No jumps in the surplus difference)**.** *The optimal utility schedule belongs to $C^1([0,1])$, with one-sided derivatives at the endpoints, and $\Delta = U^{*\prime}$ is continuous. The marginal values m, π are continuous on $[0,1]$, and $\Lambda, \widetilde{\Lambda} \in C^1([0,1])$. At every interior type where $U^* = 0$, one has $\Delta = 0$.*

Proof. Suppose $\Delta(a^-) < \Delta(a^+)$ at some $a \in (0,1)$. The zero set of the nonnegative convex function U^* is an interval, possibly empty or a point. A jump cannot lie in its interior. At least one of the following cases applies: $U^* > 0$ on $(a,1]$, with $\Delta(a^+) > 0$ if $U^*(a) = 0$; or $U^* > 0$ on $[0,a)$, with $\Delta(a^-) < 0$ if $U^*(a) = 0$. Indeed, when $U^*(a) > 0$ the zero interval lies entirely on one side of a; when $U^*(a) = 0$, the strict jump and the minimum property imply that at least one of these two slope signs is strict.

Consider the first case. The measure ν has an atom of positive mass $\Delta(a^+) - \Delta(a^-)$ at a, so the curvature of $-h_a$ is dominated by a finite multiple of ν. Its negative utility effect is also dominated by a finite multiple of U^*. If $U^*(a) > 0$, utility has a positive minimum on $[a,1]$; if $U^*(a) = 0$, convexity gives

$$U^*(\tau) \geq \Delta(a^+)(\tau - a) \qquad (\tau \geq a).$$

On $[0,a]$, $h_a = 0$. Lemma 18 therefore makes $-h_a$ admissible. Optimality gives $-\Lambda(a) \leq 0$, while Lemma 19 gives $\Lambda(a) \leq 0$. Thus $\Lambda(a) = 0$ is a maximum, and

$$\Lambda'(a^-) \geq 0 \geq \Lambda'(a^+).$$

But (62) yields

$$\begin{aligned}\Lambda'(a^+) - \Lambda'(a^-) &= \pi(a^-) - \pi(a^+) \\ &= \Psi_\delta\big(a, \Delta(a^-), U^*(a)\big) - \Psi_\delta\big(a, \Delta(a^+), U^*(a)\big) > 0.\end{aligned}$$

The strict inequality follows because the completed slope segment is interior and $\Psi(a, \cdot, U^*(a))$ is differentiable and strictly concave there. Its derivative is strictly decreasing. This contradicts the preceding one-sided inequalities.

In the second case, use the reversed mirrored kink $-\widetilde{h}_a$. At a zero-rent contact its floor bound is

$$U^*(\tau) \geq |\Delta(a^-)|(a - \tau) \qquad (\tau \leq a).$$

The same argument gives $\widetilde{\Lambda}(a) = 0$ and an upward jump in its derivative at a maximum, again a contradiction.

Thus Δ has no interior jumps. A bounded nondecreasing function has no other discontinuities, and the chosen endpoint values are its one-sided limits. The envelope relation gives $U^* \in C^1([0,1])$. Lemma 17 then gives continuity of m, π, and (61) gives continuously differentiable kink functionals. Finally, every interior zero of the nonnegative differentiable function U^* is a minimum, so its derivative is zero there. ■

Reversing kinks where the gap changes. Since Δ is now continuous, its curvature measure ν has no atoms. Write

$$\mathcal{C} \coloneqq \operatorname{supp} \nu \subset (0,1), \qquad P \coloneqq \{\tau \in [0,1] : U^*(\tau) = 0\},$$

and define the upper and lower parts of the curvature support by

$$\begin{aligned}\mathcal{C}_\uparrow &\coloneqq \overline{\{c \in \mathcal{C} : U^* > 0 \text{ on } [c,1]\}}, \\ \mathcal{C}_\downarrow &\coloneqq \overline{\{c \in \mathcal{C} : U^* > 0 \text{ on } [0,c]\}},\end{aligned} \tag{64}$$

where closures are taken in $[0,1]$. A point belongs to $\mathcal{C}$ exactly when the gap changes in every neighborhood of that point. On the complementary intervals the utility schedule is affine.

Lemma 21 (Structure of the curvature support)**.** *The relatively open set $(0,1) \setminus \mathcal{C}$ is a countable union of disjoint intervals. On the closure $[a,b]$ of each such interval, U^* is affine. Its endpoints in $(0,1)$ belong to $\mathcal{C}$, distinct components cannot share an endpoint, and $\mathcal{C}$ has no isolated points.*

Every component is either the interior of P relative to $(0,1)$, an interval on whose closure utility is strictly positive, or an affine end piece $(0,b)$ with $U^*(0) = 0 < U^*$ on $(0,b]$, or its mirror at 1. Every endpoint of P lying in $(0,1)$ belongs to the adjacent upper or lower curvature support. If $P = \{0\}$, then $0 \in \mathcal{C}_\uparrow$ unless an affine end piece begins at 0; the mirrored statement holds when $P = \{1\}$.

Proof. The support is relatively closed. Each component (a, b) of its complement has zero ν-measure, so Δ is constant there and U^* is affine on $[a, b]$. An endpoint in $(0, 1)$ must belong to the support; otherwise a neighborhood of it would extend the same component. Two components sharing an endpoint would have the same slope by continuity of Δ, making their union and common endpoint one affine interval, a contradiction. An isolated support point would have zero measure, since ν has no atoms, and a neighborhood with no other support points would also have zero measure. This contradicts its being in the support.

If an affine component contains a zero of U^*, nonnegativity forces the affine function to vanish throughout the component. The component is then the interior of P, relative to $(0, 1)$. Otherwise utility is positive on (a, b). Suppose $U^*(a) = 0 < U^*(b)$. Its slope is strictly positive, so a cannot be an interior minimum of the differentiable function U^*; hence $a = 0$. The argument at b is the mirror. This gives the stated list of components.

Let $b \in (0, 1)$ be the upper endpoint of P. Every right neighborhood of b meets $\mathcal{C}$: otherwise an affine component would meet P at b and have positive utility immediately to its right, which the preceding argument excludes at an interior endpoint. Those support points have positive utility on their upper tails, so $b \in \mathcal{C}_\uparrow$. The lower endpoint is symmetric. If $P = \{0\}$, the same reasoning says that $0 \notin \mathcal{C}_\uparrow$ exactly when some initial interval misses $\mathcal{C}$, in which case it is an affine end piece. The top endpoint follows by reflection. ■

Lemma 22 (Reversal on the curvature support)**.** *On $\mathcal{C}_\uparrow$,*

$$\Lambda = \Lambda' = 0, \qquad \pi(\tau) = -\lambda_\uparrow(\tau) = -\int_\tau^1 m(s)\, ds. \tag{65}$$

On $\mathcal{C}_\downarrow$,

$$\widetilde{\Lambda} = \widetilde{\Lambda}' = 0, \qquad \pi(\tau) = -\lambda_\downarrow(\tau) = \int_0^\tau m(s)\, ds. \tag{66}$$

Endpoint derivatives are one-sided. In particular, $\pi(1) = 0$ if $1 \in \mathcal{C}_\uparrow$, and $\pi(0) = 0$ if $0 \in \mathcal{C}_\downarrow$.

Proof. Fix $c \in \mathcal{C}$ with $U^* > 0$ on $[c, 1]$. Choose a small band $B = [c - \varepsilon, c + \varepsilon] \subset (0, 1)$ such that $U^* > 0$ on $[c - \varepsilon, 1]$. Its curvature mass $M = \nu(B)$ is positive. Average reversed kinks using the schedule's own curvature:

$$h(\tau) = -\frac{1}{M} \int_B (\tau - s)^+ \, d\nu(s).$$

This function is Lipschitz, vanishes on $[0, c - \varepsilon]$, and satisfies $-1 \le h \le 0$. Its derivative is of bounded variation and $h'' = -\nu|_B / M \ge -\nu/M$. On the remaining tail, U^* has a strictly positive minimum, so $h \ge -CU^*$ for some finite C. Lemma 18 makes h admissible.

Fubini's theorem and (57) give

$$0 \geq DJ(U^*)[h] = -\frac{1}{M}\int_B \Lambda(s)\, d\nu(s).$$

Since $\Lambda \leq 0$, this implies $\Lambda = 0$ for ν-almost every point in B. If $\Lambda(c) < 0$, continuity would give a neighborhood of c on which it is strictly negative. That neighborhood has positive curvature mass because $c \in \mathcal{C}$, a contradiction. Hence $\Lambda(c) = 0$.

By continuity, this equality extends to $\mathcal{C}_\uparrow$. The set of points $c \in \mathcal{C}$ with positive utility on $[c, 1]$ is relatively open in $\mathcal{C}$, and $\mathcal{C}$ has no isolated points. Every point of its closure is therefore approached by distinct points at which $\Lambda = 0$. Since $\Lambda \in C^1([0, 1])$, its derivative is zero on that closure, including any endpoint there. Equation (62) gives (65).

For the lower support, average $-(s - \tau)^+$ over the corresponding band. The deformation vanishes above the band, and positive utility on the lower tail gives its floor bound. The same calculation yields (66). The endpoint equalities follow because $\lambda_\uparrow(1) = \lambda_\downarrow(0) = 0$. ■

The averaging removes only curvature already present in the schedule, preserving convexity even when the curvature has no density. The balance equations equate the marginal effect of changing a gap to the cumulative effect on the utilities of the types above or below it.

Shared policies and zero-rent endpoints. At every type, implement $p(\tau)$ by the unique policy solving the inner problem in Lemma 8. This fixes the canonical optimal menu, with the limiting gaps at the two endpoints.

Lemma 23 (Affine intervals and shared policies)**.** *On a nondegenerate interval $[a, b] \subseteq [0, 1]$, U^* is affine if and only if the canonical menu assigns the same policy to every type, up to π_0-null sets. If $n = (n_1, n_2, n_\theta)$ is the outward unit normal at the common optimal statistic, then $n \cdot v > 0$ and*

$$m(\tau) = m_0 \coloneqq -\frac{n_\theta}{n \cdot v}, \qquad \pi(\tau) = \tau(1 - m_0) - \frac{n_1}{n \cdot v} \quad (\tau \in [a, b]). \tag{67}$$

Thus the marginal value of utility is constant along a shared policy, while the marginal value of the gap is affine with slope $1 - m_0$.

Proof. If U^* is affine on $[a, b]$ with slope δ_0, continuity of Δ makes $\Delta = \delta_0$ throughout that interval. The fiber over $p(\tau)$ is determined by

$$S_1 - S_2 = \delta_0, \qquad S_2 + T = U^*(\tau) - \tau\delta_0 = U^*(a) - a\delta_0.$$

Neither equality depends on τ. The fiber and its minimum-loading endpoint are therefore the same for every type in the interval. Lemma 8 gives the same uniquely implementing policy. Conversely, the utility delivered by a fixed policy is affine in the type.

Let n be the normal at the common statistic. By Lemma 12, $n \cdot v > 0$ and

$$L_\tau^\top \nabla_p T_{\min} - e_\theta = \frac{n}{n \cdot v}.$$

Since $\nabla_p T_{\min} = (-\pi, 1 - m)$, the first and third coordinates of this identity are

$$\tau(1 - m) - \pi = \frac{n_1}{n \cdot v}, \qquad -m = \frac{n_\theta}{n \cdot v}.$$

The normal is fixed across the interval, proving (67). ■

Lemma 24 (Rotation about a zero-rent endpoint)**.** *If $U^*(0) = 0$ and $\Delta(0) > 0$, both deformations $h(\tau) = \pm\tau$ are admissible, and*

$$\Lambda(0) = \int_0^1 \big[\tau m(\tau) + \pi(\tau)\big]\, d\tau = 0. \tag{68}$$

If $U^*(1) = 0$ and $\Delta(1) < 0$, both deformations $h(\tau) = \pm(1 - \tau)$ are admissible, and $\widetilde{\Lambda}(1) = 0$.

Proof. Convexity and the envelope give $U^*(\tau) \geq \Delta(0)\tau$ when $U^*(0) = 0$. The two affine deformations have zero curvature, and the negative one satisfies $-\tau \geq -U^*(\tau)/\Delta(0)$. Both therefore satisfy (58). Their first variations have opposite signs, so optimality makes both zero. The value for $h(\tau) = \tau$ is $\Lambda(0)$.

At the other endpoint, convexity gives $U^*(\tau) \geq |\Delta(1)|(1 - \tau)$ when $U^*(1) = 0$ and $\Delta(1) < 0$. Apply the same argument to $\pm(1 - \tau)$ to obtain $\widetilde{\Lambda}(1) = 0$. ■

B.3 No affine pieces outside the zero-rent pool

I now use the uniform type density to rule out affine utility intervals outside the zero-rent pool. Continue with the optimal schedule and notation of Appendix B.2: $\Delta = U^{*\prime}$, $m = \Psi_u$, $\pi = \Psi_\delta$, the zero set P, and the curvature supports $\mathcal{C}$, $\mathcal{C}_\uparrow$, and $\mathcal{C}_\downarrow$.

Lemma 25 (No affine pieces outside the pool)**.** *The optimal schedule U^* is not affine on $[0,1]$. If it is affine on a nondegenerate interval $I \subseteq [0,1]$, then $I \subseteq P$. Consequently, U^* is strictly convex on each positive-rent flank, and two distinct types receive the same policy in the canonical optimal menu if and only if both belong to P. If P is empty or a singleton, U^* is strictly convex on all of $[0,1]$. Moreover,*

$$m(\tau) < \frac{1}{2} \quad (0 < \tau < 1), \qquad m(0), m(1) \leq \frac{1}{2}. \tag{69}$$

The proof separates two implications. First, the kink conditions force any affine piece outside the pool to have marginal utility value $m = 1/2$. Second, a comparison of nearby types shows that this value would prevent the gap from changing near an interior point of its curvature support. The resulting contradiction excludes the affine piece.

What an affine piece would require.

Lemma 26 (The marginal value on an affine piece)**.** *The optimal schedule is not affine on the entire type interval. Every component (a,b) of $(0,1) \setminus \mathcal{C}$ other than the interior of P relative to $(0,1)$ has a common policy with $m \equiv 1/2$ on $[a,b]$, and at least one endpoint*

$$e \in \{a, b\} \cap (0,1)$$

belongs to $\mathcal{C}_\uparrow \cup \mathcal{C}_\downarrow$.

Proof. If U^* were affine on $[0,1]$, Lemma 23 would give $m \equiv m_0$ and $\pi(1) - \pi(0) = 1 - m_0$. The endpoint inequalities (63) imply $m_0 \geq 1$, whereas the upward utility shift gives $m_0 = \int_0^1 m \leq 0$. This is impossible. Thus every affine component has at least one endpoint in $(0,1)$.

Fix a component (a,b) outside the pool. Lemmas 21 and 23 give an affine schedule on $[a,b]$, a common policy, and a constant marginal utility value m_0. I show that one of the kink functionals $L \in \{\Lambda, \widetilde{\Lambda}\}$ satisfies

$$L(a) = L(b) = 0, \qquad L'(e) = 0 \tag{70}$$

at an interior endpoint e in the corresponding curvature support.

Suppose first that utility is positive on $[a,b]$. If the component lies above the pool, or if the pool is empty and $a > 0$, then $U^* > 0$ on $[a,1]$ and $a \in \mathcal{C}_\uparrow$. Lemma 22 gives $\Lambda(a) = \Lambda'(a) = 0$. If $b < 1$,

it also gives $\Lambda(b) = 0$; if $b = 1$, that equality holds by definition. Thus take $L = \Lambda$ and $e = a$. For a component below the pool, the reflected argument gives $L = \widetilde{\Lambda}$ and $e = b$. This reflected case also covers an empty pool with $a = 0$, since then $b < 1$.

The remaining possibility is an affine end piece. If $a = 0$ and $U^*(0) = 0 < U^*$ on $(0, b]$, then $b < 1$, $b \in \mathcal{C}_\uparrow$, and the constant slope on $[0, b]$ is positive. Lemma 22 gives $\Lambda(b) = \Lambda'(b) = 0$, while Lemma 24 gives $\Lambda(0) = 0$. Take $L = \Lambda$ and $e = b$. The end piece at 1 is the mirror, with $L = \widetilde{\Lambda}$ and $e = a$.

Along the common policy, π is affine with slope $1 - m_0$ by Lemma 23. Since $\lambda'_\uparrow = \lambda'_\downarrow = -m_0$ on this interval, the function $g = -L'$ is affine with slope $1 - 2m_0$. Equation (70) gives

$$g(e) = 0, \qquad \int_a^b g(\tau)\, d\tau = L(a) - L(b) = 0.$$

An affine function vanishing at one endpoint of a nondegenerate interval has a nonzero integral unless its slope is zero. Hence $m_0 = 1/2$, as claimed. ■

Curvature in fixed policy coordinates. A policy with statistics (S_1, S_2, T) delivers the same surplus difference $\delta = S_1 - S_2$ to every type; its utility is $y + \tau\delta$, where $y = S_2 + T$. Use these two type-independent coordinates:

$$M = L_0, \qquad E = M\mathcal{S} = \mathcal{D}_0, \qquad \widetilde{T}(\delta, y) = T_{\min}(0, \delta, y), \qquad G = \nabla\widetilde{T} \quad \text{on } \operatorname{int} E.$$

The invertible shear $A_\tau(\delta, y) = (\delta, y + \tau\delta)$ satisfies $L_\tau = A_\tau M$. Thus

$$\mathcal{D}_\tau = A_\tau E, \qquad T_{\min}(\tau, \delta, u) = \widetilde{T}(\delta, u - \tau\delta). \tag{71}$$

Along the optimal schedule put

$$\zeta(\tau) = (\Delta(\tau), y(\tau)), \qquad y(\tau) = U^*(\tau) - \tau\Delta(\tau).$$

Differentiating the identity for $T_{\min}$ in its delivery coordinates gives

$$m(\tau) = 1 - G_y(\zeta(\tau)), \qquad \pi(\tau) = -G(\zeta(\tau)) \cdot (1, -\tau). \tag{72}$$

These derivatives exist by Lemma 12. In particular, m depends on the type only through the policy coordinates ζ.

Lemma 27 (Curvature bounds for the minimum loading)**.** *There is a constant $c > 0$ such that*

$$\big(G(\zeta') - G(\zeta)\big) \cdot (\zeta' - \zeta) \geq c|\zeta' - \zeta|^2 \qquad (\zeta, \zeta' \in \operatorname{int} E). \tag{73}$$

For every $\varepsilon > 0$, there is a finite constant K_ε such that

$$|G(\zeta') - G(\zeta)| \leq K_\varepsilon |\zeta' - \zeta| \qquad (\zeta, \zeta' \in E_\varepsilon), \tag{74}$$

where $E_\varepsilon = \{\zeta \in E : \operatorname{dist}(\zeta, \partial E) \geq \varepsilon\}$. The optimal path $\zeta([0,1])$ lies in one such E_ε with $\varepsilon > 0$.

The first bound says that changing the delivered policy coordinates changes their marginal loading costs by a definite amount. The second bounds that change from above away from the frontier. Both follow from the tangent balls established in B.1.

Proof. Let $s(\zeta)$ be the statistic minimizing loading at $\zeta = (\delta, y)$. Its coordinates are

$$s(\zeta) = \big(\delta + y - \widetilde{T}(\zeta),\ y - \widetilde{T}(\zeta),\ \widetilde{T}(\zeta)\big). \tag{75}$$

Let n be its outward unit normal. Lemma 12 gives

$$w(\zeta) \coloneqq M^\top G(\zeta) - e_\theta = \frac{n}{n \cdot v}, \qquad w(\zeta) \cdot v = 1. \tag{76}$$

In particular, $|w(\zeta)|$ is bounded below by a positive constant, uniformly over interior pairs.

For two such statistics write $s = s(\zeta)$, $s' = s(\zeta')$, with normals n, n' and vectors w, w'. The outer tangent balls of Lemma 11 imply

$$n \cdot (s' - s) \leq -\frac{|s' - s|^2}{2R}, \qquad n' \cdot (s' - s) \geq \frac{|s' - s|^2}{2R}.$$

Multiplying by $|w|$ and $|w'|$, respectively, and subtracting gives

$$(w' - w) \cdot (s' - s) \geq c|s' - s|^2$$

for some uniform $c > 0$. Since $M(s' - s) = \zeta' - \zeta$ and $w' - w = M^\top(G(\zeta') - G(\zeta))$, the left side is exactly the left side of (73). The bound $|\zeta' - \zeta| \leq \|M\|\,|s' - s|$ proves that inequality after changing the constant.

For the second bound, the inner tangent balls give

$$(n' - n) \cdot (s' - s) \geq \rho|n' - n|^2, \qquad |n' - n| \leq \rho^{-1}|s' - s|.$$

Indeed, the inner ball at s' lies below the supporting plane with normal n at s, and the inner ball at s lies below the supporting plane with normal n' at s'. Adding these two supporting inequalities yields the first display; Cauchy–Schwarz gives the second.

On E_ε, G is uniformly bounded by Lemma 12 at type zero. The set E_ε is convex: convex combinations of balls of radius ε contained in E remain in E. Integrating the bounded gradient along a segment therefore makes $\widetilde{T}$ Lipschitz on E_ε, and (75) makes $s(\zeta)$ Lipschitz there as well.

The same gradient bound makes $w(\zeta)$ bounded. By (76), $n \cdot v = 1/|w(\zeta)|$ is consequently bounded away from zero on this set. Dividing the normals by $n \cdot v$ thus preserves a Lipschitz bound. Finally, $M^\top$ is injective, so (76) converts that bound for w into (74) for G.

For the path assertion, $A_\tau \zeta(\tau) = p(\tau)$, and the maps A_τ have uniformly bounded norms on $[0,1]$. They take ∂E onto $\partial \mathcal{D}_\tau$. Uniform interiority of $p(\tau)$ therefore gives a common positive lower bound on the distance from $\zeta(\tau)$ to ∂E. ■

Comparing two types on the curvature support. The envelope relation restricts how the policy coordinates can move. An increase in the gap reduces the intercept y by that increase multiplied by the type at which it occurs. The next lemma uses this restriction and the balance equations from B.2 without assuming that Δ has an ordinary derivative.

Lemma 28 (A two-point bound)**.** *There exist constants $c, \eta > 0$ such that, whenever $\tau_1 < \tau_2$ belong to the same set $\mathcal{C}_\uparrow \cap (0,1)$ or $\mathcal{C}_\downarrow \cap (0,1)$ and $\tau_2 - \tau_1 \le \eta$,*

$$c\big[\Delta(\tau_2) - \Delta(\tau_1)\big] \le (\tau_2 - \tau_1)\big[1 - 2m(\tau_i)\big], \qquad i = 1, 2. \tag{77}$$

The constants are uniform over the two types.

Proof. Write $D = \Delta(\tau_2) - \Delta(\tau_1) \ge 0$, $h = \tau_2 - \tau_1$, and $\zeta_i = \zeta(\tau_i)$. Since Δ is continuous and nondecreasing, its Stieltjes measure is ν. Integration by parts, using $U^{*\prime} = \Delta$, gives

$$y(\tau_2) - y(\tau_1) = -\int_{(\tau_1, \tau_2]} r \, d\nu(r).$$

If $D > 0$, the integral is $\bar{\tau} D$ for some $\bar{\tau} \in [\tau_1, \tau_2]$; if $D = 0$, it vanishes and any $\bar{\tau} \in [\tau_1, \tau_2]$ may be chosen. Consequently,

$$\zeta_2 - \zeta_1 = D(1, -\bar{\tau}), \qquad |\zeta(r) - \zeta_i| \le CD \quad (r \in [\tau_1, \tau_2],\ i = 1, 2) \tag{78}$$

for a uniform constant C. The second bound follows by applying the same identity between either endpoint and r, since the corresponding gap increase is at most D.

The two types lie in the same curvature support, so Lemma 22 gives

$$\pi(\tau_2) - \pi(\tau_1) = \int_{\tau_1}^{\tau_2} m(r) \, dr.$$

Substituting (72) and collecting terms at either endpoint yields

$$\begin{aligned} \big[G(\zeta_2) - G(\zeta_1)\big] \cdot (1, -\tau_2) &= \int_{\tau_1}^{\tau_2} \big[1 - m(\tau_1) - m(r)\big] \, dr, \\ \big[G(\zeta_2) - G(\zeta_1)\big] \cdot (1, -\tau_1) &= \int_{\tau_1}^{\tau_2} \big[1 - m(\tau_2) - m(r)\big] \, dr. \end{aligned} \tag{79}$$

For example, the first identity follows by writing

$$\pi(\tau_2) - \pi(\tau_1) = -\big[G(\zeta_2) - G(\zeta_1)\big] \cdot (1, -\tau_2) + h\big[1 - m(\tau_1)\big].$$

If $D = 0$, (78) makes ζ constant between the two types. Then m is constant there, and either identity in (79) gives $m(\tau_1) = m(\tau_2) = 1/2$. Both bounds in (77) hold with equality.

Now let $D > 0$. By (73),

$$\big[G(\zeta_2) - G(\zeta_1)\big] \cdot (1, -\bar{\tau}) = \frac{\big[G(\zeta_2) - G(\zeta_1)\big] \cdot (\zeta_2 - \zeta_1)}{D} \geq cD.$$

All path points lie in a common E_ε. The Lipschitz bound (74) and (78) therefore imply

$$|G(\zeta_2) - G(\zeta_1)| \leq CD, \qquad |m(r) - m(\tau_i)| \leq CD \quad (r \in [\tau_1, \tau_2],\ i = 1, 2).$$

Replacing $\bar{\tau}$ by τ_2 in the preceding lower bound costs at most ChD. The right side of the first identity in (79) is at most $h[1 - 2m(\tau_1)] + ChD$. Hence

$$cD - ChD \leq h[1 - 2m(\tau_1)] + ChD.$$

Choose a uniform $\eta > 0$ small enough to absorb the error terms into the left side whenever $h \leq \eta$. This proves the first bound in (77), after reducing c. The second identity gives the same argument with τ_1 and τ_2 interchanged in the estimates, proving the second bound. ■

Lemma 29 (Strict marginal bound on the curvature support)**.** *For every $\tau \in (\mathcal{C}_\uparrow \cup \mathcal{C}_\downarrow) \cap (0, 1)$, one has $m(\tau) < 1/2$.*

Proof. Consider $\tau_0 \in \mathcal{C}_\uparrow \cap (0, 1)$. Since $\mathcal{C}$ is relatively closed, $\mathcal{C}_\uparrow \cap (0, 1) \subseteq \mathcal{C}$. Every point of this set is approached by distinct points of the same set. Indeed, $\mathcal{C}_\uparrow$ is the closure of a relatively open subset of $\mathcal{C}$, which has no isolated points by Lemma 21. Applying (77) to τ_0 and a nearby support point, and using monotonicity of Δ, gives $1 - 2m(\tau_0) \geq 0$.

Suppose equality holds. Then (77) forces $\Delta(t) = \Delta(\tau_0)$ for every point t of the same support sufficiently close to τ_0. Choose two distinct such points on the same side, ordered $\tau_0 < t_1 < t_2$, or in the reverse order $t_2 < t_1 < \tau_0$. Monotonicity makes Δ constant on the entire interval between τ_0 and t_2. Its curvature measure is zero in that interval, contradicting $t_1 \in \mathcal{C}$. Thus $m(\tau_0) < 1/2$.

The proof for $\mathcal{C}_\downarrow$ is identical. The argument also covers an interior pool boundary: it uses nearby points on whichever side of that boundary belongs to the relevant support. ■

Proof of Lemma 25. Lemma 26 excludes an entirely affine schedule. If U^* is affine on a nondegenerate interval I not contained in P, its interior lies in an affine component outside the pool. Lemma 26 supplies an interior endpoint in $\mathcal{C}_\uparrow \cup \mathcal{C}_\downarrow$ with $m = 1/2$. This contradicts Lemma 29.

Therefore every affine interval lies in P, and

$$\mathcal{C} = (0,1) \setminus \operatorname{int}_{(0,1)} P. \tag{80}$$

A convex function fails to be strictly convex on an interval only if it is affine on a nondegenerate subinterval. Applying the result just proved gives strict convexity on each positive-rent flank, including its pool boundary. If P is empty or a singleton, it gives strict convexity on all of $[0,1]$.

If two distinct types receive the same canonical policy, their gaps are equal. Monotonicity makes Δ constant between them, so U^* is affine on that interval and both types belong to P. Conversely, a nondegenerate P has $U^* = \Delta = 0$ throughout, including its endpoints by continuity and the limiting-slope convention. All its types therefore have the same fiber and receive the same canonical policy by Lemma 23.

Finally, every point of $\mathcal{C}$ belongs to $\mathcal{C}_\uparrow \cup \mathcal{C}_\downarrow$: outside P the appropriate tail has positive utility, and interior pool boundaries belong to the corresponding support by Lemma 21. Lemma 29 gives $m < 1/2$ at these points. On the interior of a positive-length pool, m is constant because the policy is shared. Since $P \neq [0,1]$, this pool has at least one boundary in $(0,1)$, where the same policy has $m < 1/2$. This proves the strict inequality at every interior type. Continuity of m gives the weak inequalities at 0 and 1. ■

B.4 First-order conditions in the scoring weights

I translate the marginal values m and π into the weights of the optimal policies. The normal to the feasible set identifies the policy implementing each delivered pair; normalizing its two surplus weights to sum to one then gives the scoring weight σ and adjustment γ used in the main text.

From a delivered pair to its policy.

Lemma 30 (Policy weights and marginal values)**.** *Fix $\tau \in [0,1]$ and $p = (\delta, u) \in \operatorname{int} \mathcal{D}_\tau$. There is a unique pair $(a,b) \in \mathbb{R}^2$ such that the threshold policy with weights*

$$w = L_\tau^\top (a,b) - e_\theta = \big(a + b\tau,\ -a + b(1-\tau),\ b - 1\big) \tag{81}$$

delivers p to type τ. It is the principal's optimal policy for this pair, and

$$(a,b) = \nabla_p T_{\min}(\tau, p) = \big(-\Psi_\delta(\tau,p),\ 1 - \Psi_u(\tau,p)\big).$$

In particular, along the optimal schedule the canonical policy is $q_\tau = \mathbb{1}\{w(\tau)\cdot z > 0\}$, where

$$\begin{aligned} w(\tau) &= (\tau, 1-\tau, 0) - m(\tau)(\tau, 1-\tau, 1) - \pi(\tau)(1,-1,0) \\ &= \big(\tau(1-m) - \pi,\ (1-\tau)(1-m) + \pi,\ -m\big). \end{aligned} \tag{82}$$

The vector is nonzero, satisfies $w \cdot v = 1$, and is the unique exposing vector with this normalization. Its coordinates obey

$$w_\theta = -m, \qquad w_1 + w_2 = 1 - m, \qquad (1-\tau)w_1 - \tau w_2 = -\pi. \tag{83}$$

Proof. Let s^* be the minimum-loading endpoint of the fiber over p, and let n be its outward unit normal. Lemma 12 gives $n \cdot v > 0$, so $w = n/(n\cdot v)$ is an exposing vector with $w \cdot v = 1$. The two rows of L_τ span $v^\perp$, and $(-e_\theta)\cdot v = 1$. Thus w has the form (81) for a unique (a,b). By Lemma 6, its threshold policy implements s^* and delivers p; equivalently,

$$\mathbb{E}\big[(x_1 - x_2)\mathbb{1}\{w\cdot z > 0\}\big] = \delta, \qquad \mathbb{E}\big[(x^{(\tau)} - \kappa + \theta)\mathbb{1}\{w\cdot z > 0\}\big] = u.$$

If another vector $w' = L_\tau^\top(a',b') - e_\theta$ delivers the same pair, its exposed statistic lies on the same fiber. Since $w'\cdot v = 1$, its linear objective strictly increases along $+v$, so that statistic must be the minimum-loading endpoint s^*. Lemma 11 gives a unique normal ray at s^*, and the normalization then forces $w' = w$.

For any $s \in \mathcal{S}$ put $p' = L_\tau s$. The supporting inequality $w\cdot s \le w \cdot s^*$ becomes

$$T(s) \ge T_{\min}(\tau, p) + (a,b)\cdot(p' - p).$$

Minimizing over the fiber of p' shows that (a,b) is a subgradient of $T_{\min}(\tau,\cdot)$ at p. Differentiability at interior pairs identifies it with $\nabla_p T_{\min}$. Since $\Psi(\tau,\delta,u) = u - T_{\min}(\tau,\delta,u)$, substitution gives

(82). The identities (83) follow by taking the third coordinate, the sum of the first two coordinates, and their stated weighted difference. ■

Continuous scoring rules.

Lemma 31 (Scoring weights of the optimal menu)**.** *The canonical optimal menu has a unique continuous scoring representation*

$$q_\tau = \mathbb{1}\{\sigma(\tau)x_1 + [1-\sigma(\tau)]x_2 - \kappa + \gamma(\tau)\theta > 0\},$$

where $\sigma : [0,1] \to \mathbb{R}$ *and* $\gamma : [0,1] \to (-1,1)$ *are given by*

$$\sigma(\tau) = \tau - \frac{\pi(\tau)}{1-m(\tau)}, \qquad \gamma(\tau) = -\frac{m(\tau)}{1-m(\tau)}. \tag{84}$$

Equivalently,

$$H(\tau) \coloneqq \frac{\tau - \sigma(\tau)}{1-\gamma(\tau)} = \pi(\tau), \qquad -\frac{\gamma(\tau)}{1-\gamma(\tau)} = m(\tau). \tag{85}$$

Proof. Lemma 25 gives $m < 1/2$ in $(0,1)$ and $m \le 1/2$ at the endpoints. I first strengthen the endpoint inequalities to strict ones. If an endpoint belongs to a positive-length pool, its policy, and therefore its marginal value m, is shared with interior types, where the bound is strict.

Otherwise, consider the endpoint 1. If $U^*(1) > 0$, (80) puts a terminal interval of interior types in $\mathcal{C}_\uparrow$. If $U^*(1) = 0$, the pool is the singleton $\{1\}$, and every interior type belongs to $\mathcal{C}_\downarrow$. In either case fix an interior a in that terminal interval with $1 - a < \eta$, where η is from Lemma 28. For $a < t < 1$, that lemma gives

$$c[\Delta(t) - \Delta(a)] \le (t-a)[1-2m(t)].$$

If $m(1) = 1/2$, continuity and $t \uparrow 1$ imply $\Delta(1) = \Delta(a)$. Monotonicity then makes U^* affine on $[a,1]$, contrary to Lemma 25, since this interval is not contained in the pool. At 0, fix an interior b in the corresponding initial interval with $0 < b < \eta$ and let $t \downarrow 0$ in the other bound,

$$c[\Delta(b) - \Delta(t)] \le (b-t)[1-2m(t)].$$

The lower support applies when $U^*(0) > 0$, and the upper support when $P = \{0\}$. The same contradiction proves $m(0) < 1/2$.

Thus $w_1 + w_2 = 1 - m > 0$ everywhere. Dividing (82) by this sum gives (84). Moreover,

$$1 - \gamma = \frac{1}{1-m} > 0, \qquad 1 + \gamma = \frac{1-2m}{1-m} > 0,$$

so $-1 < \gamma < 1$. The inverse identities follow by substitution. Continuity of m and π from Lemma 20 gives continuity of the two weights on $[0,1]$.

For uniqueness, any other normalized scoring vector implementing the same policy exposes the same statistic. The unique normal ray at that statistic makes it a positive multiple of w. Requiring its first two coordinates to sum to one fixes that multiple, and therefore fixes σ and γ. ■

The pool and the balance equations.

Lemma 32 (The pool policy)**.** *If P has positive length, all of its types receive the unique policy*

$$\bar{q} \in \arg\max_{q:\Omega\to[0,1]} \big\{S_1(q) : S_1(q) = S_2(q),\ S_2(q) + T(q) = 0\big\}. \tag{86}$$

Its weights (σ_0, γ_0) depend only on the prior and the primitives. On P,

$$m(\tau) = m_0 = -\frac{\gamma_0}{1-\gamma_0} < \frac{1}{2}, \qquad H(\tau) = (1-m_0)(\tau - \sigma_0). \tag{87}$$

The weights on each adjacent ramp converge to (σ_0, γ_0) at the pool boundary.

Proof. On a positive-length pool, $U^* = \Delta = 0$, including its endpoints by continuity. Hence the fiber is the same at every pooled type: $S_1 = S_2$ and $S_2 + T = 0$. The principal's payoff there is $S_1 = -T$, so the minimum-loading endpoint solves (86). Lemma 6 makes its implementing policy unique up to prior-null sets. Its normalized outward normal is fixed, so its weights are constant. Equations (85) give (87); the bound follows from Lemma 25. Continuity of the weights proves the boundary statement. ■

Lemma 33 (Necessary conditions in the scoring weights)**.** *The schedules (σ, γ) of Lemma 31 induce the original schedules (Δ, U^*) and satisfy the following conditions:*

(i) Δ is nondecreasing, $U^* \geq 0$, and

$$U^*(t) - U^*(s) = \int_s^t \Delta(r)\, dr \qquad (0 \leq s < t \leq 1).$$

(ii) H is absolutely continuous on $[0,1]$, and

$$H'(\tau) \geq -\frac{\gamma(\tau)}{1-\gamma(\tau)} \quad \textit{almost everywhere}, \tag{88}$$

with equality almost everywhere on $\{U^* > 0\}$.

(iii) At the endpoints,

$$H(0) \geq 0, \qquad H(1) \leq 0, \qquad U^*(0)H(0) = U^*(1)H(1) = 0. \tag{89}$$

If $P = [\tau_-, \tau_+] \neq \emptyset$, the balance equations are

$$H(\tau) = -\int_0^\tau \frac{\gamma(s)}{1-\gamma(s)}\,ds \qquad \big(0 \le \tau \le \tau_-,\ \tau_- > 0\big), \tag{90}$$

$$H(\tau) = \int_\tau^1 \frac{\gamma(s)}{1-\gamma(s)}\,ds \qquad \big(\tau_+ \le \tau \le 1,\ \tau_+ < 1\big). \tag{91}$$

If $P = \emptyset$, both equations hold throughout $[0,1]$.

Proof. The weight normalization leaves the canonical policies unchanged. Their delivered pairs are therefore (Δ, U^*), and Lemma 7 gives part (i).

On an upper positive-rent flank, (80) puts every interior type in $\mathcal{C}_\uparrow$. Lemma 22, including its endpoint limits, gives

$$\pi(\tau) = -\int_\tau^1 m(s)\,ds.$$

Substituting $H = \pi$ and $m = -\gamma/(1-\gamma)$ proves (91), including the pool boundary by continuity. On a lower flank the same lemma gives $\pi(\tau) = \int_0^\tau m(s)\,ds$, which is (90). If there is no pool, both curvature supports are $[0,1]$, so both equations hold everywhere. In all cases $H' = m$ on the positive-rent regions.

On a positive-length pool, Lemma 32 instead gives

$$H' - m = 1 - 2m_0 > 0.$$

There are at most two positive-rent flanks and one pool interval. On each closed flank H is an integral of the continuous function m; on the pool it is affine. These finitely many pieces agree at their boundaries because H is continuous. Joining their integral representations shows that H is absolutely continuous on $[0,1]$. A singleton zero set introduces no interval of its own and has no effect on an almost-everywhere assertion. This proves part (ii).

For part (iii), (63) and $H = \pi$ give the two signs. If $U^*(0) > 0$, the lower balance equation gives $H(0) = 0$; if $U^*(1) > 0$, the upper equation gives $H(1) = 0$. At an endpoint with zero utility the complementary product vanishes directly. This proves (89). Since $1 - \gamma > 0$, the same conditions say

$$\sigma(0) \le 0, \qquad \sigma(1) \ge 1, \qquad U^*(0)\sigma(0) = U^*(1)[\sigma(1) - 1] = 0.$$

■

The marginal gain and the cost of rent. The translation also gives the payoff interpretation of the balance equations. At the optimal pair of type τ,

$$\Psi_\delta(\tau, p(\tau)) = H(\tau), \qquad -\Psi_u(\tau, p(\tau)) = \frac{\gamma(\tau)}{1-\gamma(\tau)}, \qquad \partial_u T_{\min}(\tau, p(\tau)) = \frac{1}{1-\gamma(\tau)}. \tag{92}$$

Thus H is the principal's signed marginal gain from increasing the surplus difference at fixed utility. If $H \neq 0$, a small change in the favorable direction raises her payoff at rate $|H|$ per unit of surplus difference. The marginal cost of increasing utility at a fixed surplus difference is $\gamma/(1-\gamma)$. Along this latter change, an additional marginal unit of loading changes the principal's payoff by $-\gamma$ and the agent's payoff by $1-\gamma$.

These formulas follow directly from (85) and $\Psi = u - T_{\min}$. They also separate the gain from changing the gap into two parts. Since $\partial_u T_{\min} > 0$, the implicit function theorem makes the pairs with the current loading a C^1 curve near $p(\tau)$. Its slope is

$$\left.\frac{du}{d\delta}\right|_{T_{\min}\text{ fixed}} = -\frac{\partial_\delta T_{\min}}{\partial_u T_{\min}} = \frac{\pi}{1-m} = \tau - \sigma(\tau).$$

Along this curve the principal's payoff and the agent's utility move one for one. Returning utility to its original level contributes the second term in

$$H(\tau) = (\tau - \sigma(\tau)) + (\tau - \sigma(\tau))\frac{\gamma(\tau)}{1-\gamma(\tau)}.$$

Finally, the kink identity in (61) reads

$$-\Lambda'(\tau) = H(\tau) - \int_\tau^1 \frac{\gamma(s)}{1-\gamma(s)}\, ds.$$

Its first term is the local gain from increasing the gap; the integral is the cost of the accompanying utility increase for the types above. The reversed-kink argument of Lemma 22 sets this expression to zero on the upper flank. The mirrored identity gives the balance with the types below on the lower flank. These are precisely (91) and (90).

B.5 Sufficiency and uniqueness

I now show that the conditions in Theorem 1 are sufficient for optimality. The proof compares a candidate menu with any feasible menu. At each type, the candidate policy maximizes a linear score. Integrating this pointwise comparison and using the envelope relation leaves only nonnegative terms.

Menus are taken to be jointly measurable in type and state. For a menu $q = (q_\tau)_{\tau\in[0,1]}$, write

$$\Pi(q) \coloneqq \int_0^1 V_\tau(q_\tau)\, d\tau, \qquad \delta(p) \coloneqq S_1(p) - S_2(p)$$

for the principal's expected payoff and the surplus difference of an individual policy p. Given continuous weights (σ, γ) with $\gamma < 1$, define

$$g_\tau(\omega) \coloneqq \sigma(\tau)x_1 + [1-\sigma(\tau)]x_2 - \kappa + \gamma(\tau)\theta, \qquad q_\tau = \mathbb{1}\{g_\tau > 0\}, \tag{93}$$

and set

$$U(\tau) = U_\tau(q_\tau), \qquad \Delta(\tau) = \delta(q_\tau), \qquad H(\tau) = \frac{\tau - \sigma(\tau)}{1-\gamma(\tau)}, \qquad m_\gamma(\tau) = -\frac{\gamma(\tau)}{1-\gamma(\tau)}.$$

Lemma 34 (The pointwise payoff comparison)**.** *For every type τ and measurable policy $p : \Omega \to [0,1]$,*

$$V_\tau(p) - m_\gamma(\tau)U_\tau(p) - H(\tau)\delta(p) = \frac{\mathbb{E}[g_\tau p]}{1-\gamma(\tau)}. \tag{94}$$

The policy q_τ maximizes this expression over all policies. For a competing menu $\widehat{q}$, define $\widehat{U}(\tau) = U_\tau(\widehat{q}_\tau)$, $\widehat{\Delta}(\tau) = \delta(\widehat{q}_\tau)$, and

$$\mathcal{L}(\tau) \coloneqq \frac{\mathbb{E}\big[|g_\tau|\,|q_\tau - \widehat{q}_\tau|\big]}{1-\gamma(\tau)} \geq 0. \tag{95}$$

Then

$$V_\tau(q_\tau) - V_\tau(\widehat{q}_\tau) = m_\gamma(\tau)[U(\tau) - \widehat{U}(\tau)] + H(\tau)[\Delta(\tau) - \widehat{\Delta}(\tau)] + \mathcal{L}(\tau). \tag{96}$$

Proof. Using $U_\tau(p) = V_\tau(p) + T(p)$ and $1 - m_\gamma = 1/(1-\gamma)$ gives

$$\begin{aligned} V_\tau(p) - m_\gamma U_\tau(p) - H\delta(p) &= \frac{V_\tau(p) + \gamma T(p) - (\tau - \sigma)\delta(p)}{1-\gamma} \\ &= \frac{\sigma S_1(p) + (1-\sigma)S_2(p) + \gamma T(p)}{1-\gamma}, \end{aligned}$$

which proves (94). The denominator is positive. At each state, q_τ maximizes $g_\tau p$ over $p \in [0,1]$, and

$$g_\tau(q_\tau - \widehat{q}_\tau) = |g_\tau|\,|q_\tau - \widehat{q}_\tau|.$$

Subtracting (94) for the two policies gives (96).

Continuity of the weights on the compact type interval bounds σ, γ, and $1/(1-\gamma)$. Since states are bounded, all terms in these identities are measurable and integrable for a measurable competing menu. ■

Lemma 35 (Global sufficiency)**.** *Let (σ, γ) be continuous, with $\gamma < 1$, and suppose its induced schedules satisfy conditions* (i)–(iii) *of Theorem 1. Then its menu q is feasible and globally optimal. More precisely, put*

$$r(\tau) \coloneqq H'(\tau) - m_\gamma(\tau).$$

For every feasible measurable competing menu $\widehat{q}$,

$$\begin{aligned} \Pi(q) - \Pi(\widehat{q}) = \int_0^1 \left[\mathcal{L}(\tau) + r(\tau)\widehat{U}(\tau)\right] d\tau \\ + H(0)\widehat{U}(0) - H(1)\widehat{U}(1) \geq 0. \end{aligned} \tag{97}$$

Every other optimal menu agrees with q for almost every type, up to π_0-null sets of states.

Proof. Condition (i) and Lemma 7 make the candidate menu incentive compatible and individually rational. The same lemma makes U and the utility $\widehat{U}$ of any feasible competitor Lipschitz, with

$$U' = \Delta, \qquad \widehat{U}' = \widehat{\Delta} \quad \text{almost everywhere.}$$

Condition (ii) gives $r \geq 0$ and $Ur = 0$ almost everywhere. Moreover, r is integrable because H is absolutely continuous and m_γ is bounded. Condition (iii) gives

$$H(0) \geq 0, \qquad H(1) \leq 0, \qquad U(0)H(0) = U(1)H(1) = 0.$$

Set $h = \widehat{U} - U$. Integrating (96) and applying integration by parts yields

$$\begin{aligned} \Pi(q) - \Pi(\widehat{q}) &= \int_0^1 \mathcal{L}\, d\tau - \int_0^1 \left(m_\gamma h + Hh'\right) d\tau \\ &= \int_0^1 \mathcal{L}\, d\tau + \int_0^1 rh\, d\tau + H(0)h(0) - H(1)h(1). \end{aligned}$$

Complementary slackness cancels every term involving the candidate's utility U, leaving (97). Each remaining term is nonnegative: $\mathcal{L} \geq 0$, $r \geq 0$, and $\widehat{U} \geq 0$, with the stated endpoint signs. This proves optimality against every feasible measurable menu, including menus of randomized policies.

If the competitor is also optimal, (97) forces $\mathcal{L}(\tau) = 0$ for almost every type. For each such type, g_τ is a nonzero affine function of the state, since its two match-attribute coefficients sum to one. Its zero set is therefore a plane and has prior probability zero. Equation (95) implies $\widehat{q}_\tau = q_\tau$ almost surely, proving the final assertion. ■

The payoff difference in (97) has two sources. The score loss $\mathcal{L}$ measures the competitor's failure to choose the candidate's maximizing policy at each type. The remaining terms charge

the competitor for utility supplied at types where the candidate's participation constraint binds, including either endpoint.

Proof of Proposition 3 and Theorem 1. Lemma 10 supplies an optimal utility schedule. Implementing each delivered pair by the unique inner optimizer in Lemma 8 gives its canonical optimal menu. Lemma 31 gives its continuous weights (σ^*, γ^*), with $\gamma^* \in (-1, 1)$ everywhere. This proves Proposition 3. Lemma 33 shows that these weights satisfy conditions (i)–(iii) of Theorem 1, establishing existence of a solution to that system.

Lemma 35 proves that any continuous pair satisfying the conditions induces an optimal menu. Applying its equality conclusion to the solution just constructed also shows that every optimal measurable menu agrees with the canonical menu for almost every type, up to prior-null sets of states.

Suppose another continuous pair satisfies the conditions. Its menu is optimal, so its policy coincides with the canonical policy for almost every type. At each such type, both scoring vectors expose the same statistic. The unique-normal argument in Lemma 31, with both vectors' first two coordinates summing to one, identifies their weights. Hence the two continuous weight pairs agree almost everywhere, and continuity makes them agree everywhere on $[0, 1]$. This proves uniqueness.

Finally, take any continuous pair in the range stated in Theorem 1 whose menu is feasible and optimal. The same equality and normalization arguments identify its weights everywhere with (σ^*, γ^*). It therefore satisfies conditions (i)–(iii). This proves necessity for every continuous optimal scoring pair and completes the equivalence. ■

Proof of Corollary 1.1. The optimal utility schedule is nonnegative, convex, and continuous, so its zero set P is a closed interval, possibly empty or a singleton. Lemma 25 shows that two distinct types share a policy exactly when both belong to P. If P has positive length, Lemma 32 identifies its common policy with the balanced, zero-rent maximization in part (i).

The lower and upper balance equations are (90) and (91), including the boundaries of every nonempty ramp. When P is empty both hold on $[0, 1]$. The endpoint complementary products in (89), together with $1 - \gamma > 0$, give $\sigma(0) = 0$ when $U(0) > 0$ and $\sigma(1) = 1$ when $U(1) > 0$. This proves part (ii). ■

Appendix C Additional proofs

C.1 The known-type benchmark

Proof of Proposition 1. Fix a type τ and write $M_\tau = x^{(\tau)} - \kappa$. For any $\gamma \in \mathbb{R}$, the policy $q^{\tau,\gamma} = \mathbb{1}\{M_\tau + \gamma\theta > 0\}$ uniquely maximizes

$$V_\tau(q) + \gamma T(q) = (1-\gamma)V_\tau(q) + \gamma U_\tau(q)$$

over all policies, up to prior-null sets. This follows by maximizing the score state by state; its zero plane has probability zero. In particular, if $U_\tau(q^{\tau,0}) \geq 0$, the principal's unconstrained maximizer is feasible and therefore solves the known-type problem.

Now suppose $U_\tau(q^{\tau,0}) < 0$. The function

$$u(\gamma) \coloneqq U_\tau(q^{\tau,\gamma})$$

is continuous on $[0,1]$ by dominated convergence. It is also strictly increasing. To see this, take $0 \leq \gamma_1 < \gamma_2 \leq 1$. As the adjustment increases, a newly accepted state has $\theta > 0$ and

$$-\gamma_2\theta < M_\tau \leq -\gamma_1\theta,$$

so its agent payoff $M_\tau + \theta$ is positive. A newly rejected state has $\theta < 0$ and

$$-\gamma_1\theta < M_\tau \leq -\gamma_2\theta,$$

so its agent payoff is negative, except on a prior-null boundary. Every switch thus raises utility. There is a positive-probability set of switches: choose a small nonzero θ on an available side of zero and choose x_1, x_2 near κ with M_τ strictly between the two cutoffs. These strict inequalities hold on an open set near the pivot, where (A2) gives positive density. Hence $u(\gamma_2) > u(\gamma_1)$.

Finally,

$$u(1) = \mathbb{E}[(M_\tau + \theta)^+] > 0,$$

because (A2) gives positive probability to states near the pivot with positive agent payoff. There is therefore a unique $\gamma^* \in (0,1)$ with $u(\gamma^*) = 0$.

For any policy q satisfying participation, score maximization gives

$$(1-\gamma^*)V_\tau(q) + \gamma^* U_\tau(q) \leq (1-\gamma^*)V_\tau(q^{\tau,\gamma^*}).$$

Since $U_\tau(q) \geq 0$ and $1 - \gamma^* > 0$, the principal's payoff is at most that of q^{τ,γ^*}. Equality requires equality in the score comparison, which identifies the same policy almost surely. Thus $\gamma^{\mathrm{kt}}(\tau) = \gamma^*$ is the unique binding adjustment and implements the known-type optimum. ■

C.2 The first-best menu

Write

$$q_\tau^{\mathrm{fb}} = q^{\tau,0}, \qquad B(\tau) = V_\tau(q_\tau^{\mathrm{fb}}) = \mathbb{E}[(x^{(\tau)} - \kappa)^+], \qquad \Delta^{\mathrm{fb}}(\tau) = S_1(q_\tau^{\mathrm{fb}}) - S_2(q_\tau^{\mathrm{fb}}).$$

Lemma 36 (First-best regularity)**.** *The function B is continuously differentiable, with $B' = \Delta^{\mathrm{fb}}$, and T^{fb} is Lipschitz. There is a constant $C < \infty$ such that*

$$0 \le V_\tau(q_\tau^{\mathrm{fb}}) - V_\tau(q_r^{\mathrm{fb}}) \le C|r - \tau|^2 \qquad (\tau, r \in [0,1]). \tag{98}$$

Proof. The plane $\{x^{(\tau)} = \kappa\}$ is prior-null. Dominated differentiation therefore gives

$$B'(\tau) = \mathbb{E}[(x_1 - x_2)\mathbb{1}\{x^{(\tau)} > \kappa\}] = \Delta^{\mathrm{fb}}(\tau),$$

with one-sided derivatives at the endpoints. Dominated convergence also makes this derivative continuous.

Let $h = |r - \tau|$. If the two first-best policies disagree, then

$$|x^{(\tau)} - \kappa| \le h|x_1 - x_2| \le h.$$

The probability of this slab is at most Ch, uniformly in τ. Indeed, one of the coefficients $\tau, 1-\tau$ is at least $1/2$, so fixing the other attribute and θ leaves an interval of length at most a constant times h in that coordinate. Integrating over the bounded state space and using the upper bound on the density proves the claim. Boundedness of θ consequently gives

$$|T^{\mathrm{fb}}(r) - T^{\mathrm{fb}}(\tau)| \le Ch.$$

On a disagreement state, the principal's payoff loss is $|x^{(\tau)} - \kappa| \le h$; elsewhere it is zero. Multiplying by the probability bound proves (98), after changing the constant. ■

Proof of Proposition 2. The utility assigned by the first-best menu is

$$U^{\mathrm{fb}}(\tau) = B(\tau) + T^{\mathrm{fb}}(\tau).$$

If this menu is incentive compatible, Lemma 7 makes U^{fb} absolutely continuous with derivative Δ^{fb} almost everywhere. Lemma 36 gives the same derivative for B. Their difference T^{fb} is therefore absolutely continuous with derivative zero almost everywhere, so it is constant on $[0,1]$.

Conversely, suppose $T^{\mathrm{fb}} \equiv \bar{T}$. For any report r,

$$U_\tau(q_r^{\mathrm{fb}}) = V_\tau(q_r^{\mathrm{fb}}) + \bar{T} \le V_\tau(q_\tau^{\mathrm{fb}}) + \bar{T} = U_\tau(q_\tau^{\mathrm{fb}}).$$

Thus the menu is incentive compatible. The hypothesis that every type is unconstrained gives participation. Since each assigned policy maximizes the principal's payoff for its type over all policies, the menu is globally optimal.

The same argument also shows that nonconstant T^{fb} prevents any feasible menu from attaining the integrated first-best payoff. Equality with that payoff bound would force the assigned policy to equal q_τ^{fb} almost surely for almost every type, by uniqueness of pointwise score maximization. The competing utility would then equal U^{fb} almost everywhere and hence everywhere, since both are continuous. Its IC envelope would give $(U^{\mathrm{fb}})' = \Delta^{\mathrm{fb}}$ almost everywhere, forcing T^{fb} to be constant as above. ■

C.3 The own-weight policy at the same rent

Lemma 37 (The same-rent comparison)**.** *Let $q_\tau = q^{\sigma(\tau),\gamma(\tau)}$ be the canonical optimal menu and $U^*(\tau) = U_\tau(q_\tau)$. For every type τ, there is a finite adjustment $\widehat{\gamma}(\tau) < 1$ such that*

$$U_\tau\big(q^{\tau,\widehat{\gamma}(\tau)}\big) = U^*(\tau).$$

This policy uniquely maximizes the principal's payoff among all policies delivering $U^*(\tau)$ to type τ, up to prior-null sets. In particular,

$$V_\tau\big(q^{\tau,\widehat{\gamma}(\tau)}\big) > V_\tau(q_\tau) \qquad \textit{whenever } \sigma(\tau) \neq \tau.$$

Proof. Let $q^- = \mathbb{1}\{\theta < 0\}$, and write $T^- = T(q^-)$, $\Delta^- = S_1(q^-) - S_2(q^-)$, and $\ell(\tau) = U_\tau(q^-)$. The policy q^- uniquely minimizes loading, since for every policy q,

$$T(q) - T^- = \mathbb{E}\big[|\theta|\,|q - q^-|\big] \geq 0, \tag{99}$$

with equality only when $q = q^-$ almost surely. The function ℓ is affine with slope Δ^-.

I first show that

$$U^*(\tau) > \ell(\tau) \qquad (0 \leq \tau \leq 1). \tag{100}$$

Adding q^- to the optimal menu gives each type utility $\max\{U^*, \ell\}$ and preserves incentive compatibility and participation. Where $\ell > U^*$, the principal gains

$$V_\tau(q^-) - V_\tau(q_\tau) = \ell(\tau) - U^*(\tau) + T(q_\tau) - T^- > 0.$$

Continuity would make any such strict inequality hold on a positive-measure set of types, contradicting optimality. Hence $U^* \geq \ell$ everywhere.

Suppose equality holds at some type τ. Evaluate the supporting inequality for the assigned policy from Lemma 30 at the statistic of q^-. Because the two policies deliver equal utility at this type, it gives

$$0 \leq T(q_\tau) - T^- \leq \pi(\tau)[\Delta^- - \Delta(\tau)]. \tag{101}$$

At an interior contact, differentiability of $U^* - \ell$ gives $\Delta(\tau) = \Delta^-$. At a contact at 0, its right

derivative gives $\Delta(0) \geq \Delta^-$; at a contact at 1, its left derivative gives $\Delta(1) \leq \Delta^-$. Together with $\pi(0) \geq 0$ and $\pi(1) \leq 0$, these make the last term in (101) nonpositive in every case. Thus $T(q_\tau) = T^-$, and (99) forces $q_\tau = q^-$ almost surely. Its normalized exposing vector is $(0, 0, -1)$. Uniqueness of the normal and (83) then give $m(\tau) = 1$, contradicting Lemma 25. This proves (100).

Fix τ and put $Y_\tau = x^{(\tau)} - \kappa + \theta$. For $b > 0$, consider

$$q_b = \mathbb{1}\{bY_\tau - \theta > 0\}.$$

The function $b \mapsto U_\tau(q_b)$ is continuous, because every score-zero plane is prior-null. Dominated convergence gives

$$\lim_{b \downarrow 0} U_\tau(q_b) = \ell(\tau), \qquad \lim_{b \to \infty} U_\tau(q_b) = \mathbb{E}[Y_\tau^+].$$

Uniform interiority permits a small increase in delivered utility at the fixed gap, so $U^*(\tau) < \mathbb{E}[Y_\tau^+]$, the largest utility any policy can deliver. Together with (100), the intermediate value theorem gives a finite $b > 0$ with $U_\tau(q_b) = U^*(\tau)$. Dividing its score by b identifies

$$q_b = q^{\tau, \widehat{\gamma}(\tau)}, \qquad \widehat{\gamma}(\tau) = 1 - \frac{1}{b} < 1.$$

The policy q_b uniquely maximizes $bU_\tau(q) - T(q)$ over all policies. At fixed utility this minimizes loading and therefore maximizes $V_\tau = U_\tau - T$, proving the comparison and uniqueness claims. If $\sigma(\tau) \neq \tau$, the assigned policy cannot equal q_b almost surely: both scoring vectors would expose the same statistic and hence be proportional by the unique-normal argument in Lemma 31. Since their first two coordinates sum to one, the vectors would coincide, forcing $\sigma(\tau) = \tau$. The payoff improvement is consequently strict. ■

C.4 Blind menus

Proof of Corollary 1.2. Suppose first that the optimal adjustment is identically zero. Then $m_\gamma = 0$ and $H(\tau) = \tau - \sigma(\tau)$. Theorem 1 makes H absolutely continuous with $H' \geq 0$ almost everywhere, while $H(0) \geq 0$ and $H(1) \leq 0$. Hence

$$0 \leq H(0) \leq H(\tau) \leq H(1) \leq 0,$$

so $H \equiv 0$ and $\sigma(\tau) = \tau$. The optimal menu is therefore the first-best menu. It is incentive compatible and individually rational, so Proposition 2 implies that T^{fb} is constant. By contraposition, nonconstant T^{fb} requires $\gamma(\tau) \neq 0$ somewhere, and continuity makes this true on a positive-measure set of types.

I next show that a scoring policy with $\gamma \neq 0$ cannot agree almost surely with any blind policy

$p(x_1, x_2) \in [0, 1]$. Fix its weights (σ, γ). Choose distinct θ_0, θ_1 in the interior of Θ, arbitrarily close to zero, and let $\theta_m = (\theta_0 + \theta_1)/2$. At

$$x_1 = x_2 = \kappa - \gamma\theta_m,$$

the score at θ_j is $\gamma(\theta_j - \theta_m)$, with opposite signs for $j = 0, 1$. By taking the two values close enough to zero, these states lie in the pivotal neighborhood of (A2). The signs persist on two rectangles $A \times J_0$ and $A \times J_1$ in that neighborhood, where A is the same positive-area set of match attributes and J_0, J_1 are positive-length intervals of private benefits. The threshold policy equals zero throughout one rectangle and one throughout the other.

Because the density is bounded below there, almost-sure agreement with $p(x_1, x_2)$ would imply agreement for Lebesgue-almost every state in both rectangles. Fubini's theorem would then require $p = 0$ and $p = 1$ almost everywhere on the same set A, a contradiction. This argument also applies when Θ is one-sided: both chosen values may lie on the available side of zero.

By Lemma 35, every optimal feasible menu agrees with the canonical optimal menu almost surely in states for almost every type. A blind menu cannot do so on the positive-measure set where $\gamma \neq 0$. Thus every feasible blind menu has strictly lower expected payoff. This includes the first-best menu offered for free choice: with the outside option included, its chosen policies induce an incentive compatible, individually rational direct menu, and all of those policies are blind. ■

C.5 First-best benefits in the admissions example

Derivation of (7). Only applicants who need aid contribute to expected private benefits, so

$$T^{\text{fb}}(\tau) = -cp \Pr\{\tau x_1 + (1 - \tau)x_2 > \kappa \mid \text{aid}\}.$$

For the original type weights in the example, $\tau \geq \max\{\kappa, 1 - \kappa\}$. Thus the cutoff

$$\frac{\kappa - (1 - \tau)x_2}{\tau}$$

lies in $[0, 1]$ for every $x_2 \in [0, 1]$. Conditional on x_2 and aid status, x_1 remains uniform, giving acceptance probability $1 - [\kappa - (1 - \tau)x_2]/\tau$. Averaging over aid applicants, whose mean extracurricular score is m, yields

$$T^{\text{fb}}(\tau) = -cp\left[1 - \frac{\kappa - (1 - \tau)m}{\tau}\right] = -cp\left[(1 - m) - \frac{\kappa - m}{\tau}\right].$$

Differentiation gives $(T^{\text{fb}})'(\tau) = -cp(\kappa - m)/\tau^2 < 0$, since $m < \kappa$ in the example. ■

Appendix D Ordered private benefits

I prove Theorem 2. Let q_τ be the canonical optimal menu and retain the notation

$$p(\tau) = (\Delta(\tau), U^*(\tau)), \qquad m(\tau) = \Psi_u(\tau, p(\tau)), \qquad \pi(\tau) = \Psi_\delta(\tau, p(\tau)).$$

There is no pool, so $U^* > 0$ on $[0,1]$. Appendix B.4 gives

$$\pi \in C^1([0,1]), \qquad \pi' = m, \qquad \pi(0) = \pi(1) = 0, \tag{102}$$

where endpoint derivatives are one-sided. The scoring weights satisfy

$$\sigma(\tau) = \tau - \frac{\pi(\tau)}{1 - m(\tau)}, \qquad \gamma(\tau) = -\frac{m(\tau)}{1 - m(\tau)}, \qquad 1 - m(\tau) > 0. \tag{103}$$

By Lemma 25, $\Delta = U^{*\prime}$ is continuous and strictly increasing. Its derivative measure $\nu = D\Delta$ is finite, nonnegative, and atomless, and assigns positive mass to every nonempty open subinterval. No endpoint atoms are attached to ν.

Use the first-best objects from Appendix C.2 and put

$$\alpha(\tau) = B(\tau) + T^{\mathrm{fb}}(\tau), \qquad p^{\mathrm{fb}}(\tau) = (\Delta^{\mathrm{fb}}(\tau), \alpha(\tau)), \qquad e(\tau) = U^*(\tau) - \alpha(\tau).$$

Lemma 36 gives $B' = \Delta^{\mathrm{fb}}$ and absolute continuity of T^{fb}. In particular,

$$e' = \Delta - \Delta^{\mathrm{fb}} - (T^{\mathrm{fb}})' \qquad \text{almost everywhere.} \tag{104}$$

D.1 The direction of scoring distortion

Lemma 38 (A concavity inequality)**.** *Define the principal's loss relative to the first best by*

$$\mathcal{L}(\tau) = B(\tau) - \Psi(\tau, p(\tau)).$$

Then $\mathcal{L}$ is continuous and nonnegative, and

$$\mathcal{L}(\tau) = 0 \quad \Longleftrightarrow \quad p(\tau) = p^{\mathrm{fb}}(\tau) \quad \Longleftrightarrow \quad \pi(\tau) = m(\tau) = 0.$$

Moreover,

$$\pi(\Delta - \Delta^{\mathrm{fb}}) + me \le -\mathcal{L}, \tag{105}$$

and therefore

$$(\pi e)' + \pi(T^{\mathrm{fb}})' + \mathcal{L} \le 0 \qquad \textit{almost everywhere}. \tag{106}$$

Proof. For each type, the fibers defining Ψ partition the feasible statistics. Maximizing $\Psi(\tau, \cdot)$ over those fibers is therefore the same as maximizing the principal's payoff over all policies. The unique maximizer is the first-best policy, with delivered pair $p^{\mathrm{fb}}(\tau)$ and value $B(\tau)$. This proves nonnegativity of $\mathcal{L}$ and its first equality condition. At the interior path point $p(\tau)$, a zero gradient

of the differentiable concave function $\Psi(\tau, \cdot)$ is equivalent to global maximality. This proves the second equality condition. Continuity follows from continuity of the canonical policy statistics and B.

The supporting inequality for $\Psi(\tau, \cdot)$ at $p(\tau)$ gives

$$B(\tau) = \Psi(\tau, p^{\mathrm{fb}}(\tau)) \leq \Psi(\tau, p(\tau)) + \pi(\tau)[\Delta^{\mathrm{fb}}(\tau) - \Delta(\tau)] + m(\tau)[\alpha(\tau) - U^*(\tau)].$$

Rearranging proves (105). Substitute (104) and use $\pi' = m$. The product rule for the absolutely continuous functions π and e gives (106). ■

Lemma 39 (One-sided scoring distortion)**.** *If T^{fb} is nonincreasing, then*

$$\pi(\tau) \geq 0, \qquad \sigma(\tau) \leq \tau \qquad (0 \leq \tau \leq 1).$$

At the endpoints,

$$\sigma(0) = 0, \qquad \sigma(1) = 1, \qquad \gamma(0) \leq 0 \leq \gamma(1). \tag{107}$$

If T^{fb} is strictly decreasing, π cannot vanish on a nonempty open interval.

Proof. Suppose $\pi < 0$ somewhere, and let (a, b) be a connected component of its negative set, allowing $a = 0$ or $b = 1$. Continuity and (102) give $\pi(a) = \pi(b) = 0$. Integrating (106) over this interval yields

$$\int_a^b \pi(\tau)(T^{\mathrm{fb}})'(\tau)\, d\tau + \int_a^b \mathcal{L}(\tau)\, d\tau \leq 0.$$

The first integral is nonnegative because $(T^{\mathrm{fb}})' \leq 0$ almost everywhere. The second is strictly positive: $\pi < 0$ on (a, b) excludes the first-best pair by Lemma 38. This is a contradiction. Thus $\pi \geq 0$, and (103) gives $\sigma \leq \tau$ and the endpoint equalities.

Since $\pi \geq 0$ and its endpoint values are zero, its one-sided derivatives satisfy $m(0) = \pi'(0) \geq 0$ and $m(1) = \pi'(1) \leq 0$. The adjustment formula in (103) gives the remaining endpoint signs.

Finally, if π vanished on an open interval, then $m = \pi' = 0$ there. Lemma 38 would give $U^* = \alpha$ and $\Delta = \Delta^{\mathrm{fb}}$ throughout that interval. Equation (104) then implies $(T^{\mathrm{fb}})' = 0$ almost everywhere there. Absolute continuity makes T^{fb} constant on that interval, contradicting strict decrease. ■

D.2 Variation of private benefits along the menu

Lemma 40 (The private-benefit identity)**.** *Write $T^{\mathrm{opt}}(\tau) = T(q_\tau)$. This function is continuous and of bounded variation, and*

$$DT^{\mathrm{opt}} = -\pi\nu. \tag{108}$$

Equivalently, for $0 \le s < t \le 1$,

$$T^{\mathrm{opt}}(t) - T^{\mathrm{opt}}(s) = -\int_{(s,t]} \pi(r)\,\nu(dr). \tag{109}$$

In terms of the scoring weights,

$$(1-\gamma)DT^{\mathrm{opt}} = (\sigma - \tau)D\Delta. \tag{110}$$

Proof. Use the fixed policy coordinates from Appendix B.3:

$$\zeta(\tau) = (\Delta(\tau), y(\tau)), \qquad y(\tau) = U^*(\tau) - \tau\Delta(\tau), \qquad T^{\mathrm{opt}}(\tau) = \widetilde{T}(\zeta(\tau)).$$

The function $\widetilde{T}$ is continuously differentiable on $\operatorname{int} E$ by Lemmas 12 and 17. Uniform interiority and the invertible shear in (71) place the continuous path ζ in a compact convex subset K of $\operatorname{int} E$. Equation (72) gives

$$\nabla\widetilde{T}(\zeta(\tau)) \cdot (1, -\tau) = -\pi(\tau).$$

Since Δ is continuous and monotone, it is of bounded variation. The envelope identity and the product rule give

$$Dy = DU^* - D(\tau\Delta) = \Delta\,d\tau - [\Delta\,d\tau + \tau\nu] = -\tau\nu, \qquad D\zeta = (1, -\tau)\nu.$$

The chain rule along this continuous path therefore yields

$$DT^{\mathrm{opt}} = \nabla\widetilde{T}(\zeta) \cdot D\zeta = -\pi\nu.$$

For completeness, this chain rule follows directly from partitions. The gradient is bounded and uniformly continuous on K, so $\widetilde{T}$ is Lipschitz there and T^{opt} has bounded variation. On each segment between successive path points, the increment of $\widetilde{T}$ equals its gradient at the initial point times the path increment, plus a remainder. The sum of the absolute remainders is bounded by the total variation of ζ times the largest oscillation of the gradient on these segments. That oscillation tends to zero with the partition mesh, by continuity of ζ and uniform continuity of the gradient. The main sums converge to the Stieltjes integral of $\nabla\widetilde{T}(\zeta)$ against $D\zeta$. This proves (108) and (109), including any singular continuous variation of Δ.

Finally, (103) gives

$$1 - \gamma = \frac{1}{1-m}, \qquad \sigma - \tau = -\frac{\pi}{1-m}.$$

Multiplying (108) by $1/(1-m)$ proves (110). ∎

D.3 Ordering and compression

Proof of Theorem 2. Lemma 39 proves part (a). Since $\pi \geq 0$ and $\nu \geq 0$, (109) makes T^{opt} nonincreasing.

To obtain the range bounds, fix a scoring weight σ. Raising γ can only add accepted states with $\theta > 0$ or remove accepted states with $\theta < 0$. Thus, for $\gamma_1 \leq \gamma_2$,

$$\theta\big(q^{\sigma,\gamma_2} - q^{\sigma,\gamma_1}\big) \geq 0 \qquad \text{state by state,}$$

and $T(q^{\sigma,\gamma_1}) \leq T(q^{\sigma,\gamma_2})$. Applying this observation to the undistorted endpoint weights and the adjustment signs in (107) gives

$$T(q_0) = T(q^{0,\gamma(0)}) \leq T(q^{0,0}) = T^{\mathrm{fb}}(0), \qquad T(q_1) = T(q^{1,\gamma(1)}) \geq T(q^{1,0}) = T^{\mathrm{fb}}(1).$$

Together with monotonicity, these are precisely

$$T^{\mathrm{fb}}(1) \leq T(q_1) \leq T(q_0) \leq T^{\mathrm{fb}}(0),$$

proving part (b).

If T^{fb} is strictly decreasing, fix $s < t$. Lemma 39 supplies a point in (s,t) where $\pi > 0$. Continuity gives a nonempty open interval $J \subset (s,t)$ and a constant $c > 0$ with $\pi \geq c$ on J. Since the optimal utility has no affine pieces, $\nu(J) > 0$. Hence

$$\int_{(s,t]} \pi(r)\,\nu(dr) \geq c\nu(J) > 0,$$

and (109) gives $T(q_t) < T(q_s)$.

If T^{fb} is constant, integrate (106) over $[0,1]$. The endpoint terms vanish and $(T^{\mathrm{fb}})' = 0$ almost everywhere, so $\int_0^1 \mathcal{L} \leq 0$. Nonnegativity and continuity give $\mathcal{L} \equiv 0$. Lemma 38 identifies the first-best pair at every type, and uniqueness of the maximizing policy gives $q_\tau = q_\tau^{\mathrm{fb}}$ almost surely. Finally, exchanging x_1 and x_2 and replacing τ by $1-\tau$ gives the conclusions when x_1 is favored. ■

A singleton pool at the lowest type. The one-sided distortion and benefit-ordering conclusions also hold when $U^*(0) = 0 < U^*(\tau)$ for $\tau > 0$, as stated after Theorem 2. In this case Appendix B.4 gives $\pi' = m$ on $(0,1)$, $\pi(0) \geq 0$, and $\pi(1) = 0$. Every negative component of π still has zero values at both ends: if its left endpoint is 0, continuity forces $\pi(0) = 0$. The proof of Lemma 39 therefore still gives $\pi \geq 0$ and $\sigma \leq \tau$. The private-benefit identity is unchanged, so $T(q_\tau)$ is nonincreasing, and strictly decreasing when T^{fb} is strictly decreasing. At the top, $\sigma(1) = 1$ and $\gamma(1) \geq 0$ still imply $T(q_1) \geq T^{\mathrm{fb}}(1)$. The lower boundary condition allows $\sigma(0) < 0$.

Appendix E Less informative attributes

We prove the remaining results of Section 4.2. Let G_1, G_2, and Π be the fixed marginal laws of x_1, x_2, and θ. Write H_η for the joint law of (x_2, θ) under P_η, so

$$H_0 = G_2 \otimes \Pi, \qquad H_\eta = (1-\eta)H_0 + \eta H_1, \qquad P_\eta = G_1 \otimes H_\eta.$$

For a policy q, denote its statistics and the agent's payoff by

$$s_\eta(q) = \big(S_{1,\eta}(q), S_{2,\eta}(q), T_\eta(q)\big), \qquad S_{i,\eta}(q) = \mathbb{E}_\eta[(x_i - \kappa)q], \qquad U_\tau^\eta(q) = V_\tau^\eta(q) + T_\eta(q).$$

As in Appendix B.5, menus are jointly measurable in type and state. These three statistics determine both parties' payoffs for every true type, regardless of the policy's reported type.

E.1 The benchmark and preliminary identities

Fixed match quality and the garbling comparison. Under every P_η, the match attributes have joint law $G_1 \otimes G_2$. Hence the first-best policy $q_\tau^{\mathrm{fb}} = q^{\tau,0}$ and its principal payoff $V^{\mathrm{fb}}(\tau)$ do not vary with η. Under P_0, this policy depends only on variables independent of the mean-zero private benefit, so

$$T_0(q_\tau^{\mathrm{fb}}) = 0.$$

Linearity in the prior then gives

$$T_\eta(q_\tau^{\mathrm{fb}}) = (1-\eta)T_0(q_\tau^{\mathrm{fb}}) + \eta T_1(q_\tau^{\mathrm{fb}}) = \eta T_1^{\mathrm{fb}}(\tau),$$

which is (9).

For $0 \le \eta_1 < \eta_2$, put $\lambda = \eta_1/\eta_2$. Retaining x_2 with probability λ and otherwise drawing it independently from G_2 transforms its joint law with θ into

$$\lambda H_{\eta_2} + (1-\lambda)(G_2 \otimes \Pi) = H_{\eta_1}.$$

The replacement uses no information about θ, so this is a garbling of x_2 as a signal about θ. It also leaves x_1 independent of the pair and preserves the joint law of the match attributes.

Proof of Lemma 4. Under P_0, every first-best rule has zero loading. For any true type τ and report r,

$$U_\tau^0(q_r^{\mathrm{fb}}) = V_\tau^0(q_r^{\mathrm{fb}}) \le V^{\mathrm{fb}}(\tau) = U_\tau^0(q_\tau^{\mathrm{fb}}).$$

The inequality follows from pointwise maximization of the principal's payoff. Thus the first-best menu satisfies (IC). Its truthful utility is $V^{\mathrm{fb}}(\tau) = \mathbb{E}[(x^{(\tau)} - \kappa)^+] \ge 0$, which gives (IR). Since B bounds every menu's payoff, the first-best menu attains $V(0) = B$, and $C(0) = 0$. ■

When the first best is the known-type benchmark. We also verify the sufficient condition in the footnote defining C. Put

$$h(x) = \mathbb{E}_1[\theta \mid x_2 = x], \qquad r_\tau(x) = \Pr_{G_1}\{\tau x_1 + (1-\tau)x > \kappa\}.$$

The function r_τ is nondecreasing. If h is nondecreasing as well, then, for independent copies X, X' with law G_2,

$$T_1^{\mathrm{fb}}(\tau) = \mathbb{E}[h(X)r_\tau(X)] = \frac{1}{2}\mathbb{E}\big[(h(X) - h(X'))(r_\tau(X) - r_\tau(X'))\big] \geq 0,$$

where the second equality uses $\mathbb{E}[h(X)] = \mathbb{E}[\theta] = 0$. Equation (9) gives $T_\eta^{\mathrm{fb}}(\tau) \geq 0$ for every η. Consequently, the first-best rule satisfies participation for each known type:

$$U_\tau^\eta(q_\tau^{\mathrm{fb}}) = V^{\mathrm{fb}}(\tau) + T_\eta^{\mathrm{fb}}(\tau) \geq 0.$$

It attains the principal's unrestricted payoff bound and is therefore the known-type optimum. In this case $C(\eta)$ is exactly the loss due to private type information.

For completeness, likelihood-ratio ordering implies the required ordering of conditional means. For $x < x'$, write $f_x, f_{x'}$ for the two conditional densities of θ with respect to their sum as a dominating measure μ. Likelihood-ratio ordering gives $f_x(a)f_{x'}(b) \geq f_x(b)f_{x'}(a)$ for $a < b$. Thus

$$\mathbb{E}_1[\theta \mid x_2 = x'] - \mathbb{E}_1[\theta \mid x_2 = x] = \int_{a<b} (b-a)\big[f_x(a)f_{x'}(b) - f_x(b)f_{x'}(a)\big]\, d\mu(a)\, d\mu(b) \geq 0.$$

E.2 Combining menus across priors

Lemma 41 (Mixing feasible menus)**.** *Let $\eta_a, \eta_b \in [0,1]$, $t \in [0,1]$, and $\eta_t = (1-t)\eta_a + t\eta_b$. If menus $(q_r^a)_r$ and $(q_r^b)_r$ satisfy (IC) and (IR) under P_{η_a} and P_{η_b}, respectively, there is a menu $(q_r^t)_r$ satisfying these constraints under P_{η_t} with*

$$s_{\eta_t}(q_r^t) = (1-t)s_{\eta_a}(q_r^a) + t s_{\eta_b}(q_r^b) \qquad \textit{for every report } r. \tag{111}$$

Proof. Take the common dominating measure $\mu = P_0 + P_1$, choose nonnegative density versions ρ_0, ρ_1, and put $\rho_\eta = (1-\eta)\rho_0 + \eta\rho_1$. Then $\rho_\eta = dP_\eta/d\mu$, and the affine identity holds pointwise:

$$\rho_{\eta_t} = (1-t)\rho_{\eta_a} + t\rho_{\eta_b}.$$

Define

$$q_r^t(\omega) = \frac{(1-t)\rho_{\eta_a}(\omega)q_r^a(\omega) + t\rho_{\eta_b}(\omega)q_r^b(\omega)}{\rho_{\eta_t}(\omega)} \quad \text{when } \rho_{\eta_t}(\omega) > 0, \tag{112}$$

and set $q_r^t = 0$ elsewhere. This is a measurable policy with values in $[0,1]$. Integrating against $x_1 - \kappa, x_2 - \kappa$, and θ proves (111). In particular, for every true type τ and report r,

$$U_\tau^{\eta_t}(q_r^t) = (1-t)U_\tau^{\eta_a}(q_r^a) + tU_\tau^{\eta_b}(q_r^b). \tag{113}$$

Each component menu gives type τ at least as much utility from reporting τ as from reporting r. Taking the same convex combination of these inequalities proves (IC) for the new menu. Taking the convex combination of the two truthful participation inequalities proves (IR). ■

Proof of Theorem 3. The identity $P_{\eta_1} = (1-\lambda)P_0 + \lambda P_{\eta_2}$ allows us to apply Lemma 41 to the first-best menu under P_0 and the given menu under P_{η_2}. The first component is feasible by Lemma 4. The resulting menu $(\tilde{q}^1_\tau)_\tau$ is feasible under P_{η_1} and satisfies

$$V^{\eta_1}_\tau(\tilde{q}^1_\tau) = (1-\lambda)V^{\mathrm{fb}}(\tau) + \lambda V^{\eta_2}_\tau(q^2_\tau) \qquad (0 \le \tau \le 1).$$

Subtracting from $V^{\mathrm{fb}}(\tau)$ gives (11). ■

Proof of Corollary 3.1. Write $V(\eta)$ as the supremum of the principal's expected payoff over feasible menus under P_η. It is finite: the zero policy is feasible, and the pointwise first-best bound gives $0 \le V(\eta) \le B$.

Apply Theorem 3 to a feasible menu whose payoff is within ε of $V(\eta_2)$. Integrating its pointwise payoff identity and letting $\varepsilon \downarrow 0$ yields

$$V(\eta_1) \ge (1-\lambda)B + \lambda V(\eta_2) \ge V(\eta_2). \tag{114}$$

Equivalently,

$$C(\eta_1) \le \lambda C(\eta_2) \le C(\eta_2).$$

These inequalities give the claimed monotonicity and proportional bound. If $C(\eta_2) > 0$, the first bound also gives

$$V(\eta_1) - V(\eta_2) \ge (1-\lambda)C(\eta_2) > 0.$$

For concavity, fix η_a, η_b and $t \in [0,1]$. Choose feasible menus within ε of the optimal values at these two priors and apply Lemma 41. Its statistic identity gives a feasible menu at $\eta_t = (1-t)\eta_a + t\eta_b$ with the corresponding convex combination of payoffs. Letting $\varepsilon \downarrow 0$ gives

$$V(\eta_t) \ge (1-t)V(\eta_a) + tV(\eta_b).$$

Thus V is concave, and $C = B - V$ is convex. ■

E.3 Correcting acceptance decisions

The construction below retains part of each interval of mistaken decisions. Matching both its probability and its first-score integral preserves all payoff components when x_2 and θ are fixed.

Lemma 42 (Retaining an interval with the same mean)**.** *Let G be an atomless probability law on $[0,1]$. For every interval $I \subseteq [0,1]$ and $w \in [0,1]$, there is an interval $J \subseteq I$, possibly empty, such that*

$$G(J) = wG(I), \qquad \int_J x\, dG(x) = w \int_I x\, dG(x). \tag{115}$$

The intervals can be chosen measurably as the endpoints of I and w vary. When G is uniform, retaining the midpoint of I and multiplying its length by w gives such an interval.

Proof. Take $J = \varnothing$ if $G(I) = 0$ or $w = 0$, and take $J = I$ if $w = 1$. Otherwise, let F_G be the continuous distribution function and Q_G its generalized inverse. In probability coordinates, the interval I occupies $[A, B]$, up to null endpoints. Put

$$M = B - A = G(I) > 0, \qquad m = wM \in (0, M), \qquad S = \int_A^B Q_G(u)\, du = \int_I x\, dG(x).$$

Slide a probability interval of length m across $[A, B]$. Its first moment is

$$L(z) = \int_z^{z+m} Q_G(u)\, du, \qquad A \le z \le B - m.$$

This function is continuous and nondecreasing. The bottom interval has mean no larger than the mean over $[A, B]$, and the top interval has mean no smaller. Hence

$$L(A) \le wS \le L(B - m).$$

The intermediate value theorem supplies a solution to $L(z) = wS$. Choose the leftmost solution and set

$$J = I \cap \{x : z \le F_G(x) \le z + m\}.$$

The probability integral transform gives (115). The intersection with I ensures literal containment even across gaps in the support and at null endpoints.

To verify measurability, parameterize the sliding interval by $z = A + r(B - m - A)$, $r \in [0,1]$. Its first moment is measurable in the interval endpoints and w, and continuous and nondecreasing in r. The leftmost crossing of wS is therefore measurable, as can be checked using rational values of r. The resulting set J is measurable in these parameters and x. For a uniform law, a centered subinterval preserves the mean directly. ■

Proof of Corollary 3.2. If $\eta_1 = 0$, take $q_\tau^1 = q_\tau^{\text{fb}}$. This is the constructed menu when $\lambda = 0$, and its error sets are empty. Hence assume $\eta_1 > 0$.

Write $z = (x_2, \theta)$. The identity

$$H_{\eta_1} = (1 - \lambda)H_0 + \lambda H_{\eta_2}$$

allows a measurable likelihood weight

$$w(z) = \lambda \frac{dH_{\eta_2}}{dH_{\eta_1}}(z) \in [0, 1], \qquad w\, dH_{\eta_1} = \lambda\, dH_{\eta_2}. \tag{116}$$

Choose its values in $[0, 1]$ on any null exceptional set. Independence of x_1 means that its conditional law is always G_1, and that w does not depend on x_1. The randomized menu constructed for Theorem 3 can therefore be written, almost surely, as

$$\tilde{q}_\tau^1(x_1, z) = q_\tau^{\text{fb}}(x_1, z) + w(z)\big[q_\tau^2(x_1, z) - q_\tau^{\text{fb}}(x_1, z)\big]. \tag{117}$$

For each type τ and pair z, separate mistaken acceptances from mistaken rejections:

$$I_\tau^+(z) = \{x_1 : q_\tau^2(x_1, z) = 1,\ q_\tau^{\text{fb}}(x_1, z) = 0\},$$
$$I_\tau^-(z) = \{x_1 : q_\tau^2(x_1, z) = 0,\ q_\tau^{\text{fb}}(x_1, z) = 1\}.$$

Each set is an interval, possibly empty, since both policies are affine thresholds in x_1. This remains true when the old rule's coefficient on x_1 is zero or negative. Discard sections with zero G_1-mass. Every remaining interval has positive length, so its endpoints are the infimum and supremum of the rational points it contains. Membership of each rational point is measurable in (τ, z) by joint measurability of the menu; hence these endpoints are measurable. Apply Lemma 42 separately to these intervals, with law G_1 and weight $w(z)$, to obtain $J_\tau^+(z) \subseteq I_\tau^+(z)$ and $J_\tau^-(z) \subseteq I_\tau^-(z)$. Define

$$q_\tau^1(x_1, z) = q_\tau^{\text{fb}}(x_1, z) + \mathbb{1}\{x_1 \in J_\tau^+(z)\} - \mathbb{1}\{x_1 \in J_\tau^-(z)\}. \tag{118}$$

This rule keeps the old decision on the retained error intervals and uses the first-best decision elsewhere. It is jointly measurable and takes values in $\{0, 1\}$. Its error set is contained in the old error set for every state.

The two moment equalities in (115) imply

$$\int \begin{pmatrix} 1 \\ x_1 \end{pmatrix} \big[q_\tau^1(x_1, z) - \tilde{q}_\tau^1(x_1, z)\big]\, dG_1(x_1) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \text{for } H_{\eta_1}\text{-almost every } z.$$

Conditional on z, the coefficients $x_2 - \kappa$ and θ are constant. Integrating over H_{η_1} thus preserves all three policy statistics:

$$s_{\eta_1}(q_\tau^1) = s_{\eta_1}(\tilde{q}_\tau^1) \qquad \text{for every } \tau.$$

Thus all payoffs, and hence (IC) and (IR), are preserved for every true type and report.

Finally, the two signed error intervals are disjoint, and trimming retains the fraction $w(z)$ of each interval's probability. Writing $E_\tau^j(z)$ for the section of the error set at z gives

$$G_1(E_\tau^1(z)) = w(z)G_1(E_\tau^2(z)).$$

Integration using (116) yields

$$P_{\eta_1}(E_\tau^1) = \int w(z)G_1(E_\tau^2(z))\, dH_{\eta_1}(z) = \lambda P_{\eta_2}(E_\tau^2).$$

Together with $E_\tau^1 \subseteq E_\tau^2$, this proves (12). ■